

Supplemental Materials - Model MCMC Diagnostic Plots

For the parameters of interest for each model, we show the posterior trace, histogram, expected cumulative distribution (ECDF), and autocorrelation distribution. Following those are histograms of the autocorrelation (AC.####), potential scale reduction factor (PSRF), and effective sample size (ESS) values for all parameters of interest in the model. Finally for each model, we show the posterior predictive check distribution (PPD) expressed as the distribution of the deviation of the observed values from the PPD values.

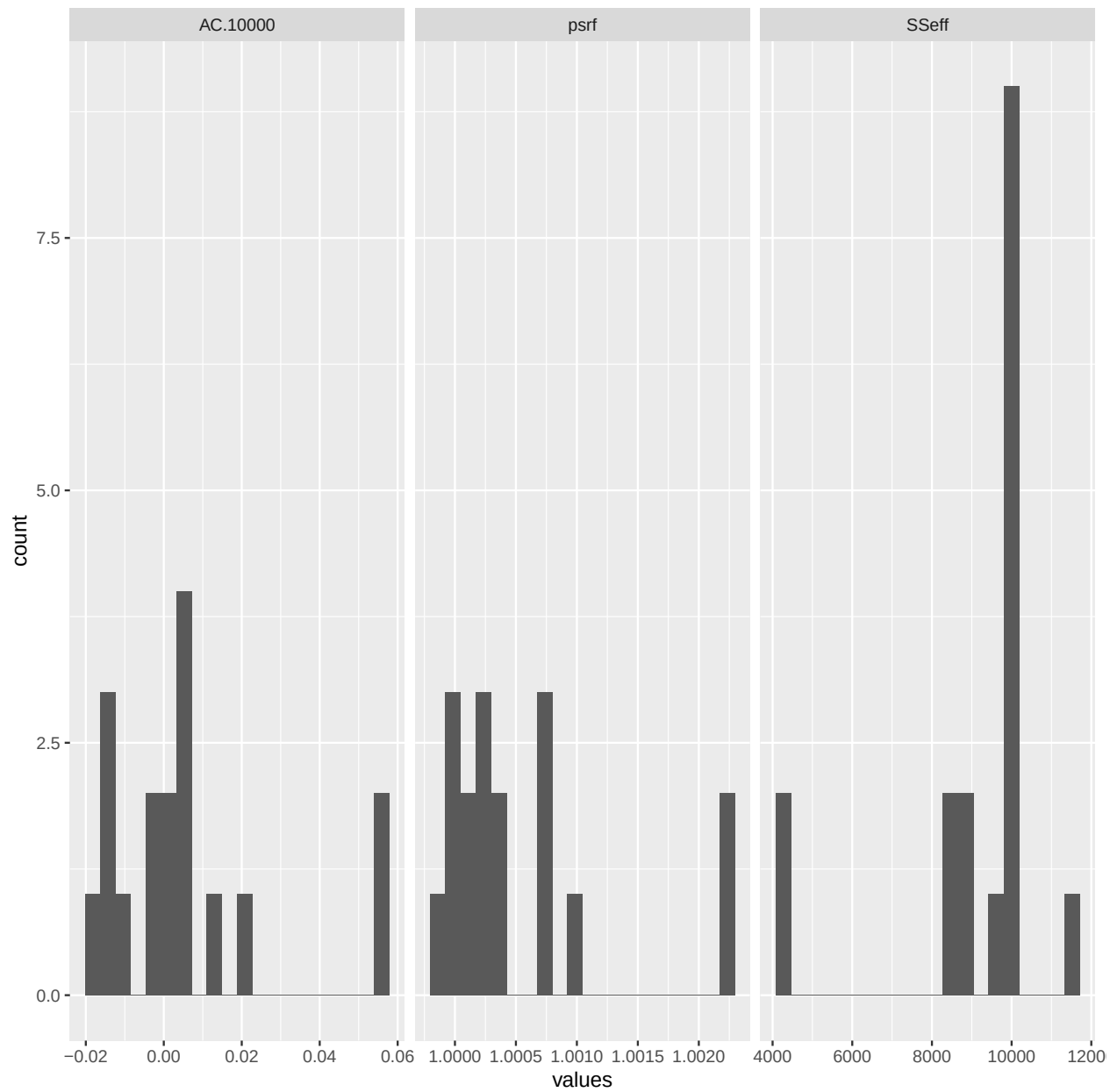
Model I

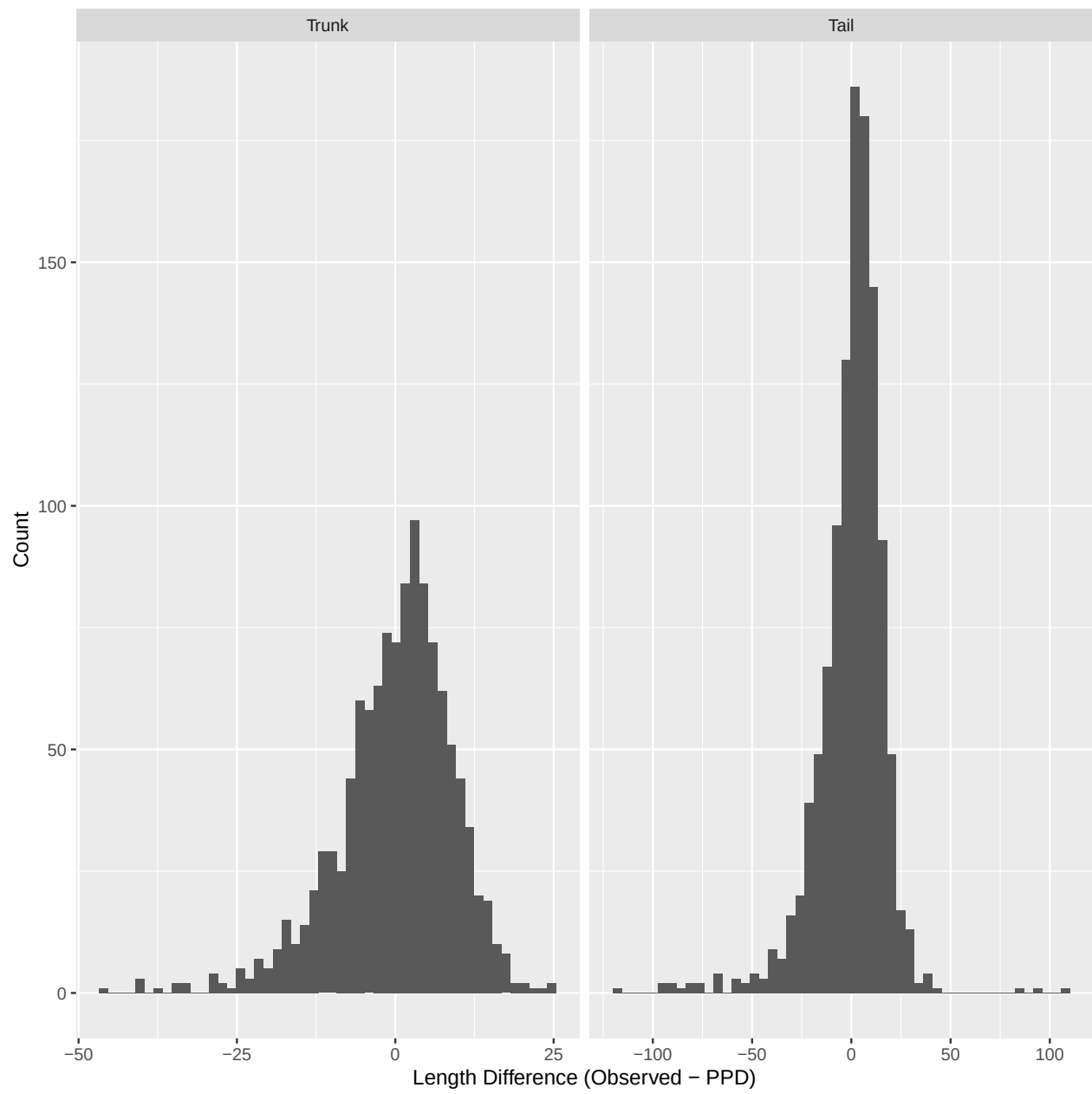
Number of chains: 10

Burnin: 1.001×10^5

Samples per chain: 1000

Total samples: 10000



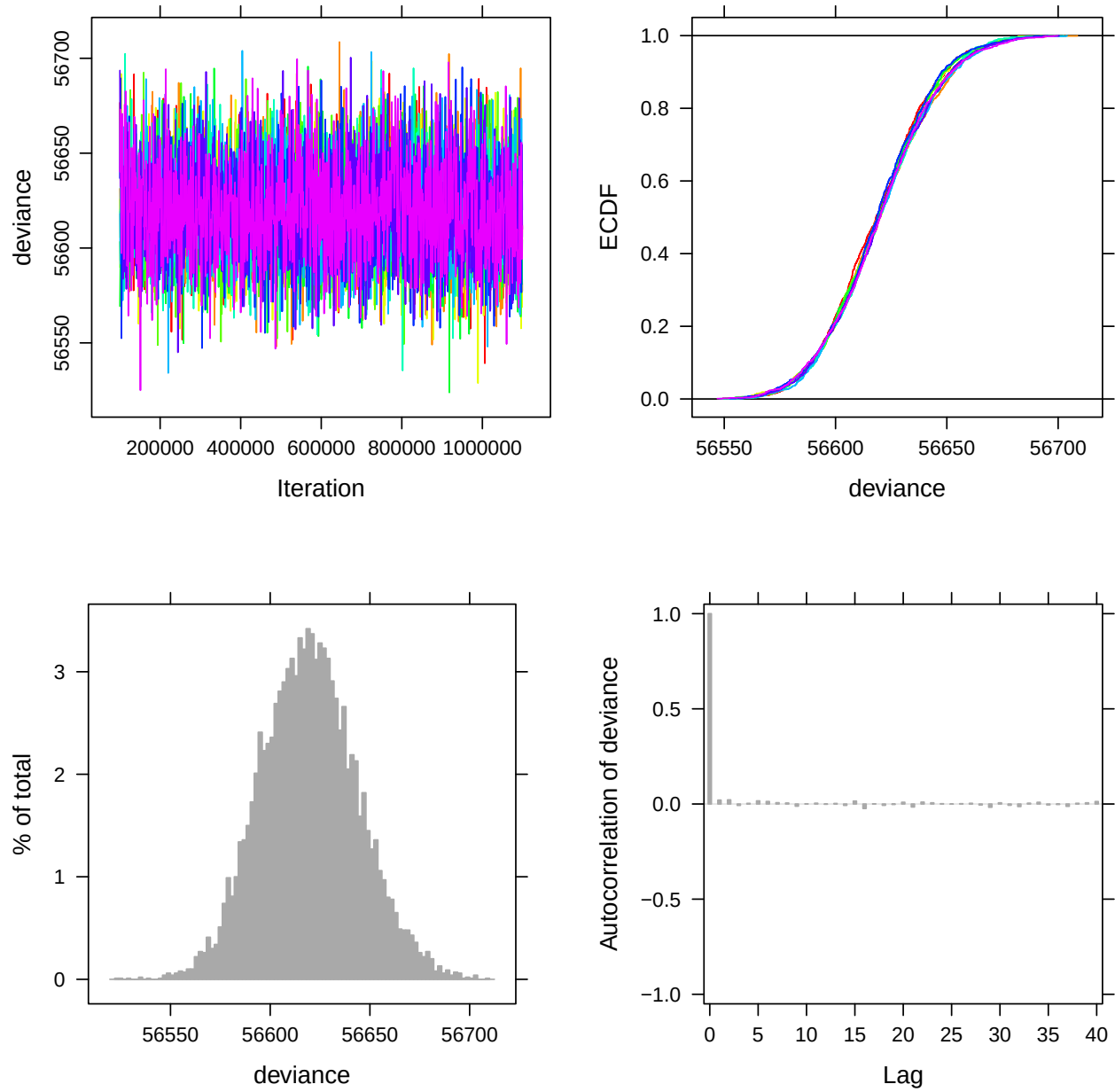


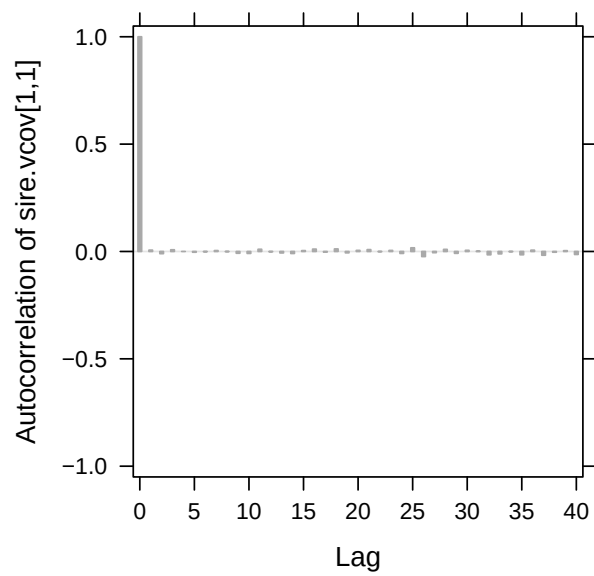
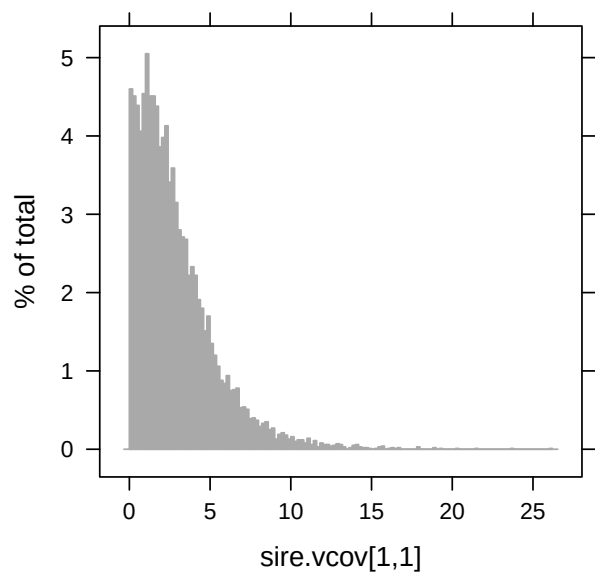
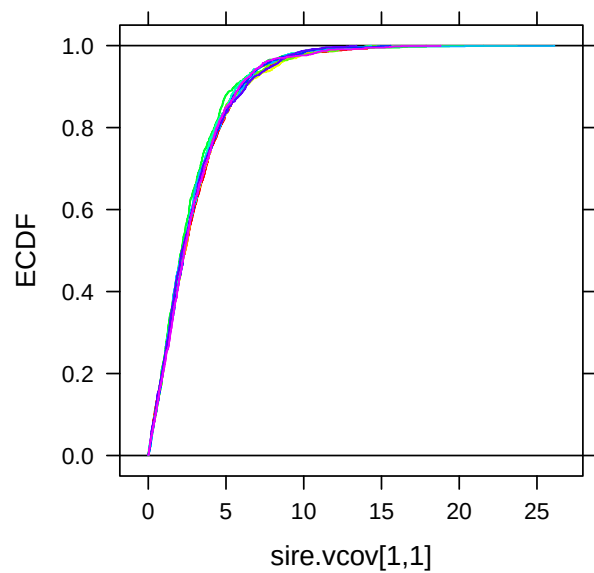
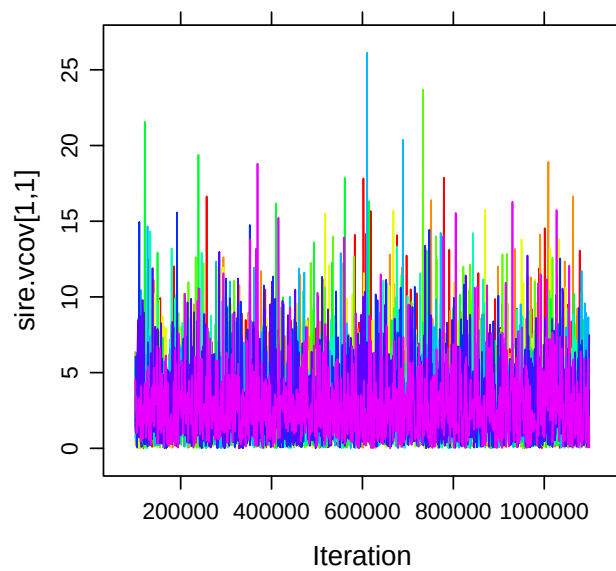
Generating summary statistics and plots (these will NOT be saved for reuse)...

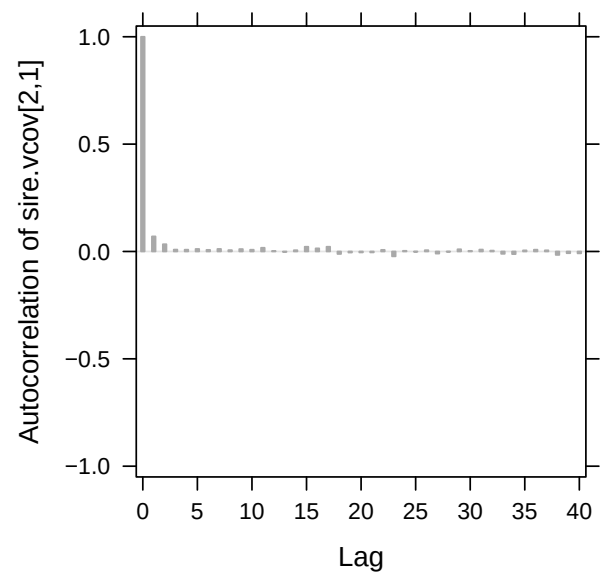
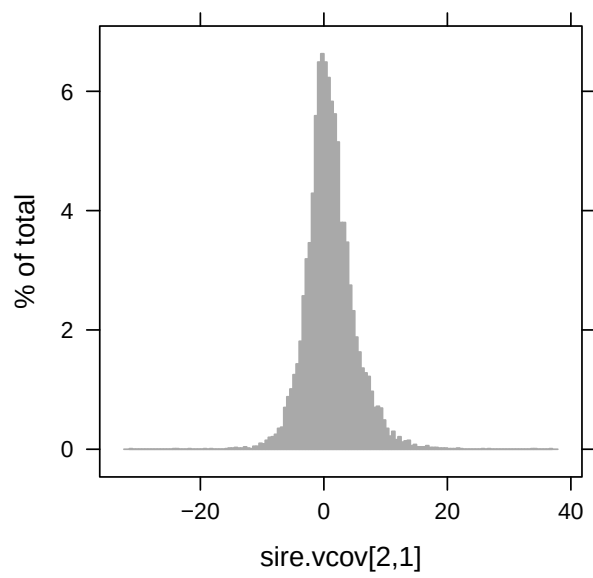
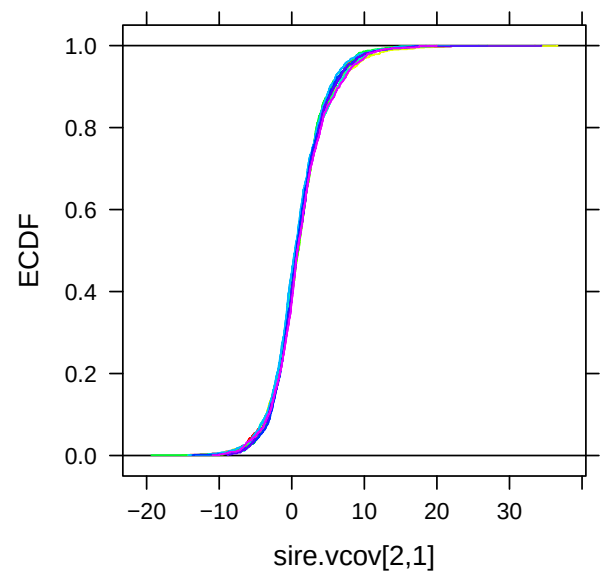
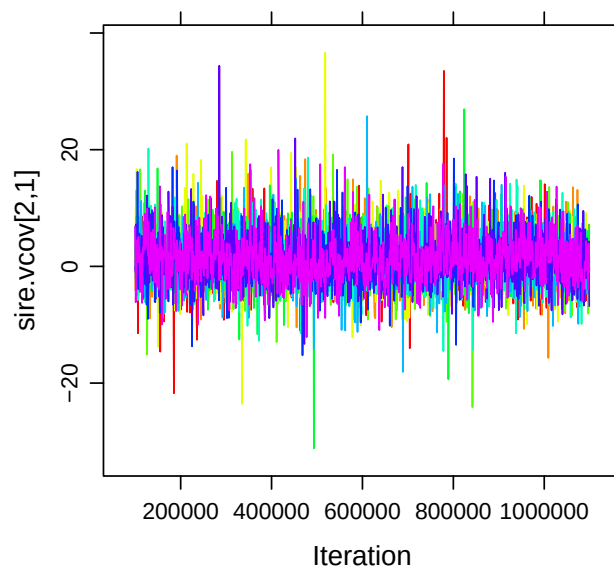
Calculating summary statistics...

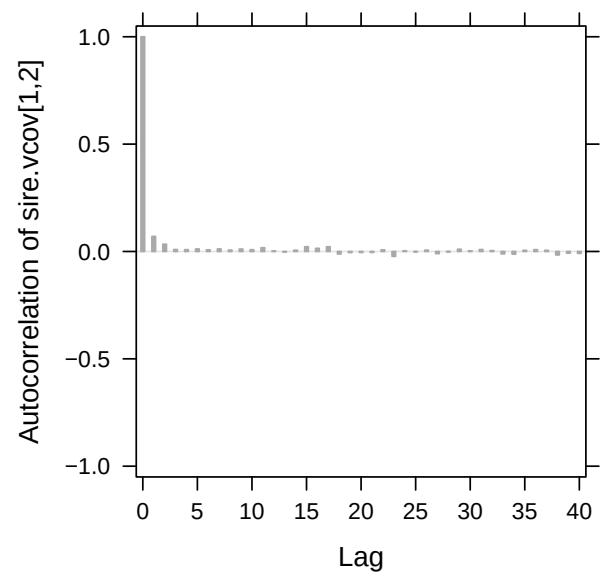
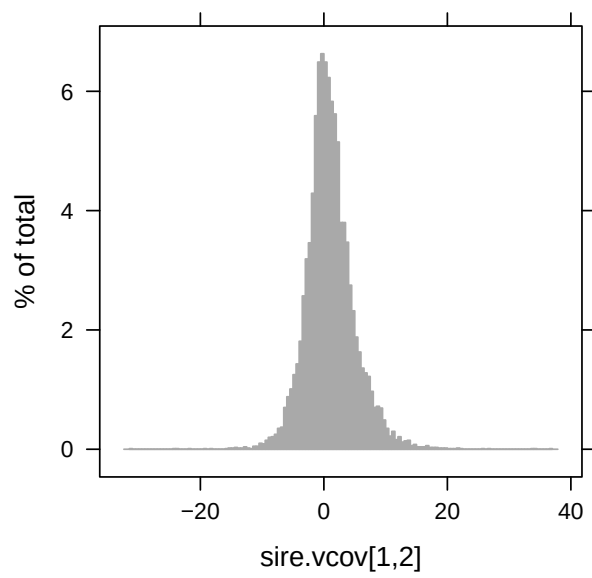
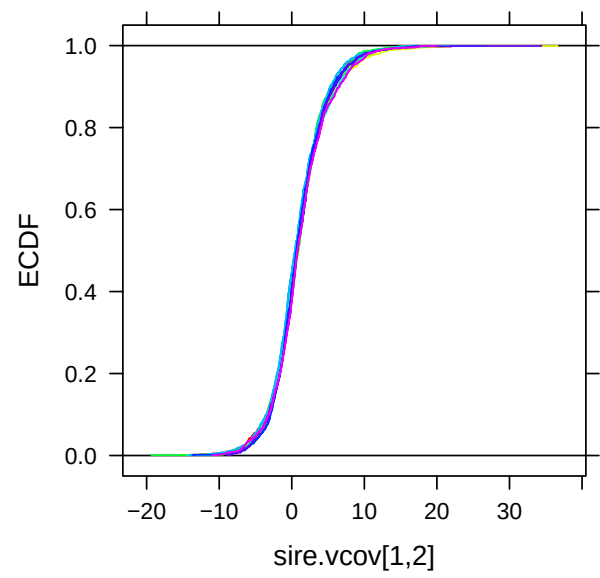
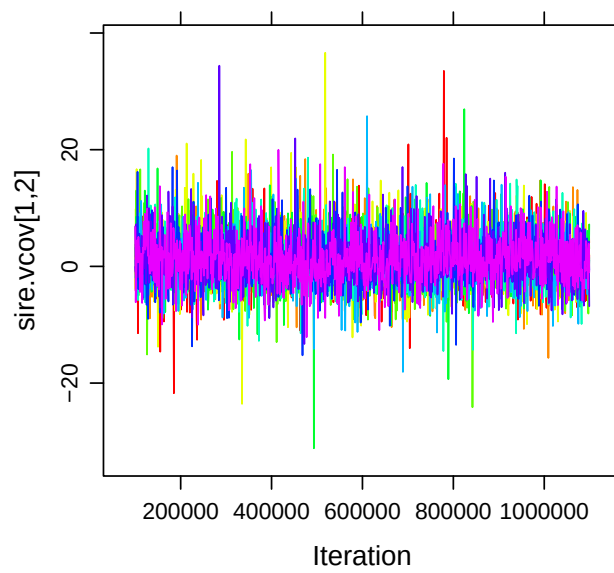
Calculating the Gelman-Rubin statistic for 17 variables....

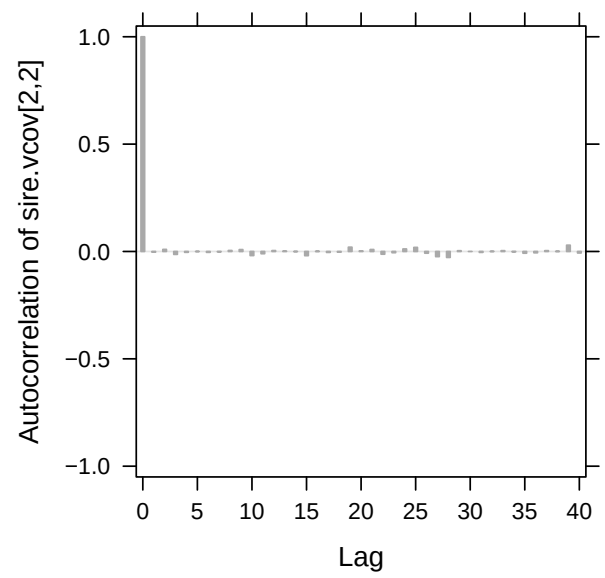
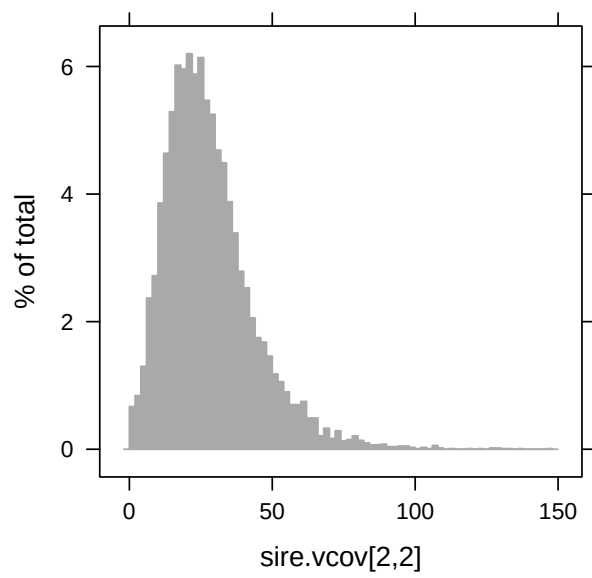
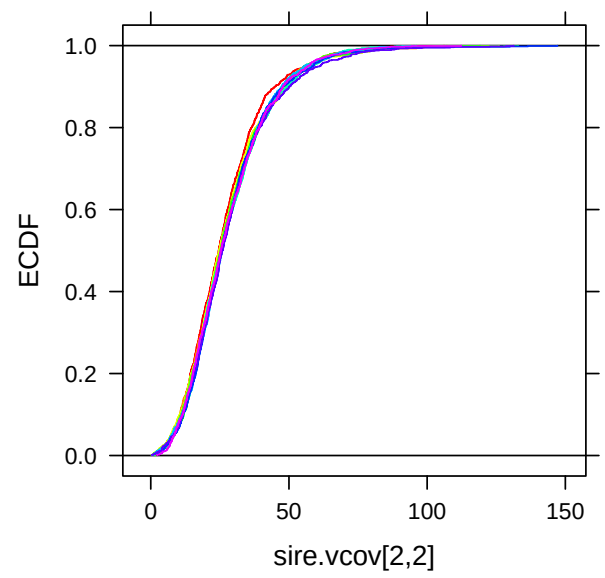
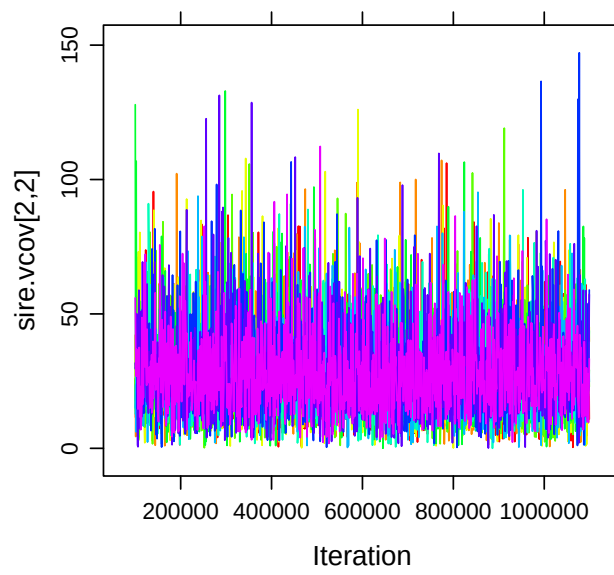
Note: Unable to calculate the multivariate psrf

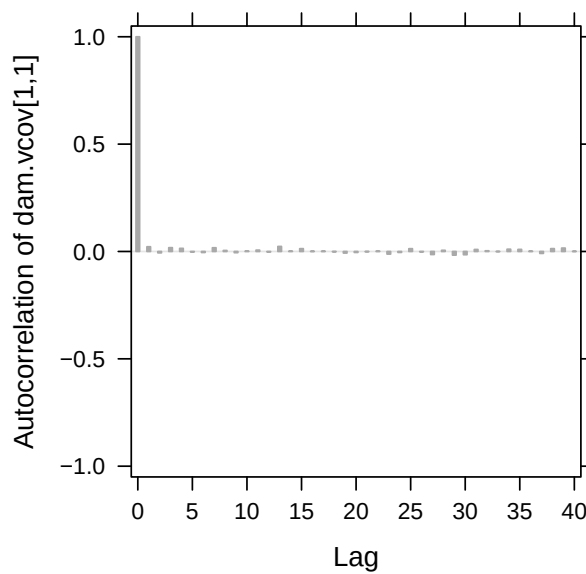
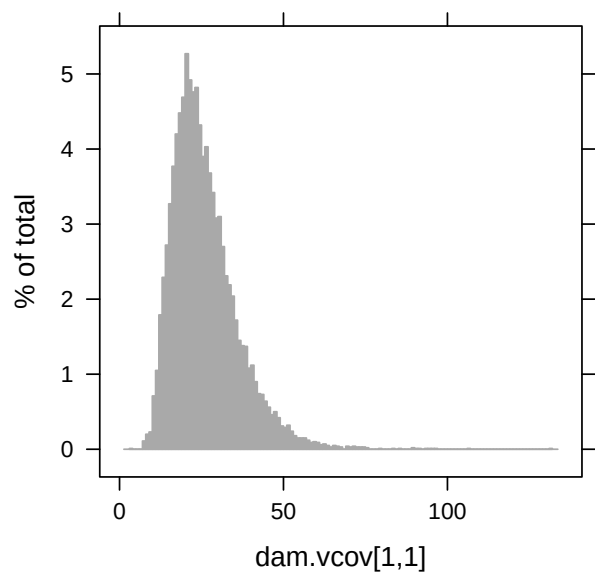
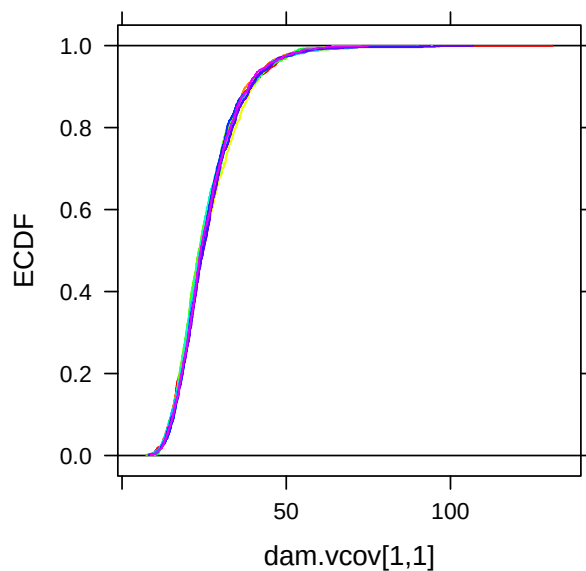
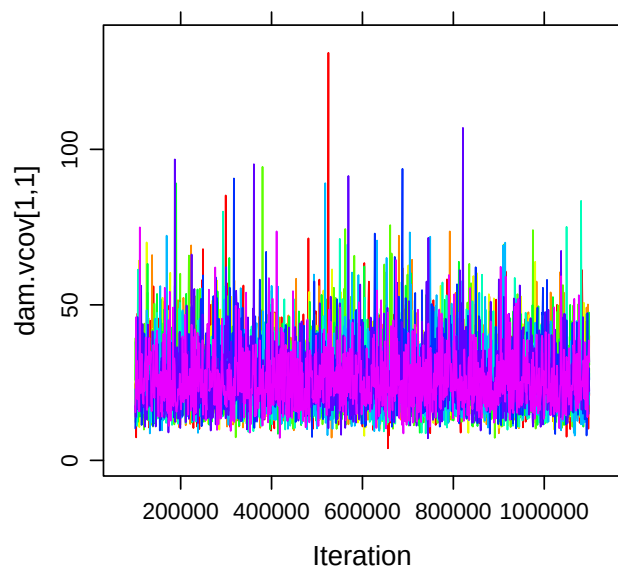


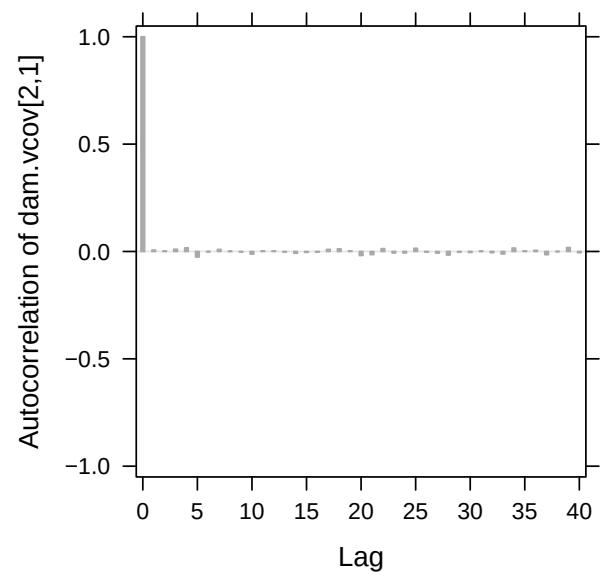
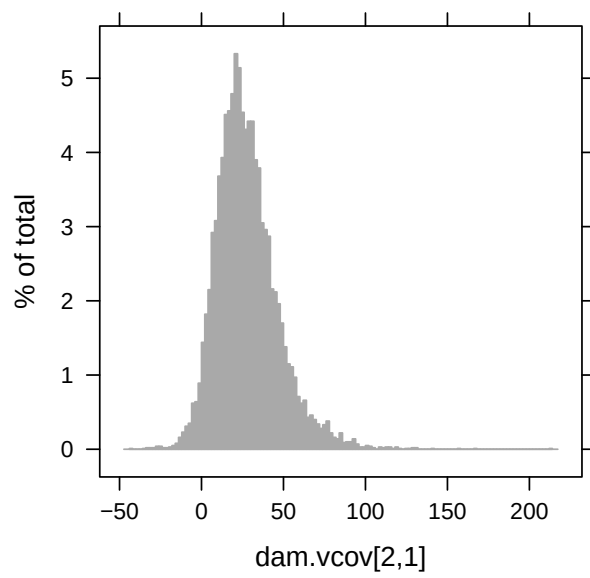
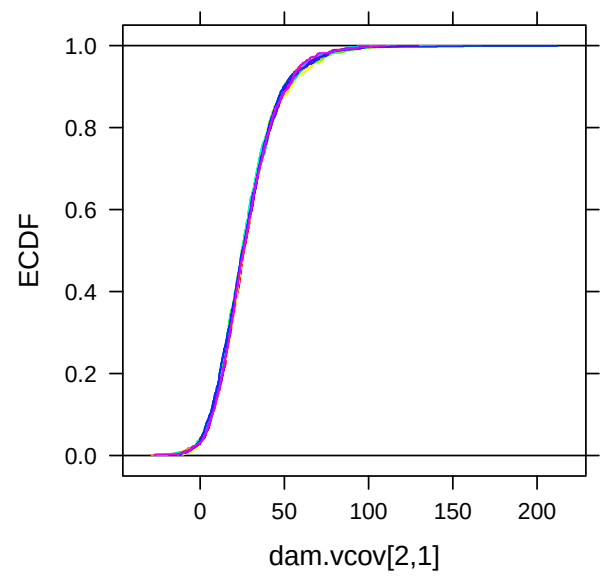
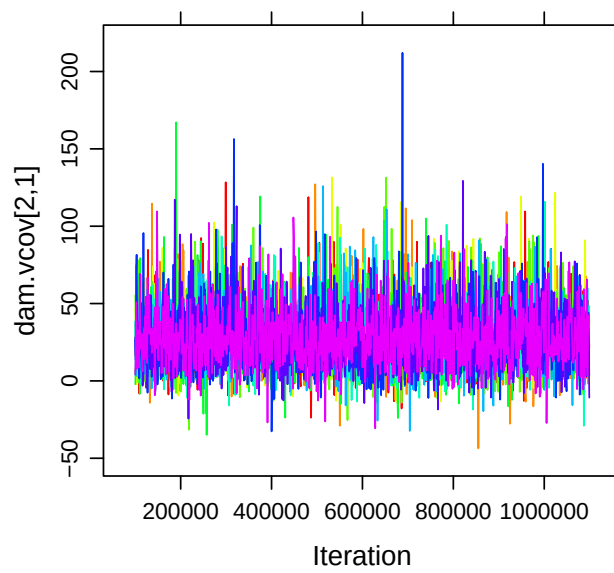


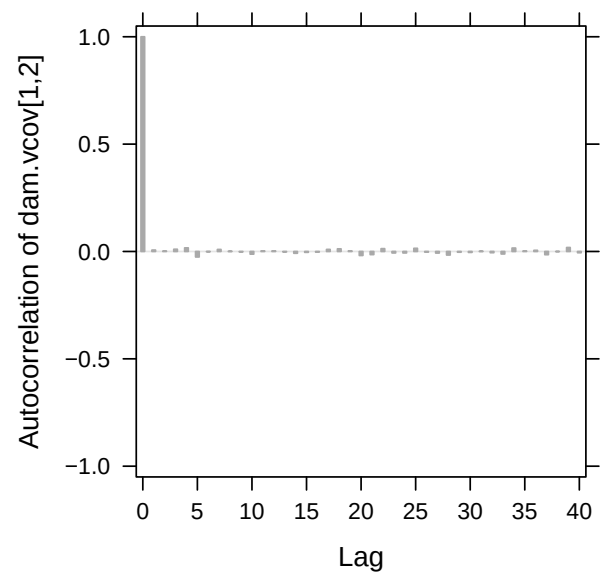
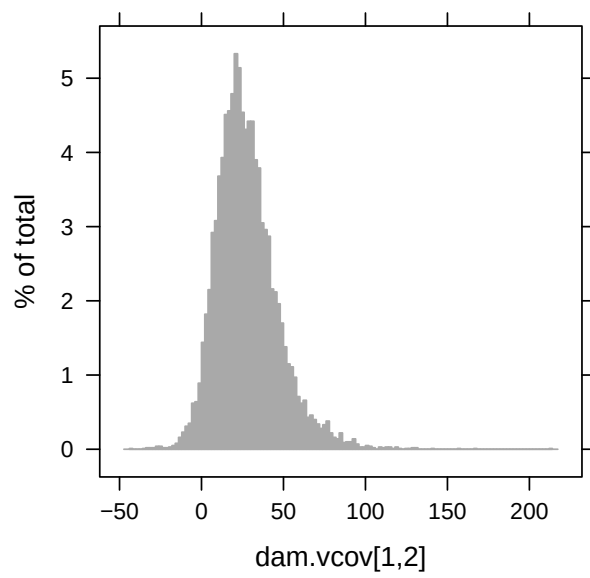
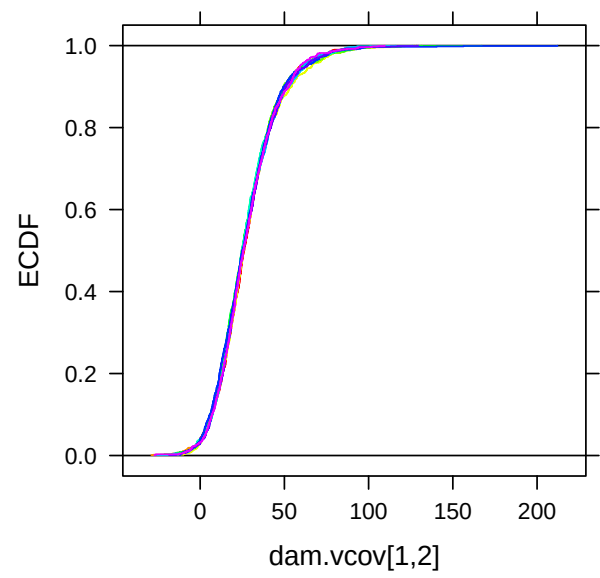
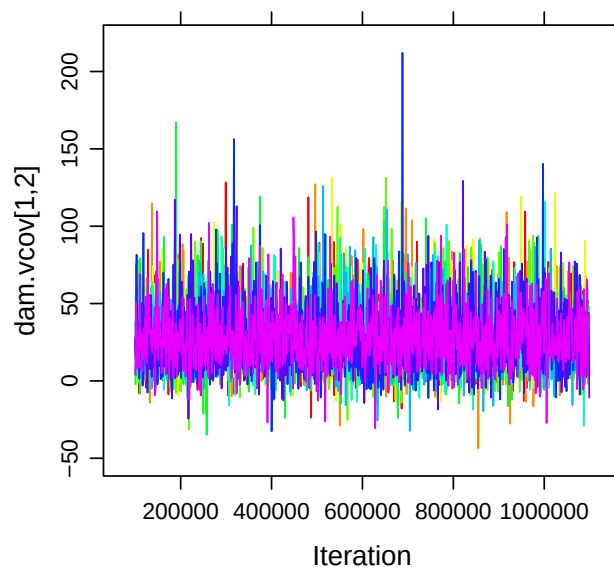


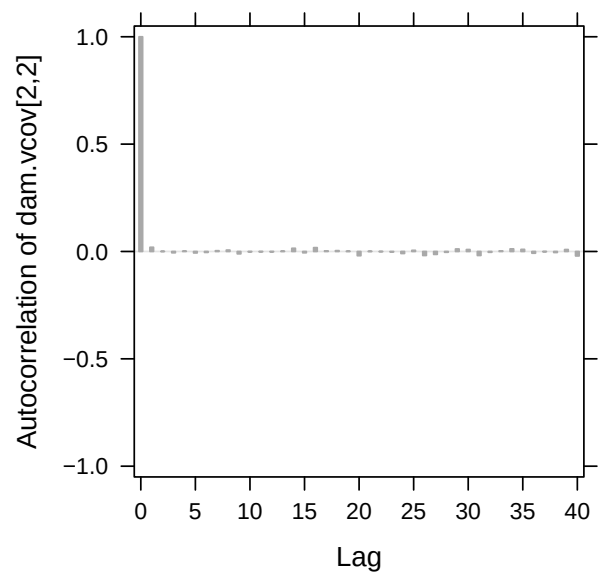
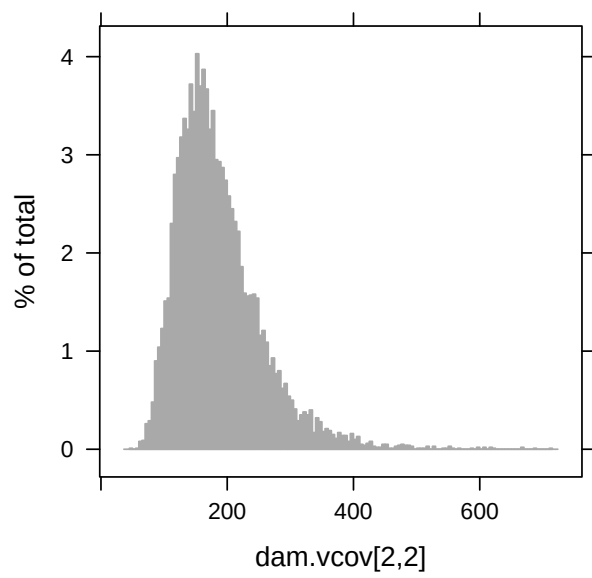
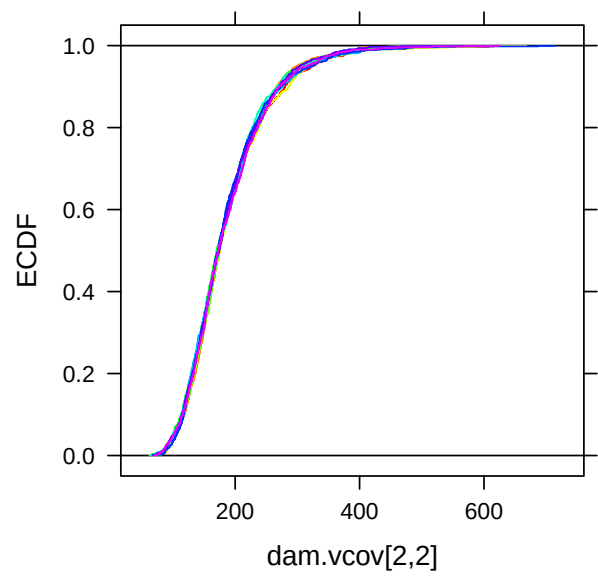
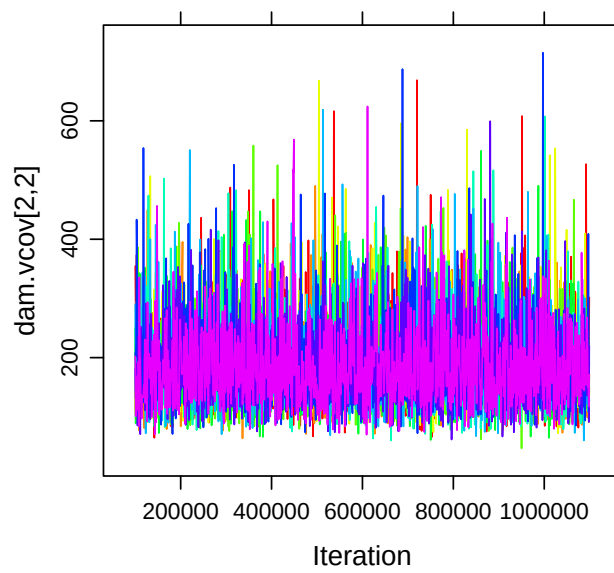


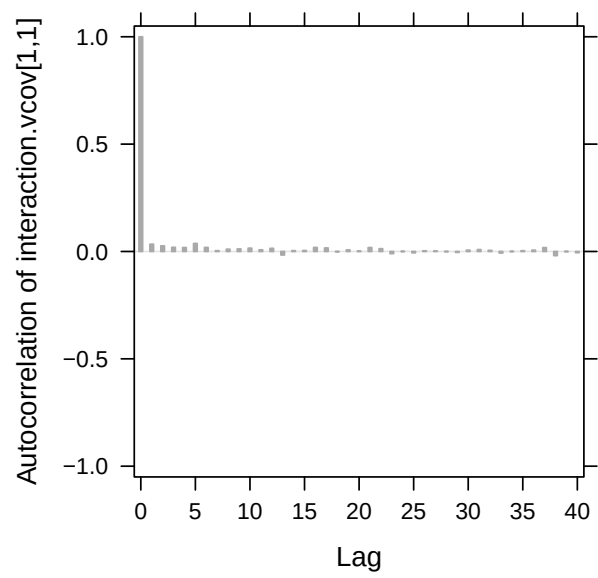
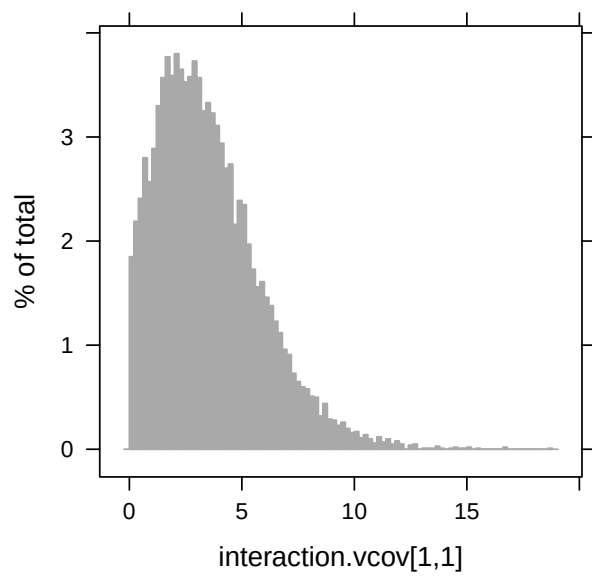
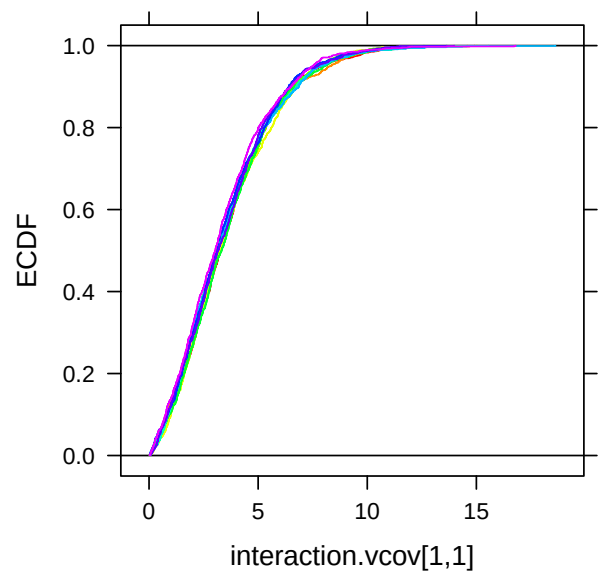
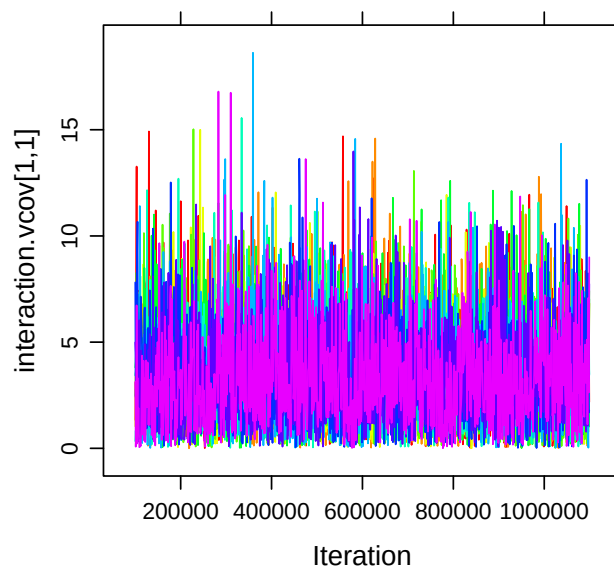


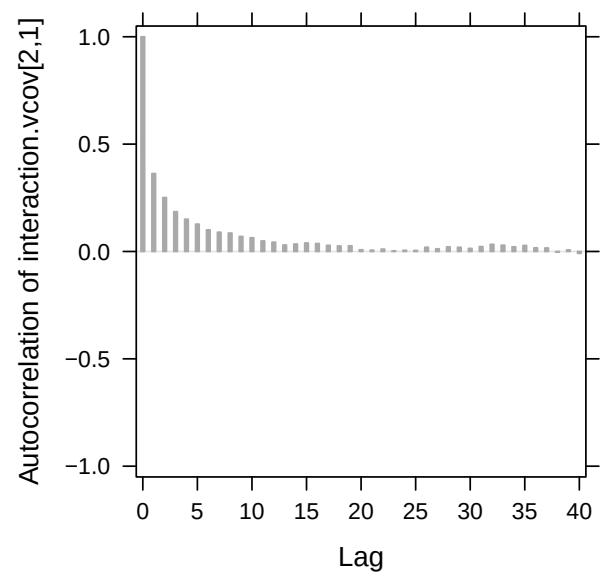
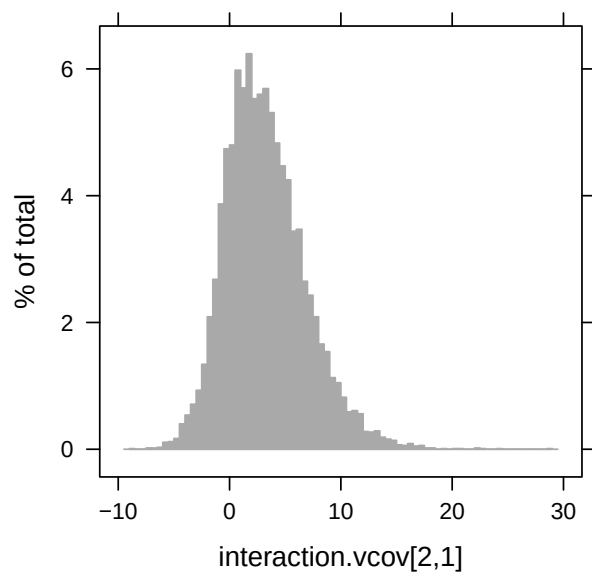
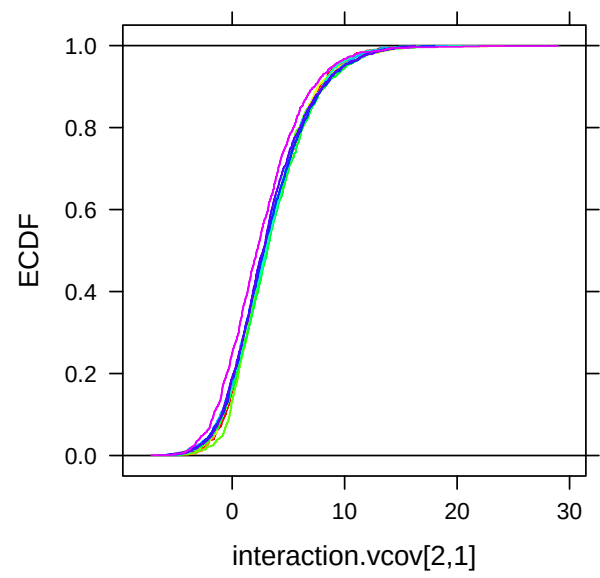
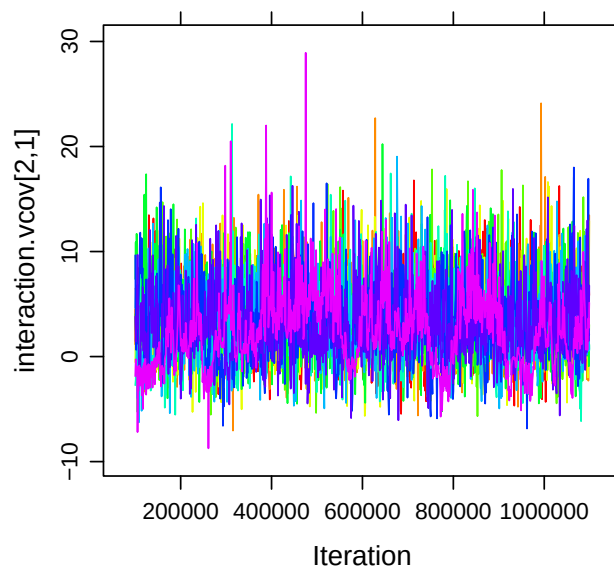


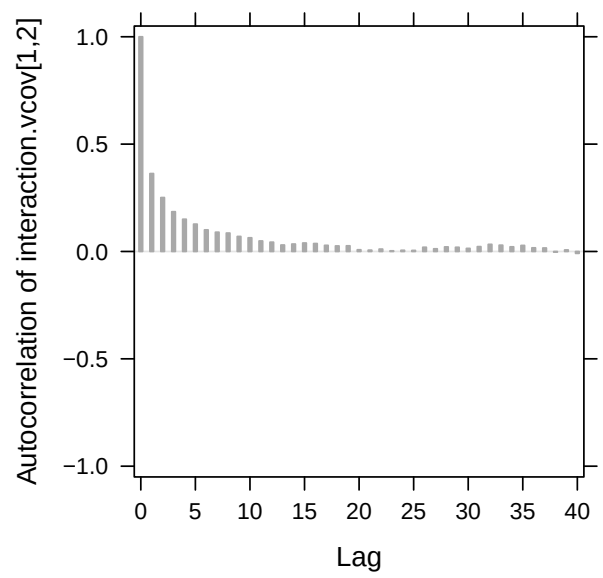
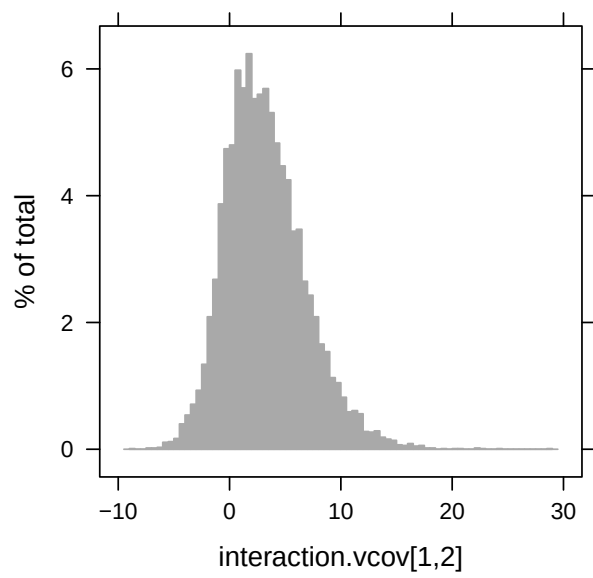
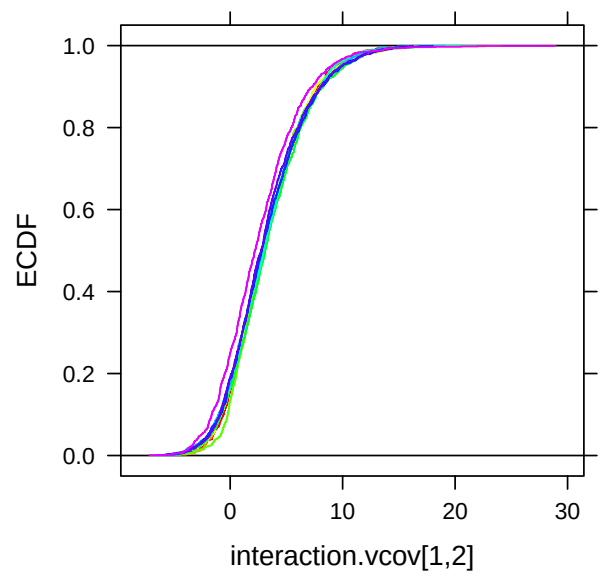
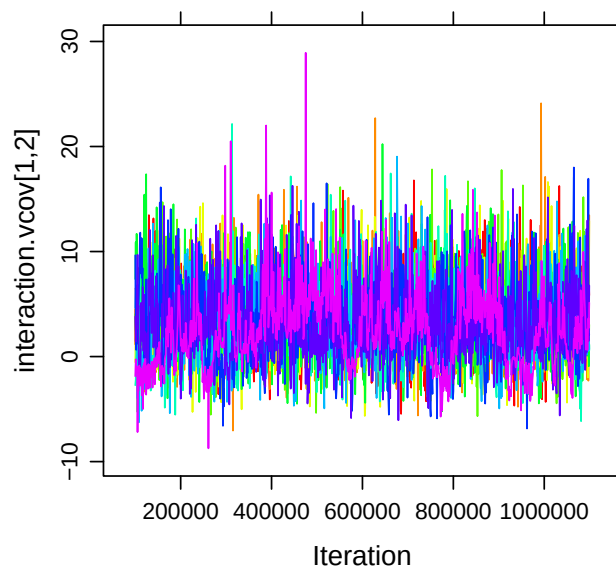


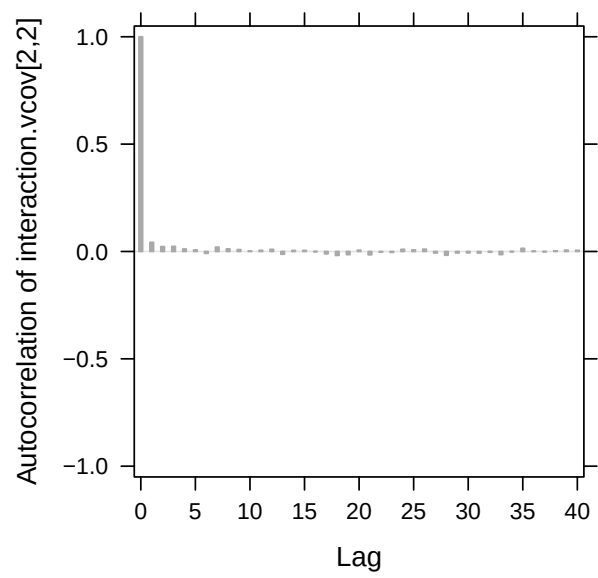
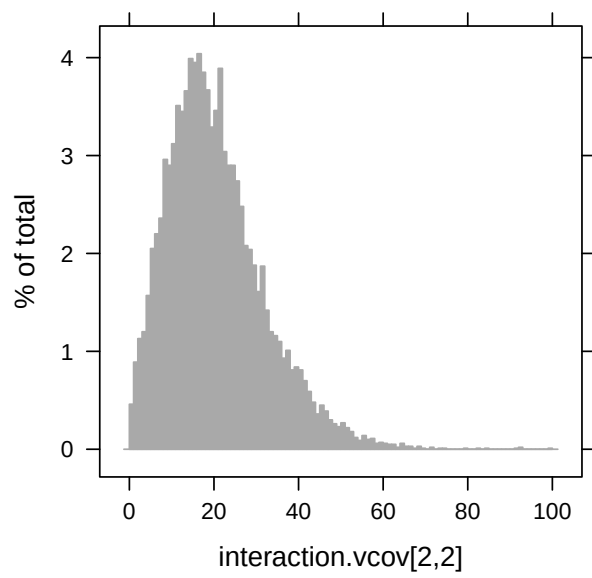
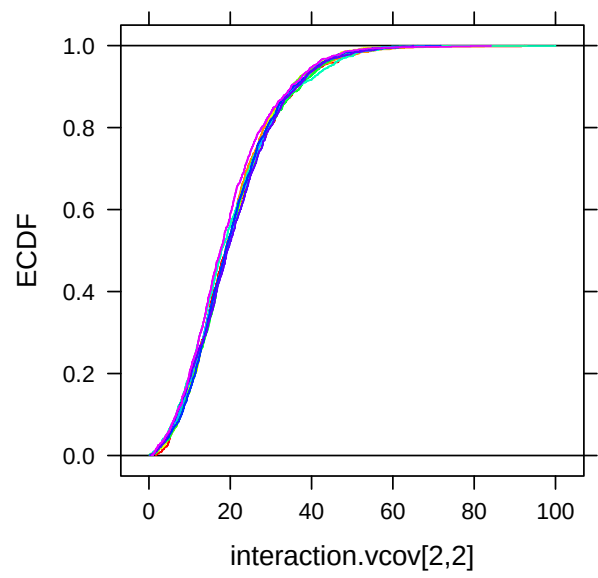
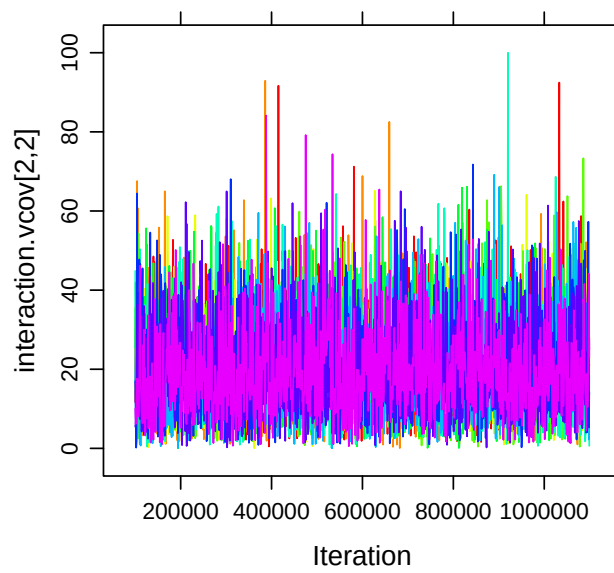


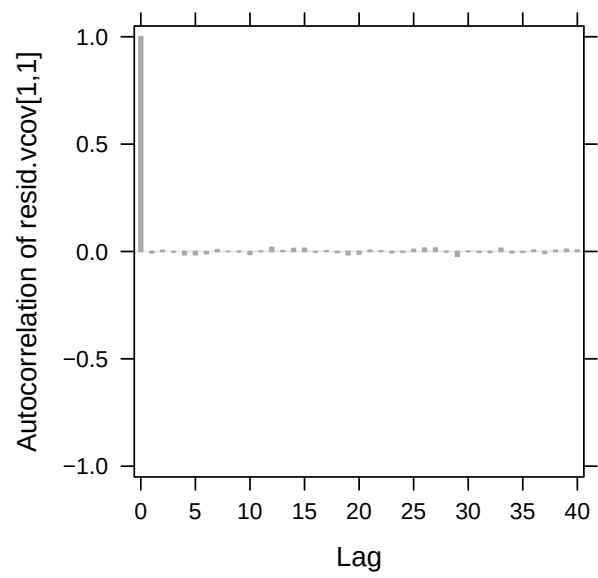
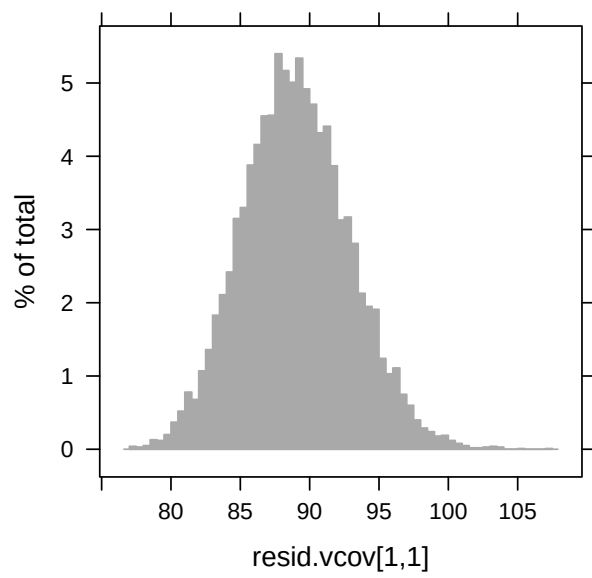
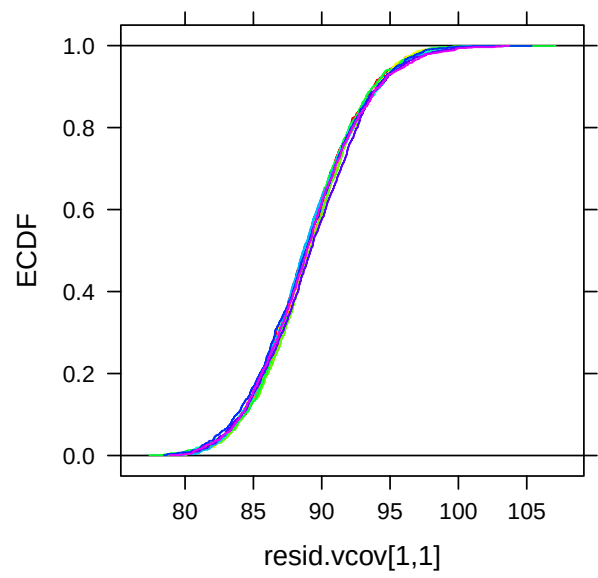
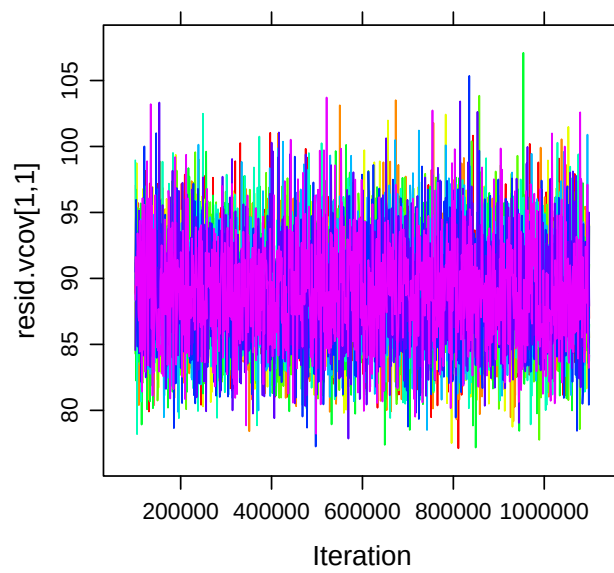


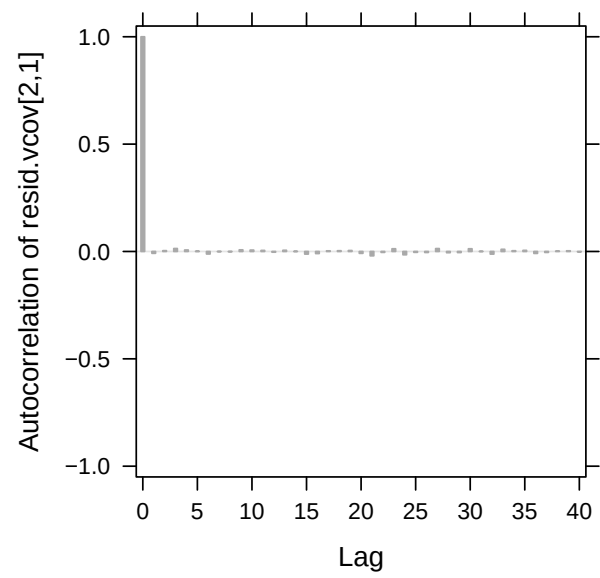
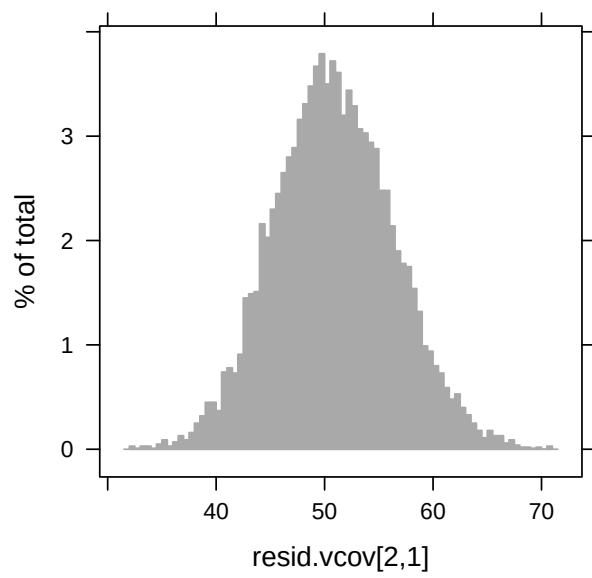
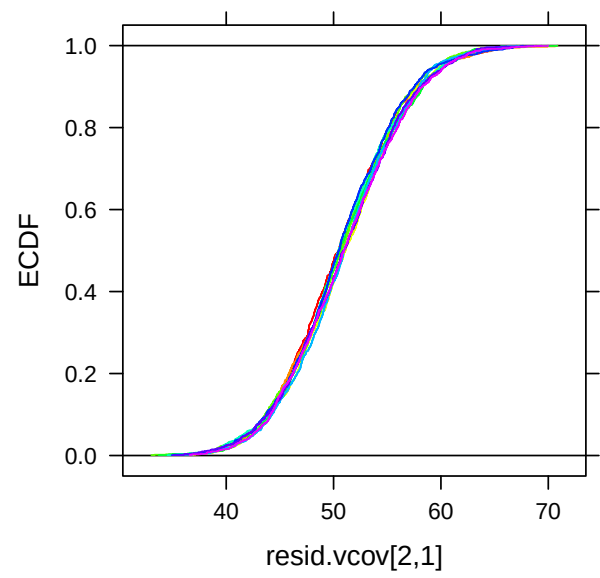
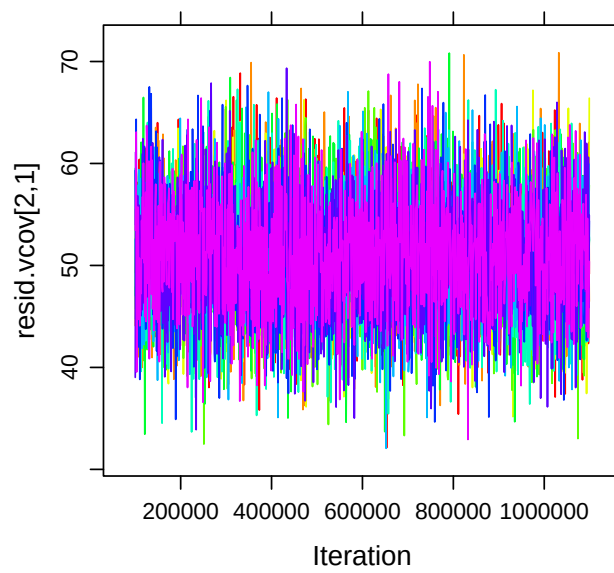


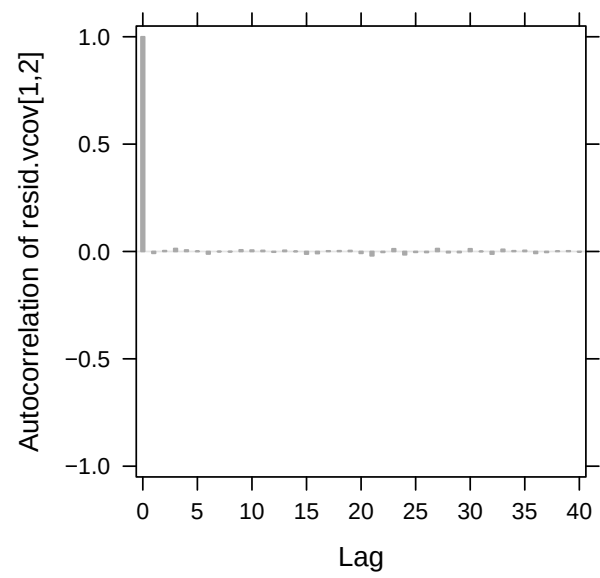
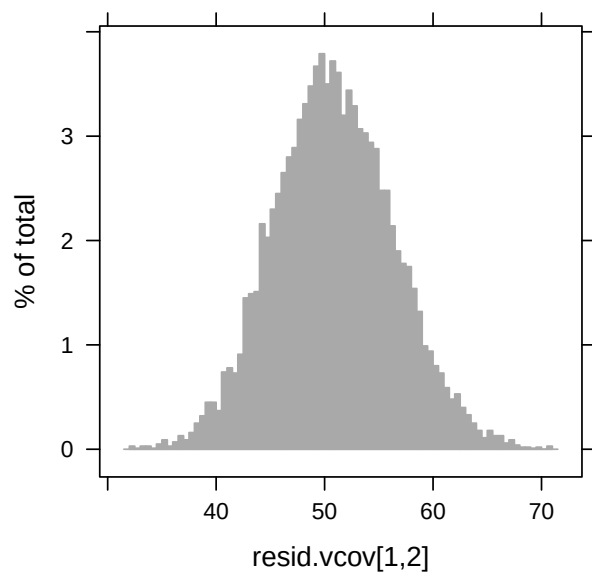
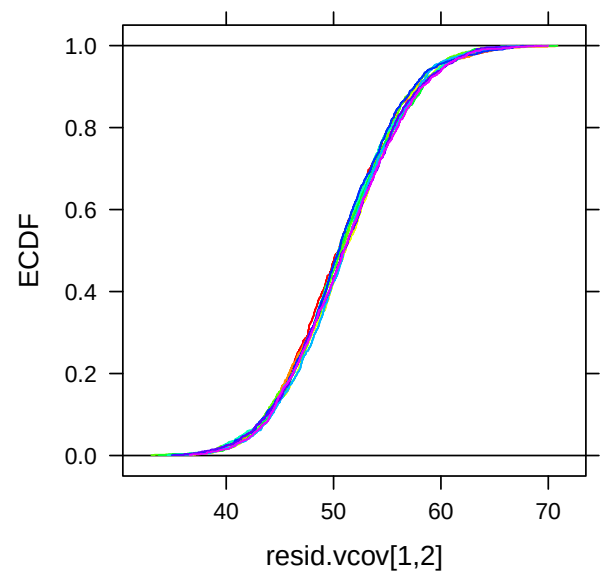
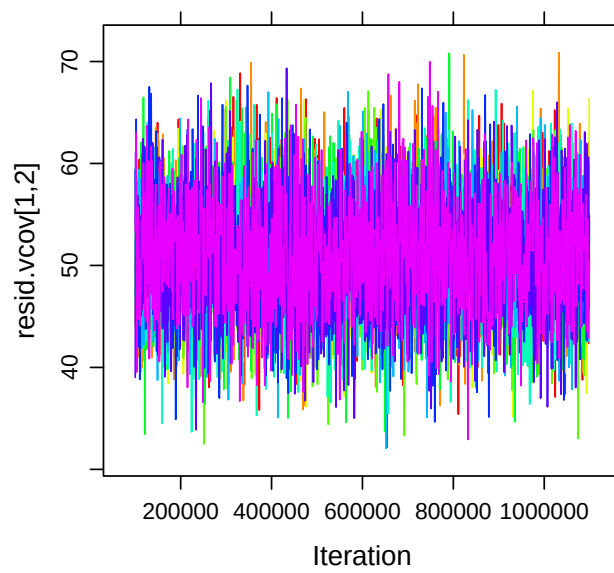


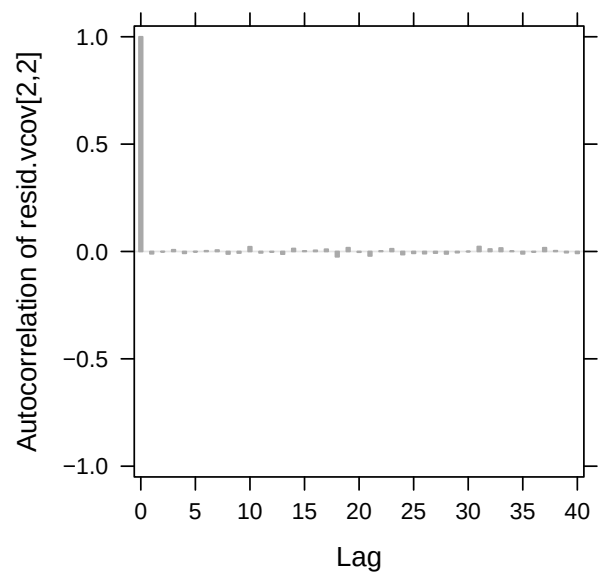
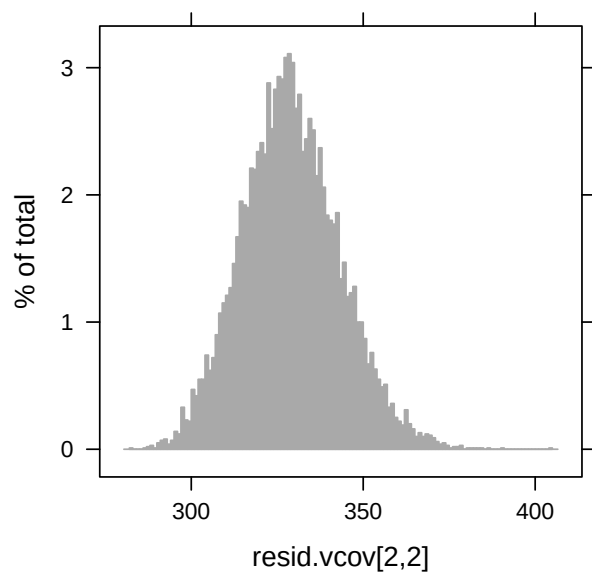
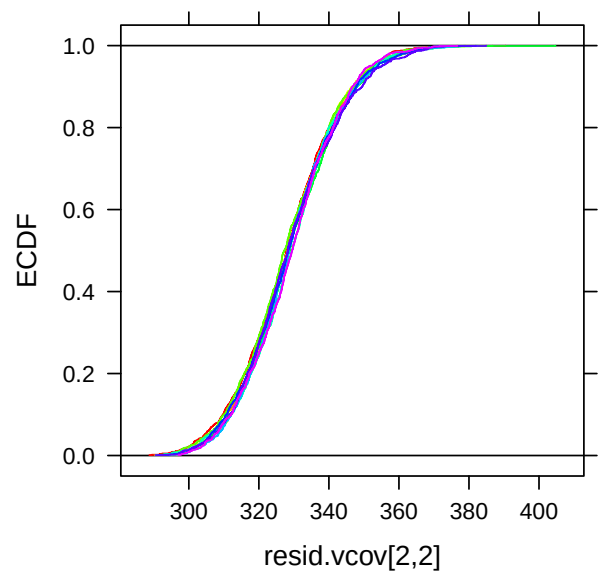
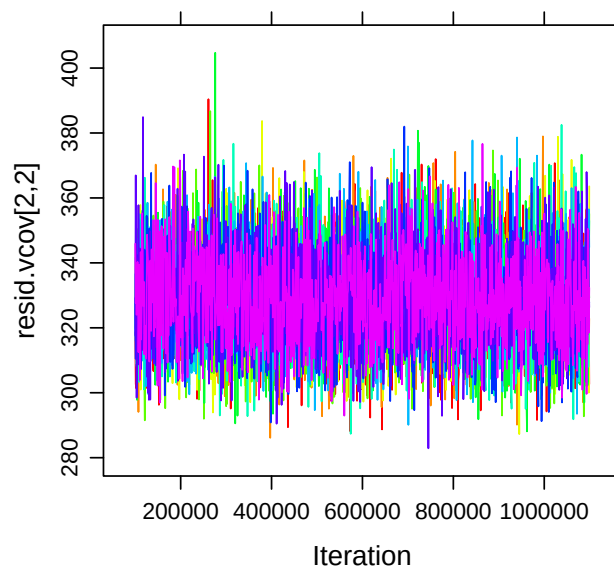


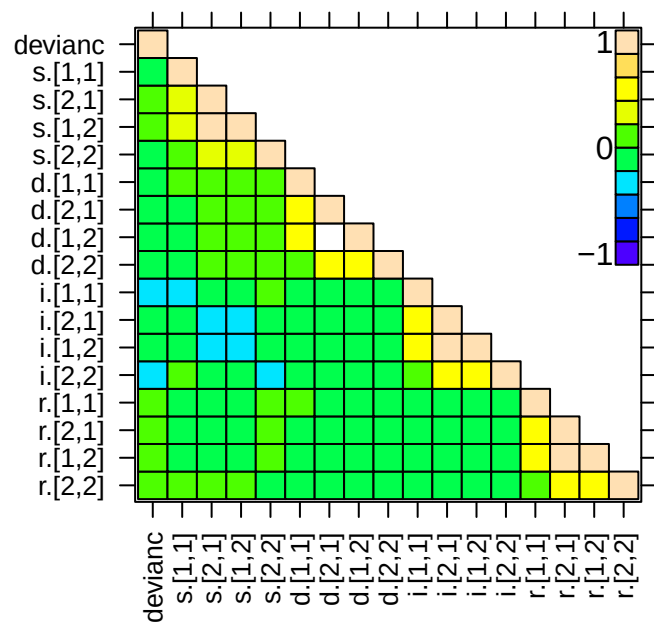












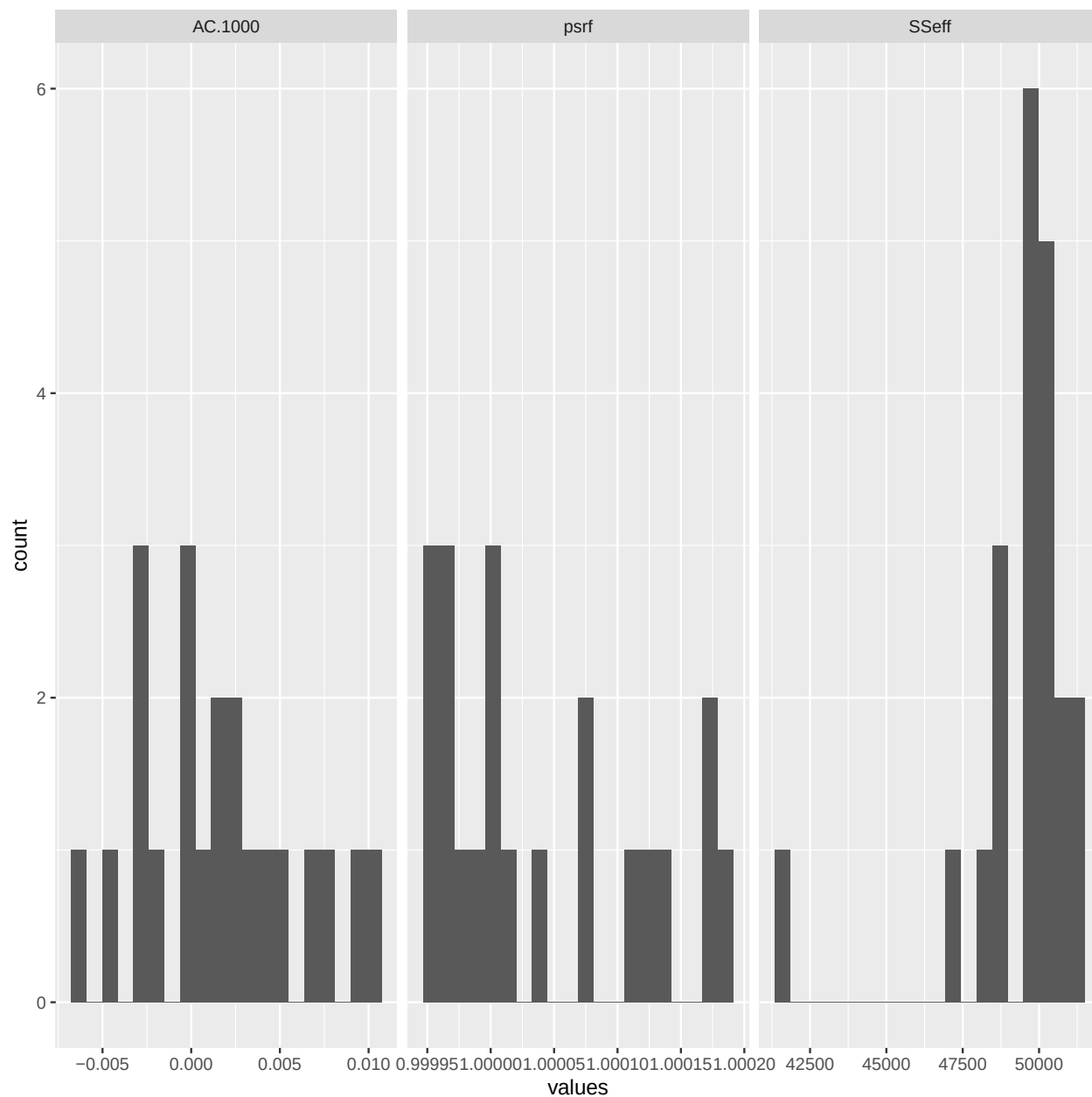
Model II

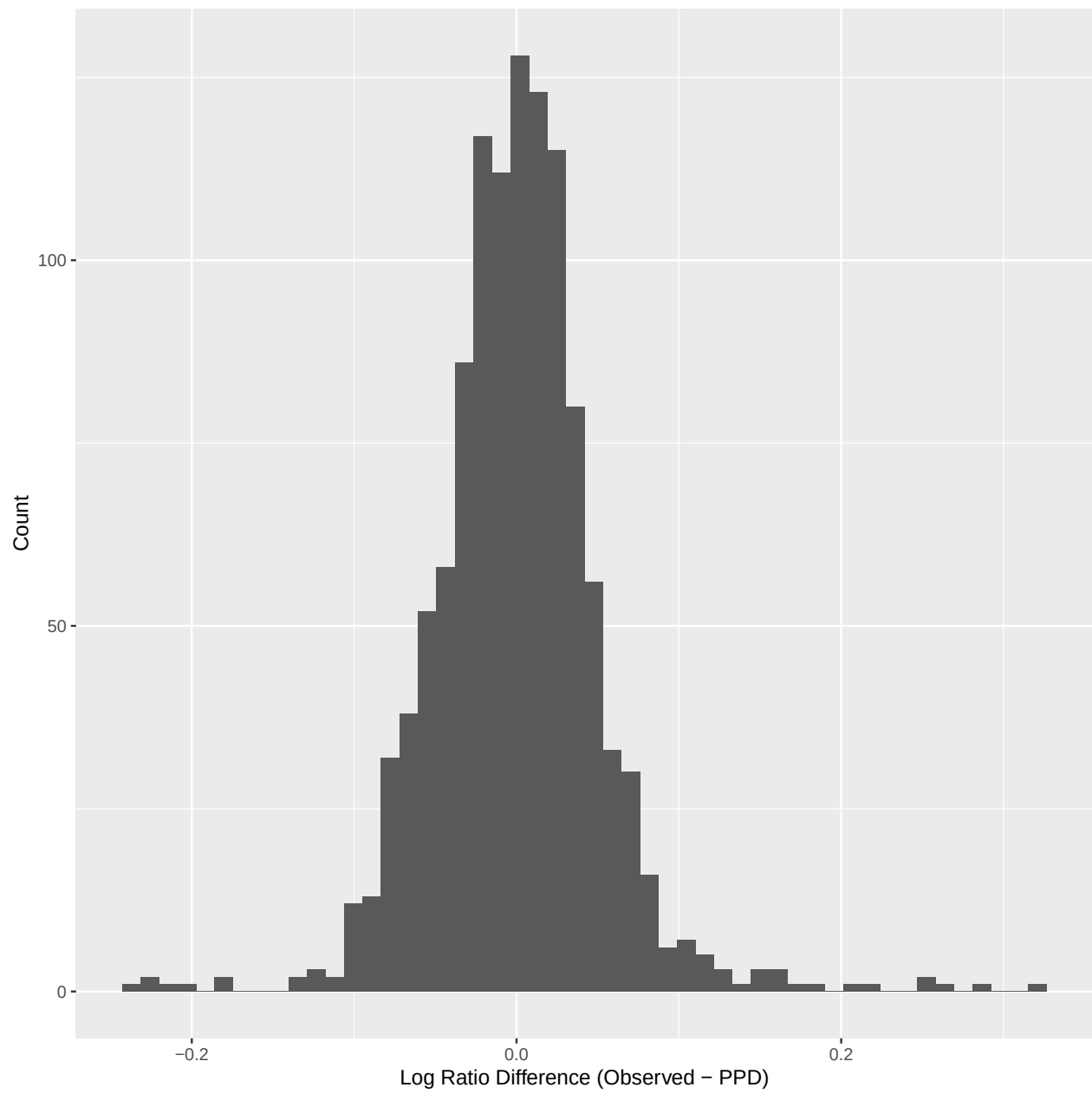
Number of chains: 10

Burnin: 5.01×10^4

Samples per chain: 5000

Total samples: 50000

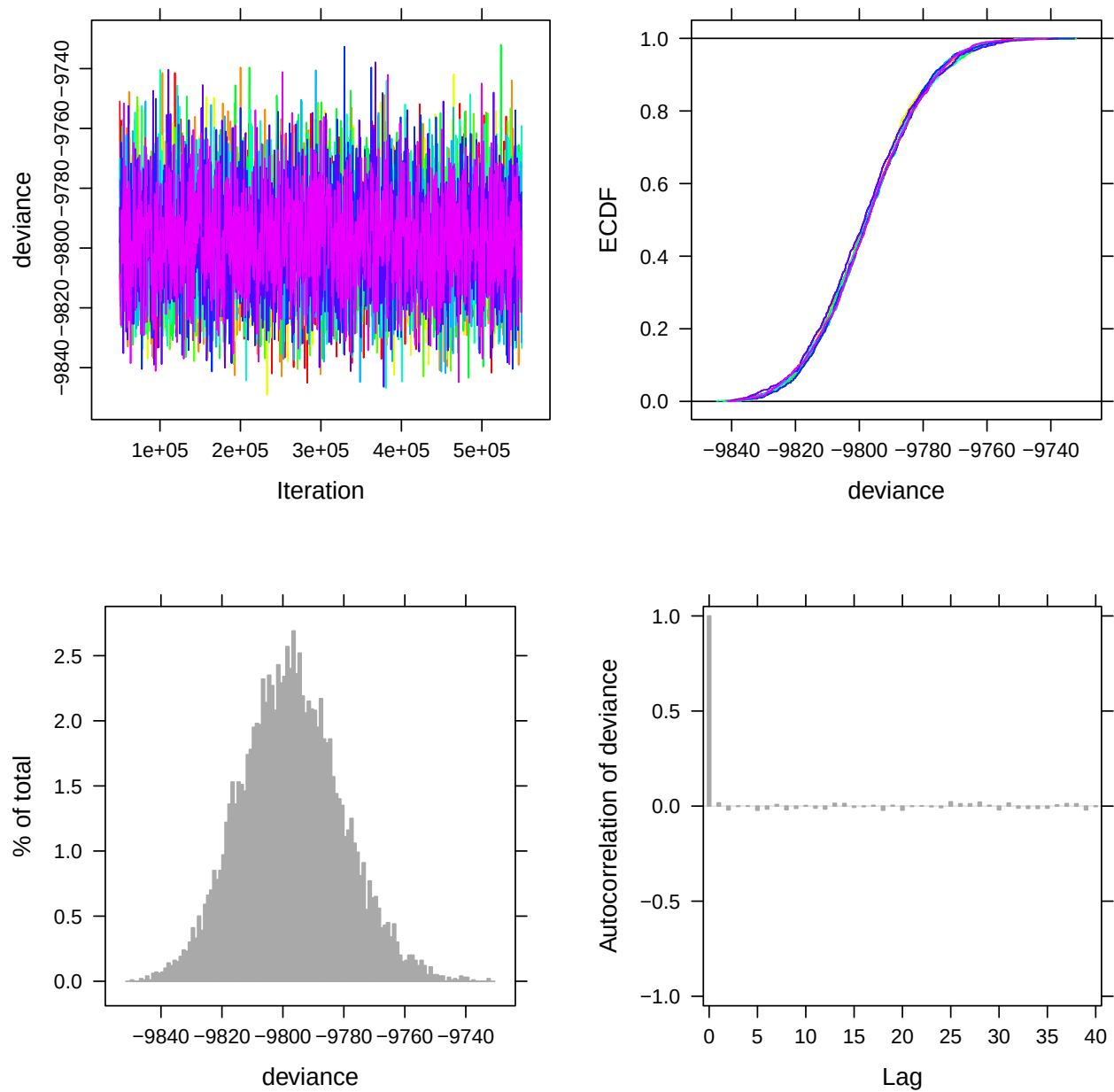


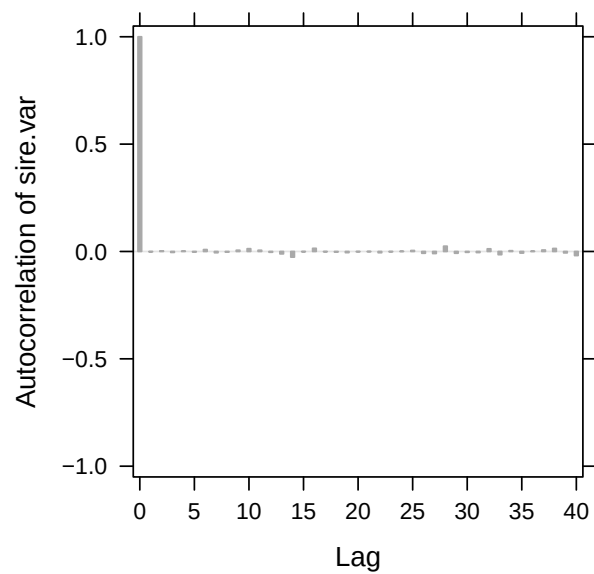
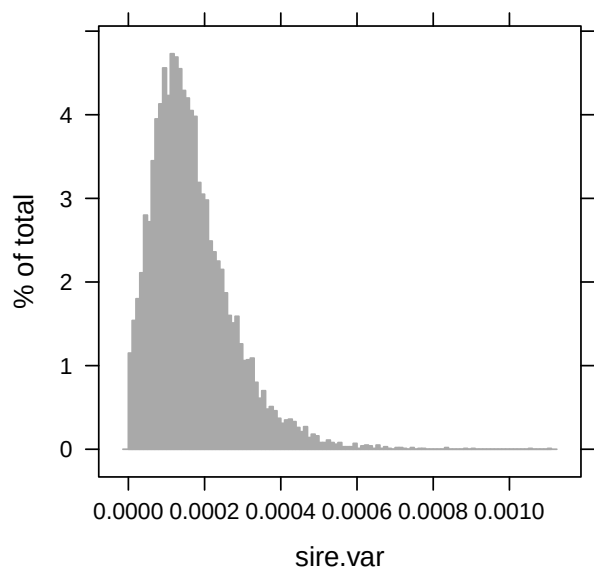
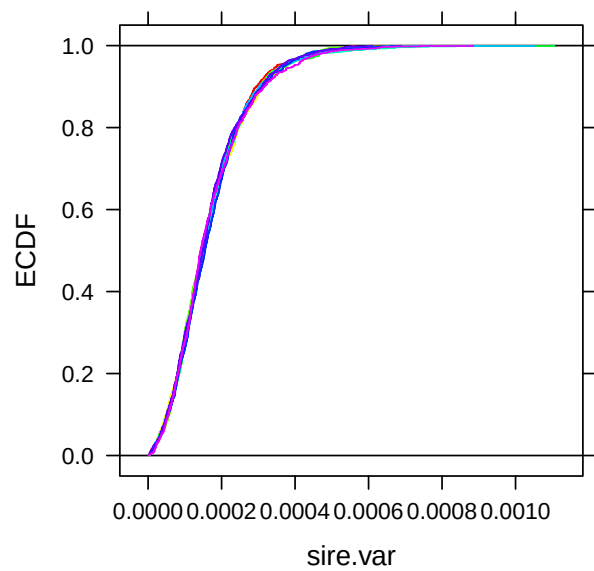
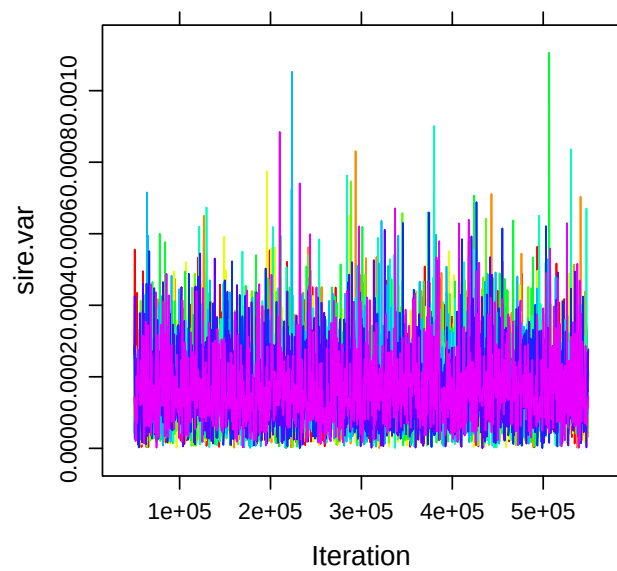


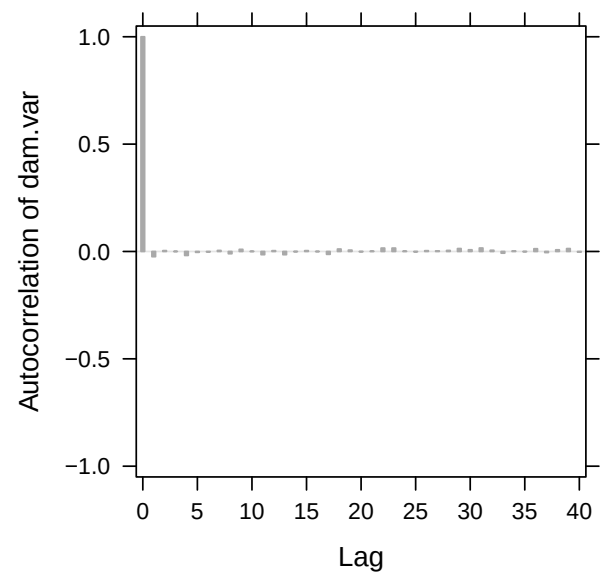
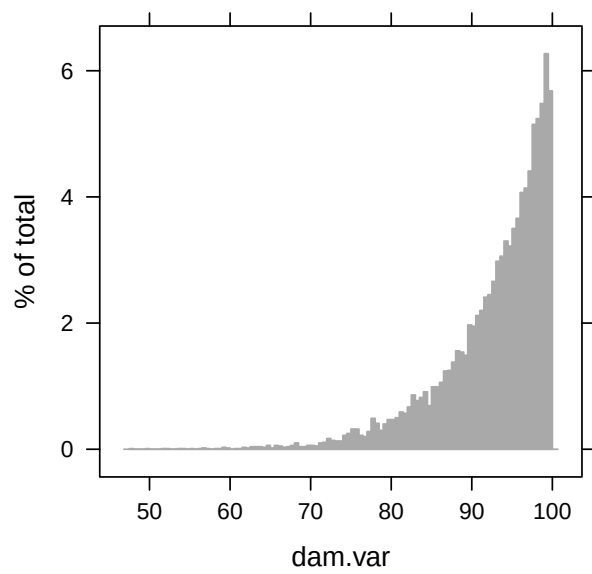
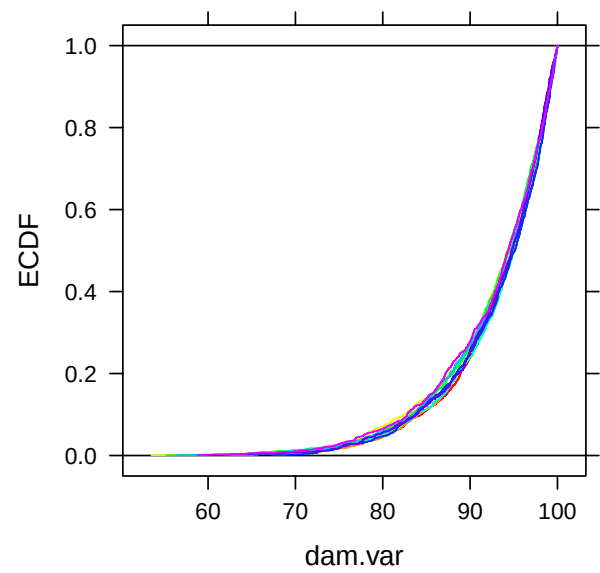
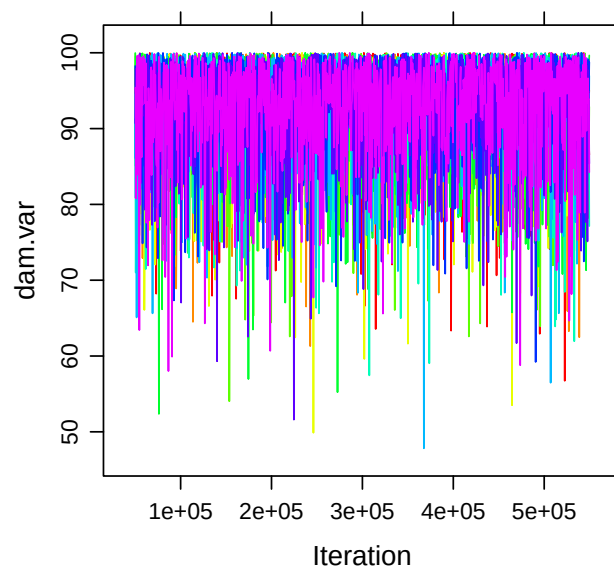
Generating summary statistics and plots (these will NOT be saved for reuse)...

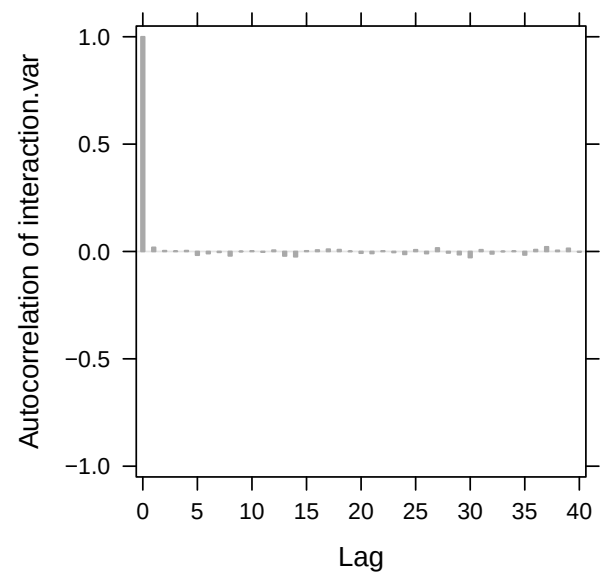
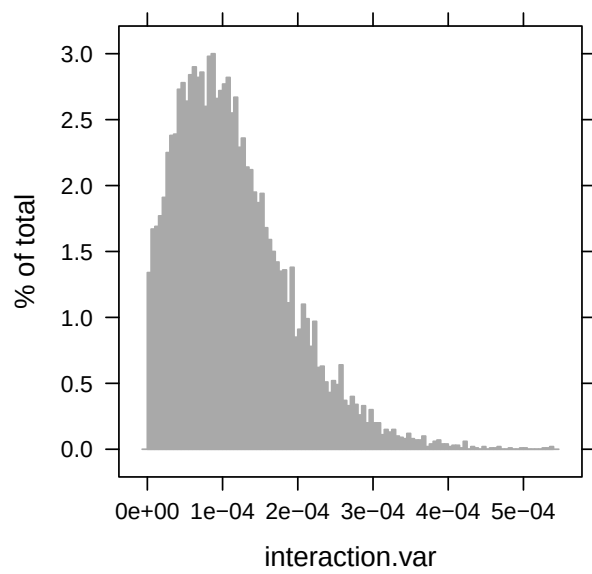
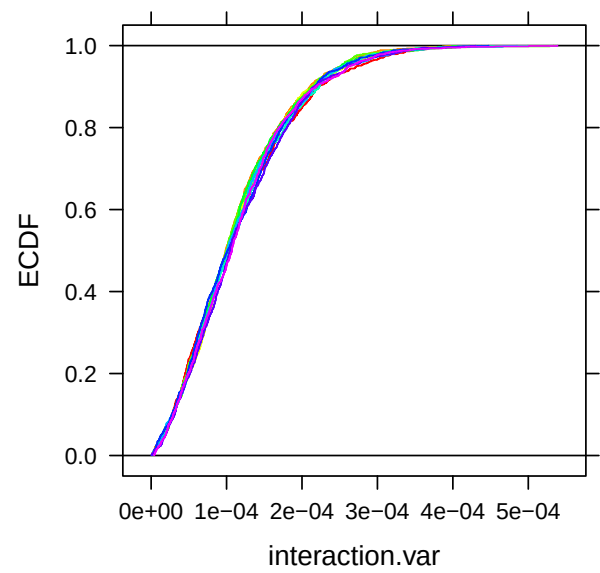
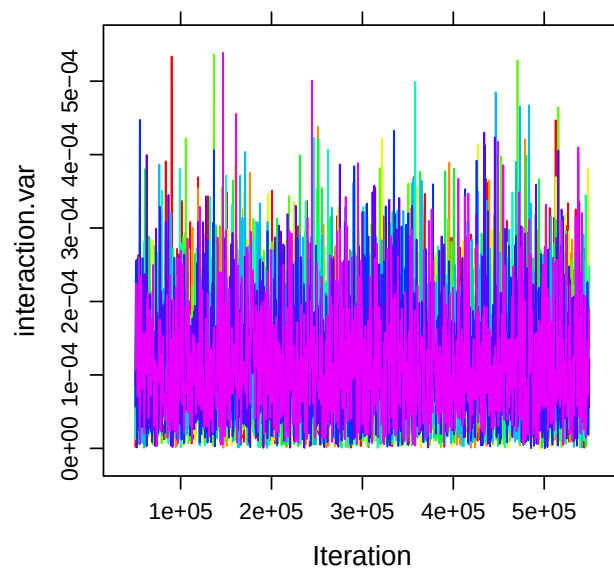
Calculating summary statistics...

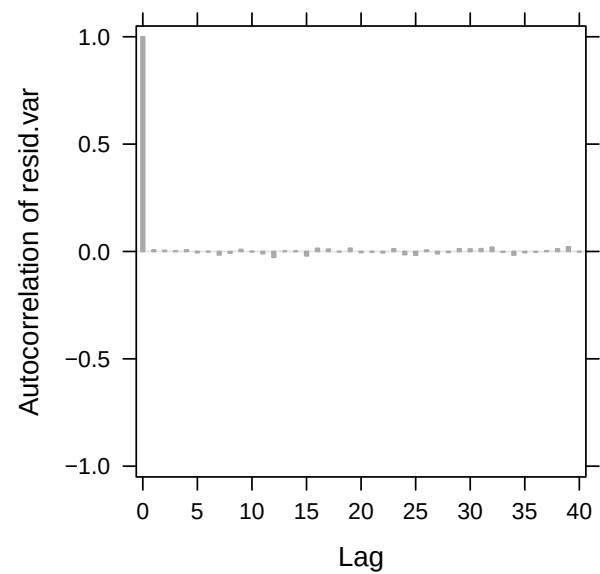
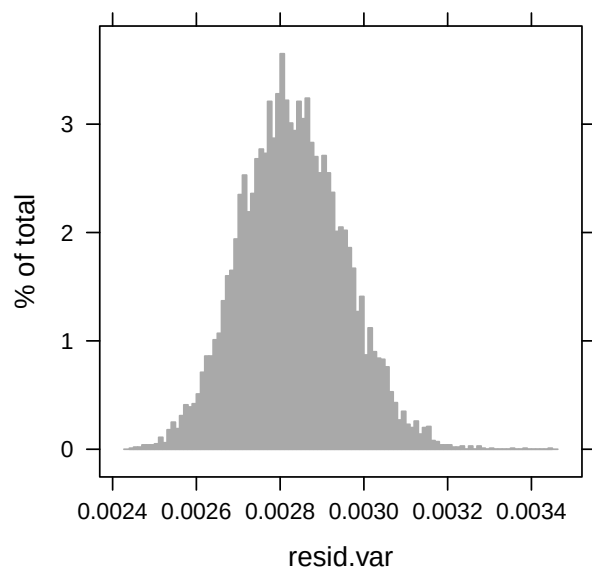
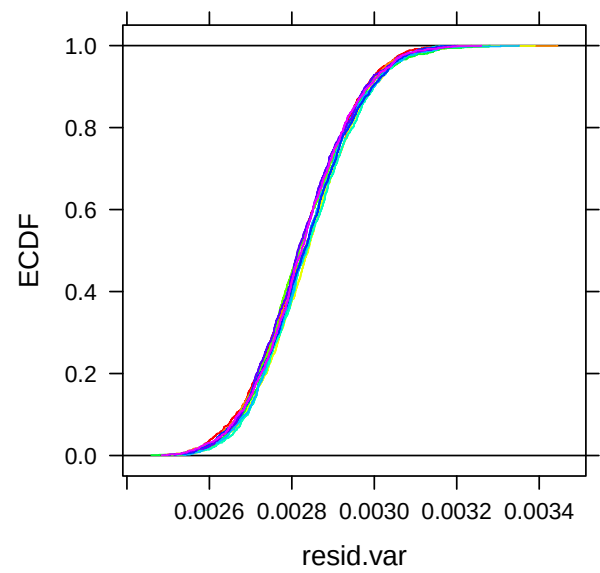
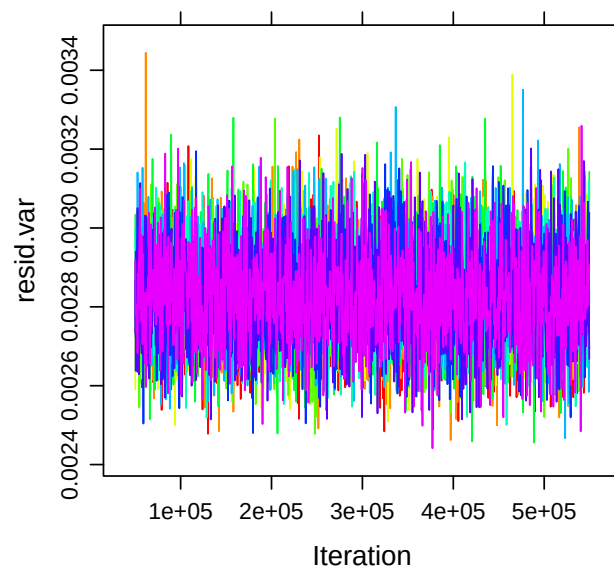
Calculating the Gelman-Rubin statistic for 21 variables....

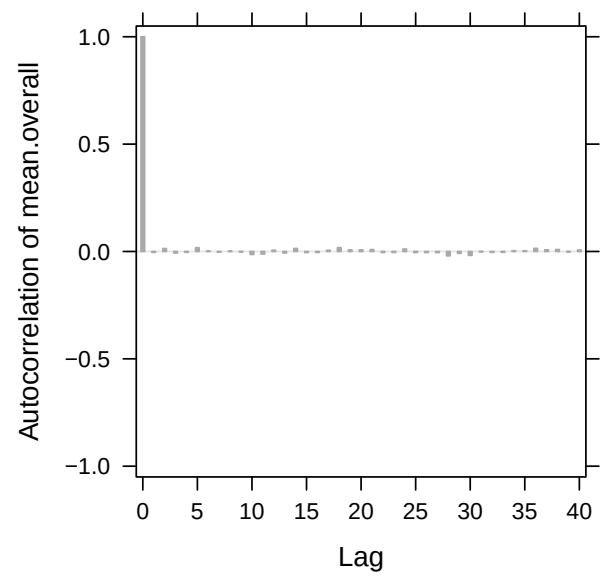
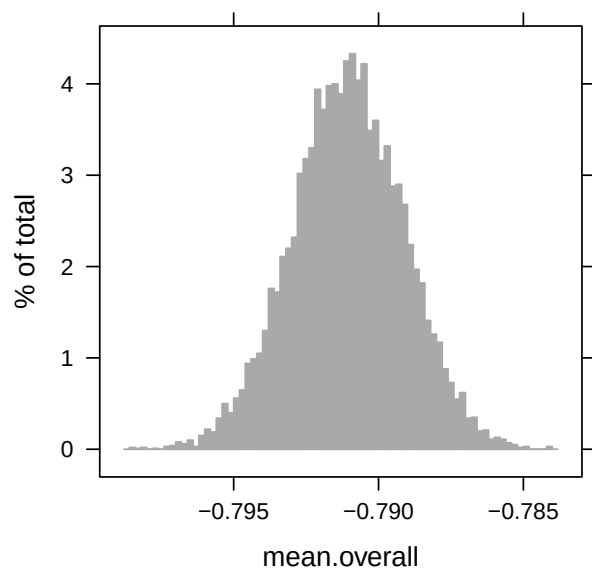
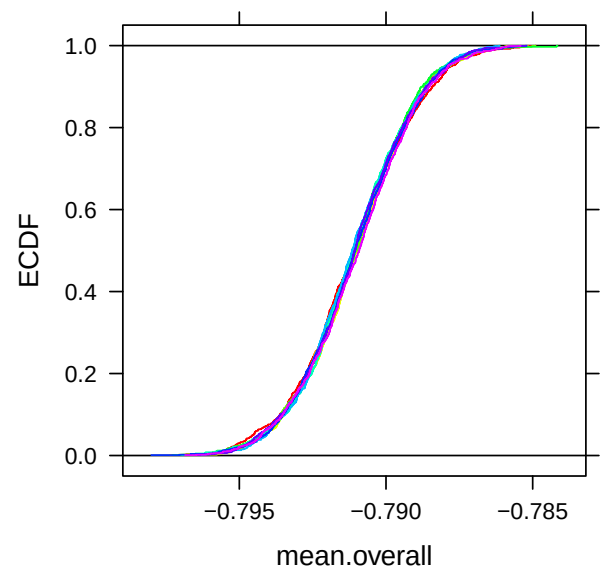
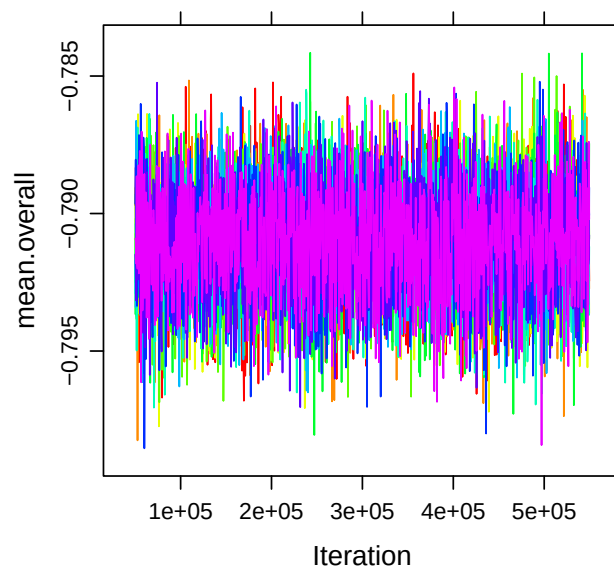


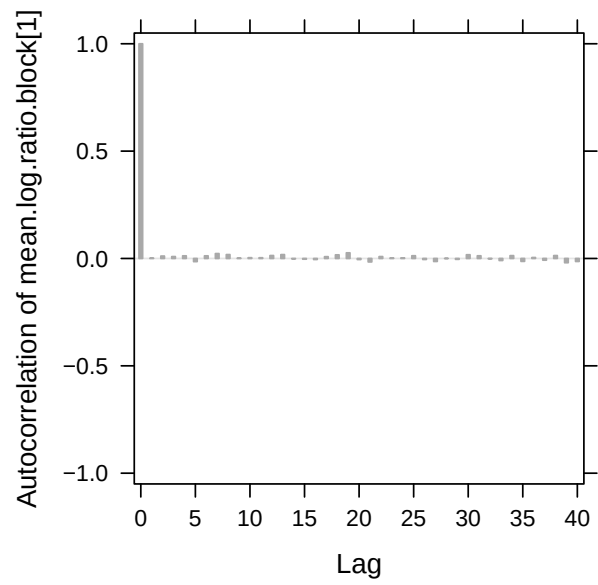
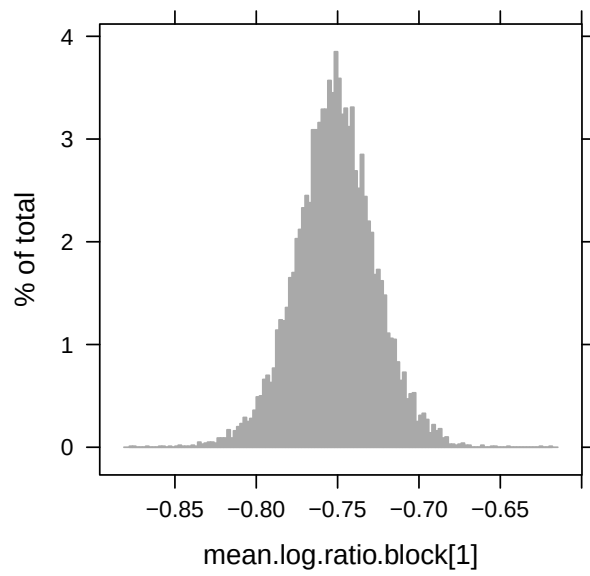
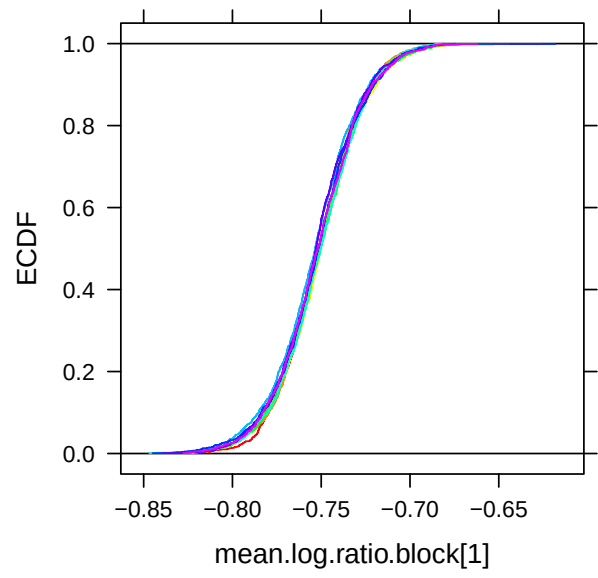
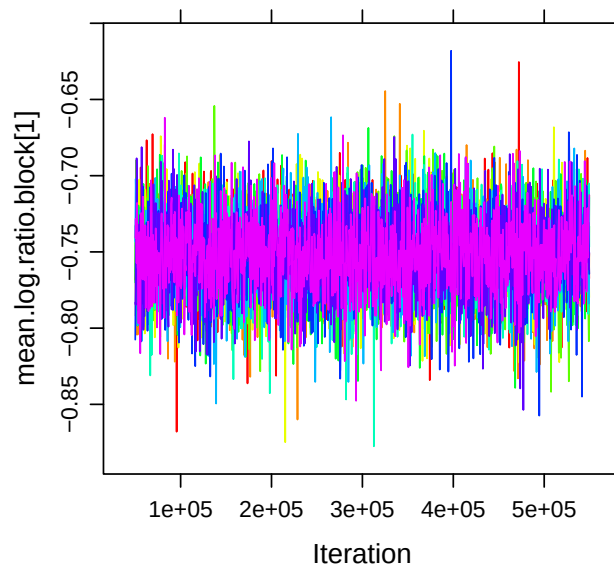


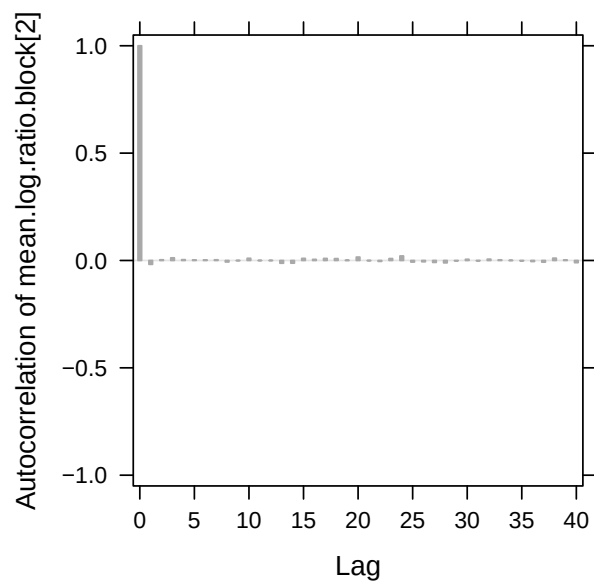
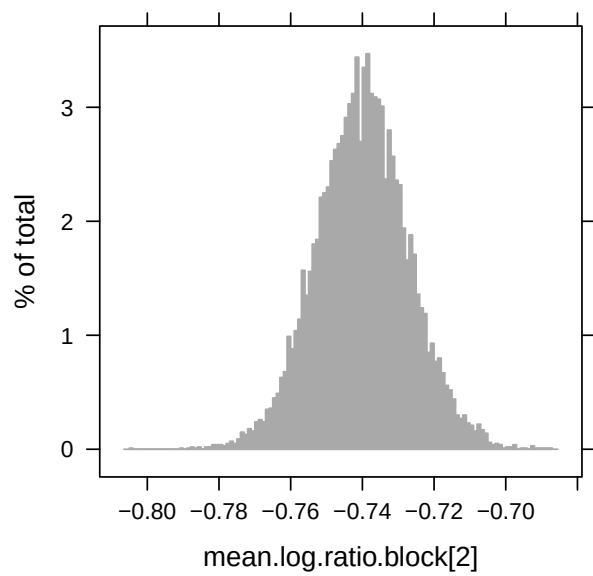
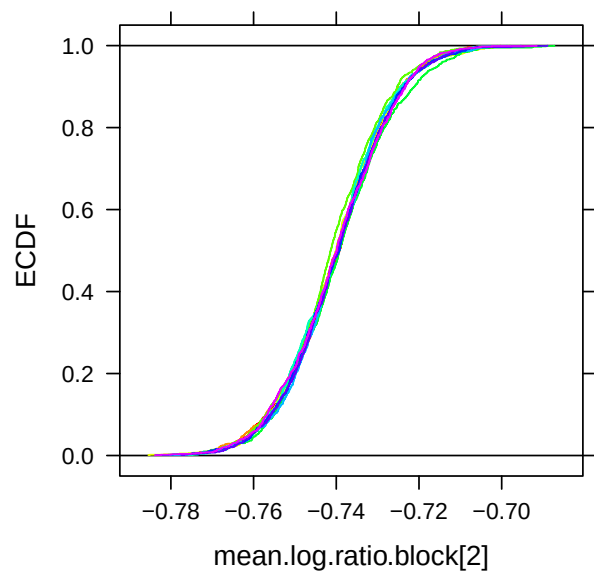
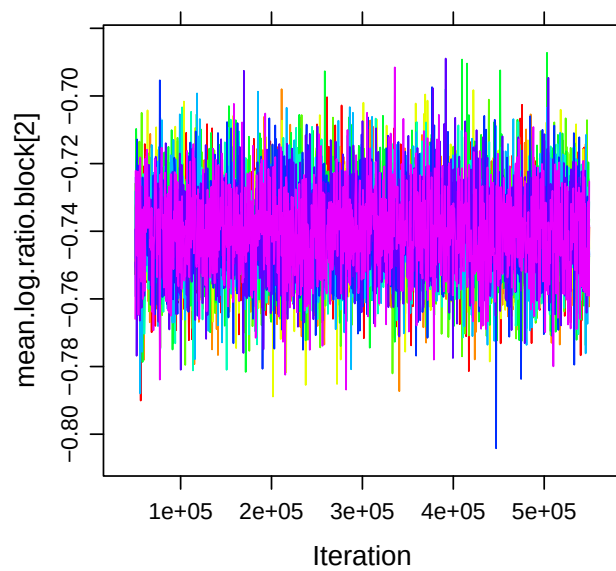


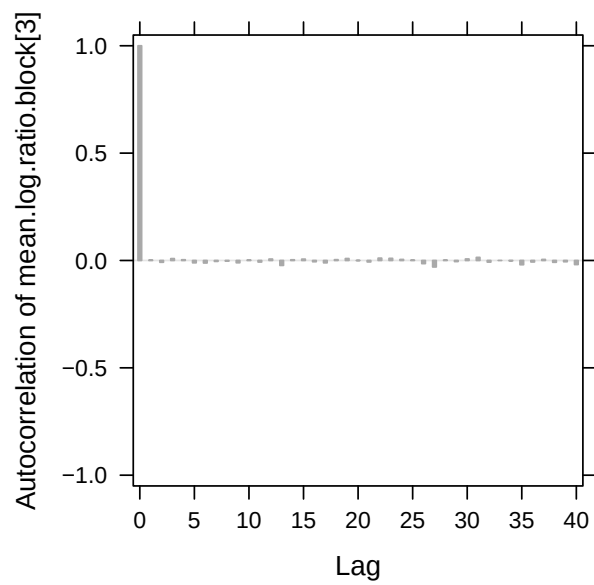
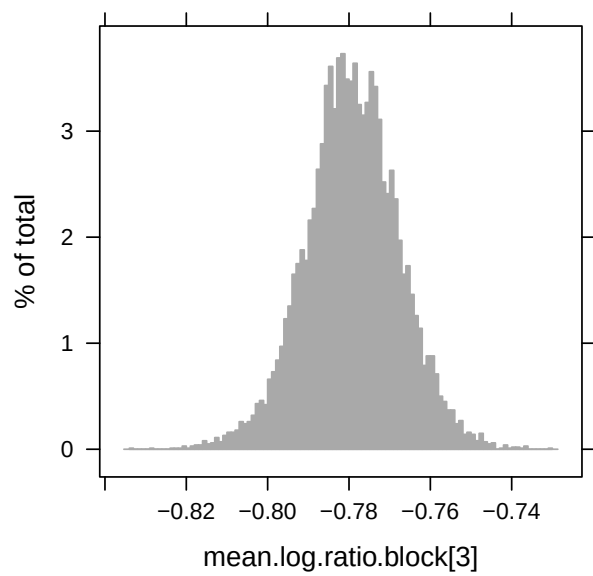
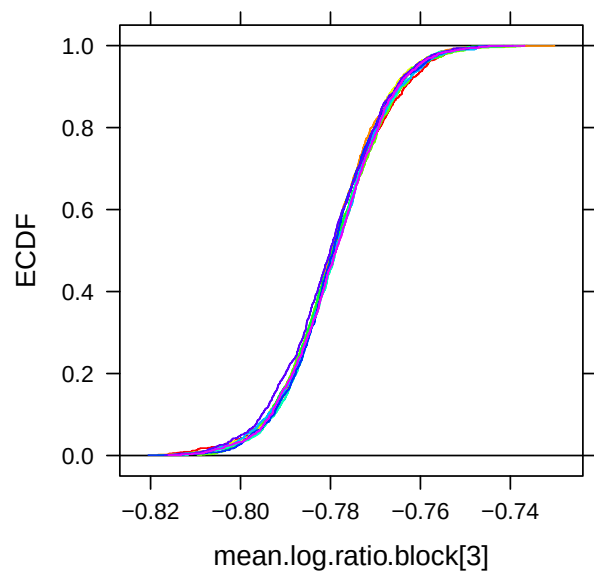
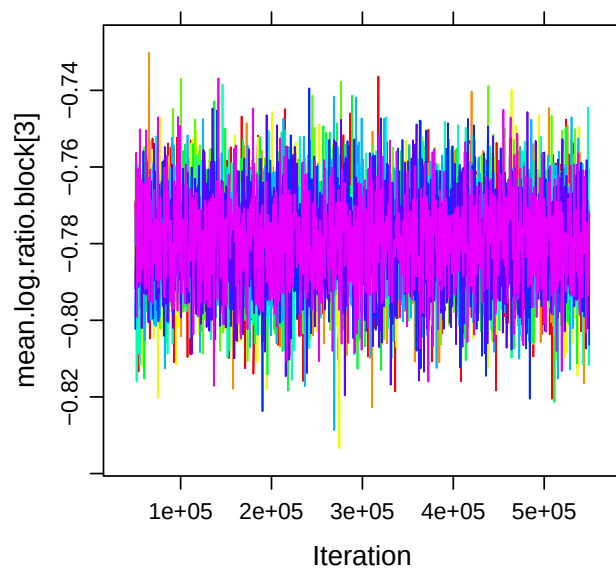


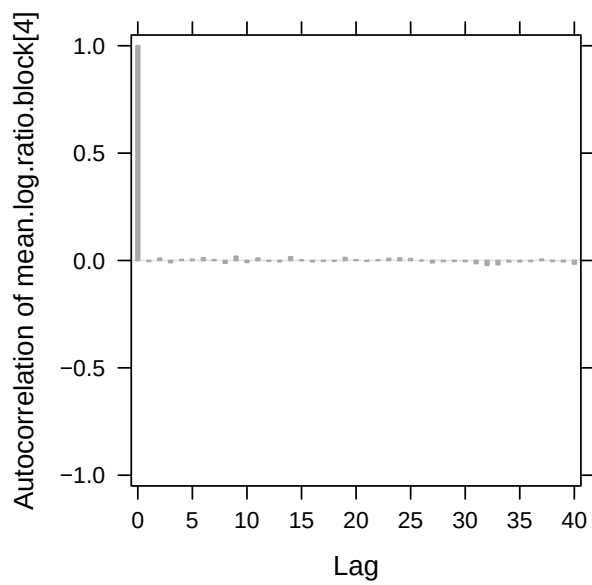
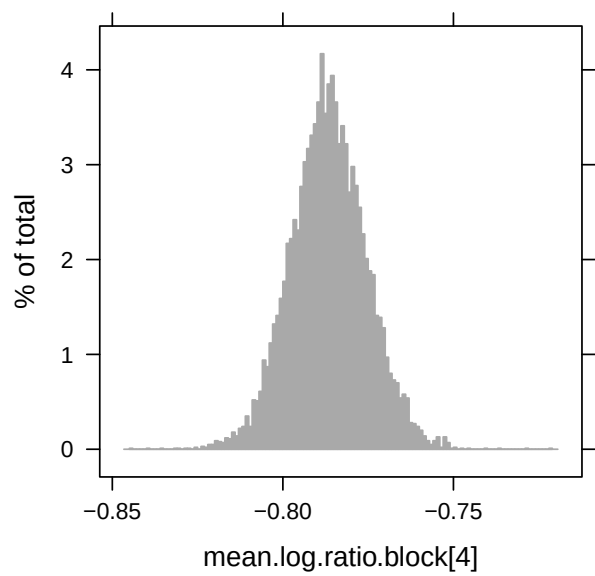
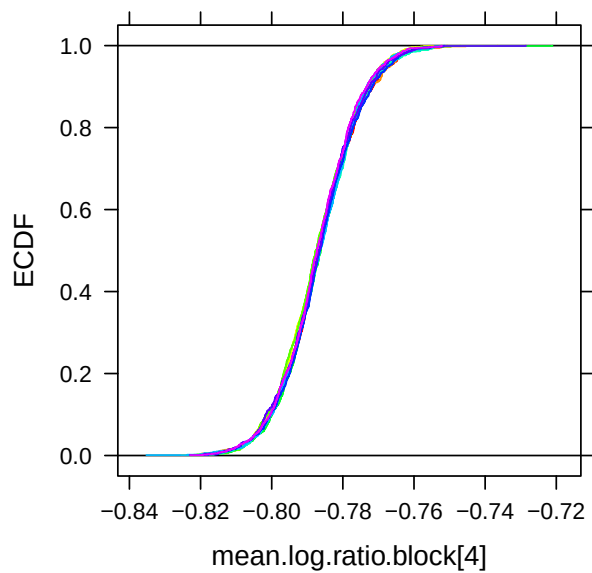
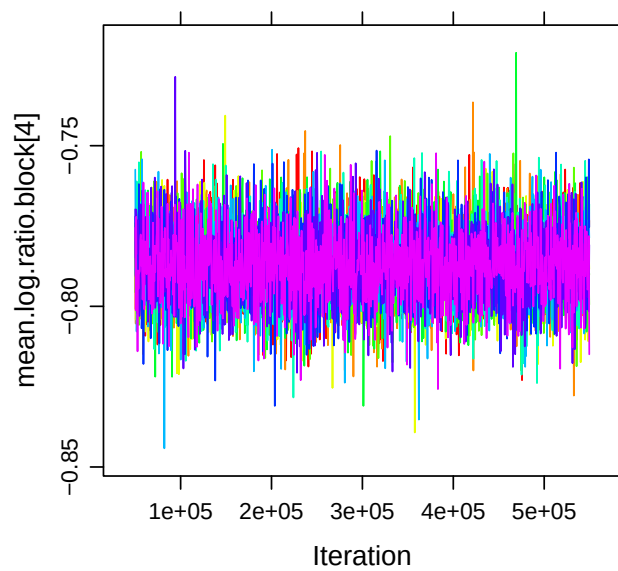


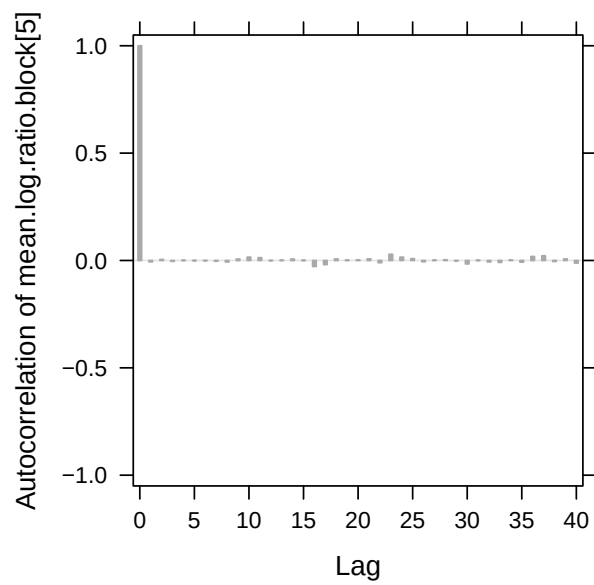
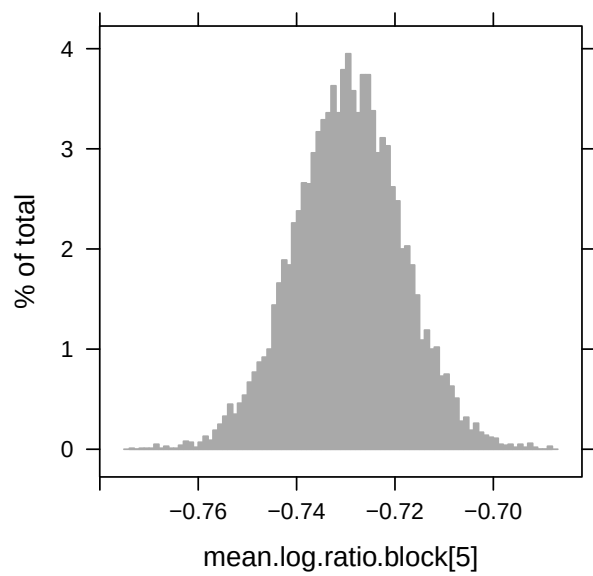
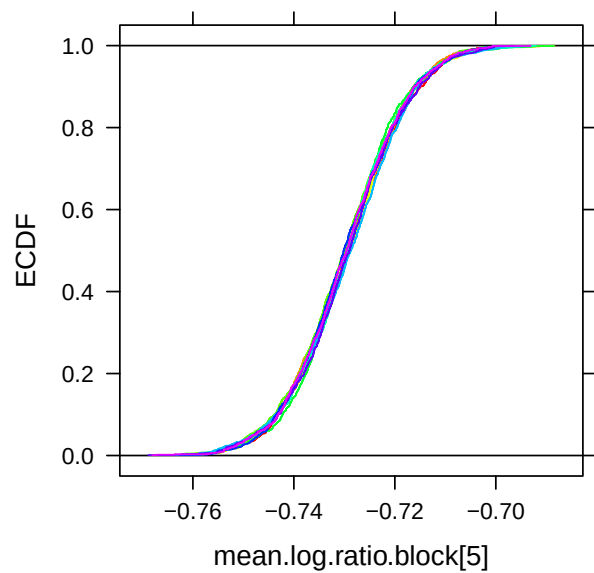
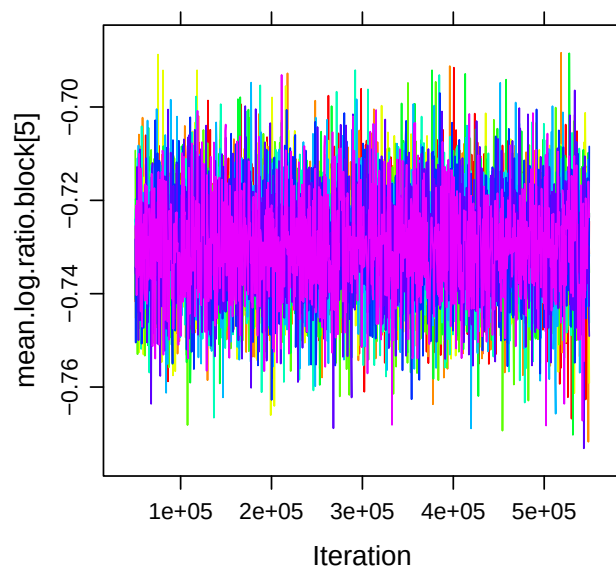


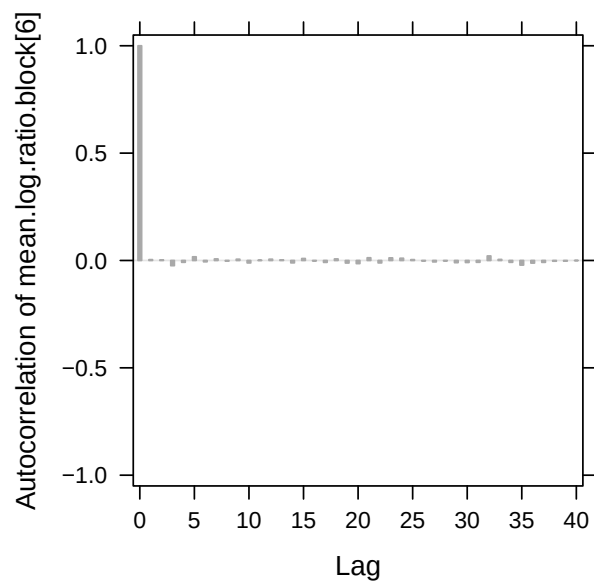
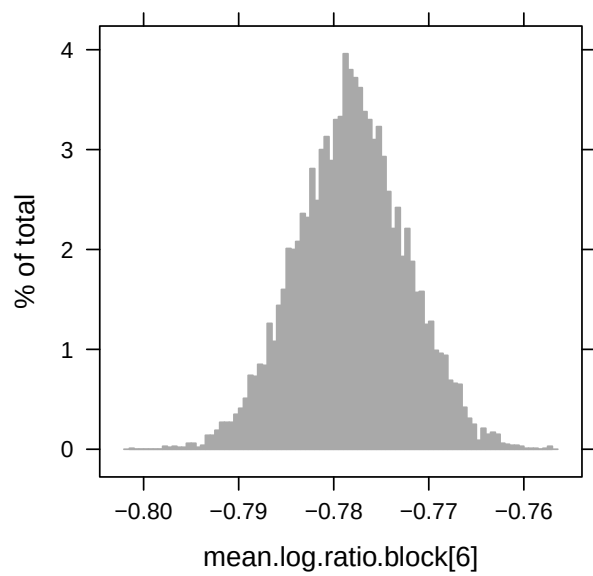
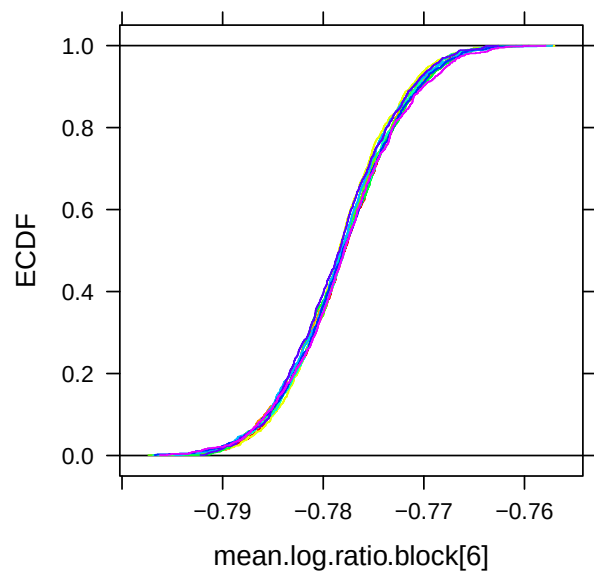
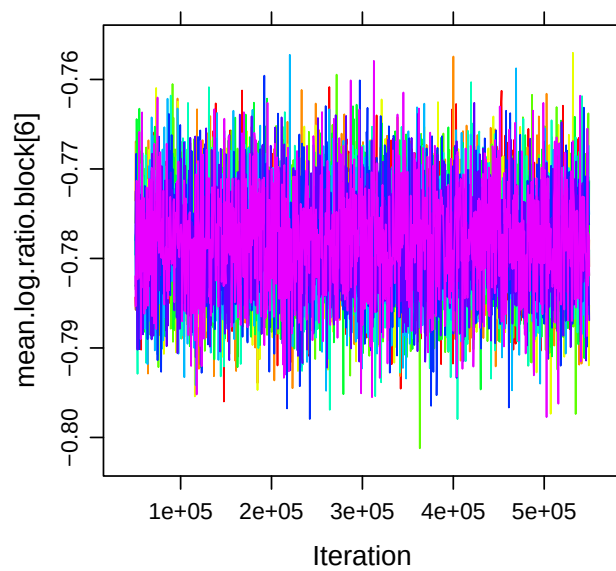


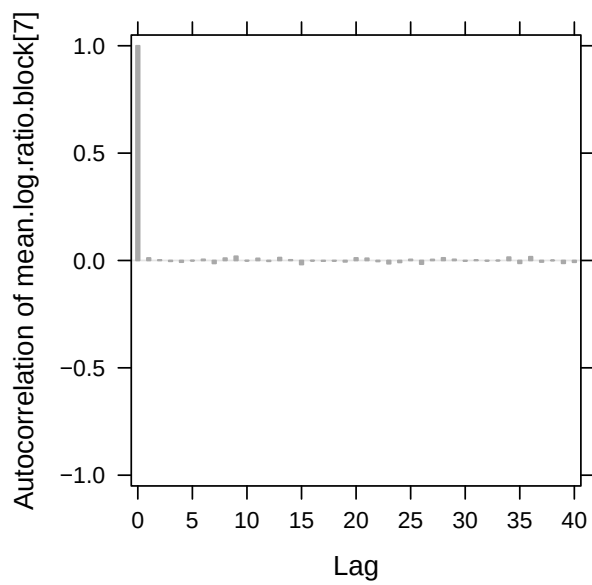
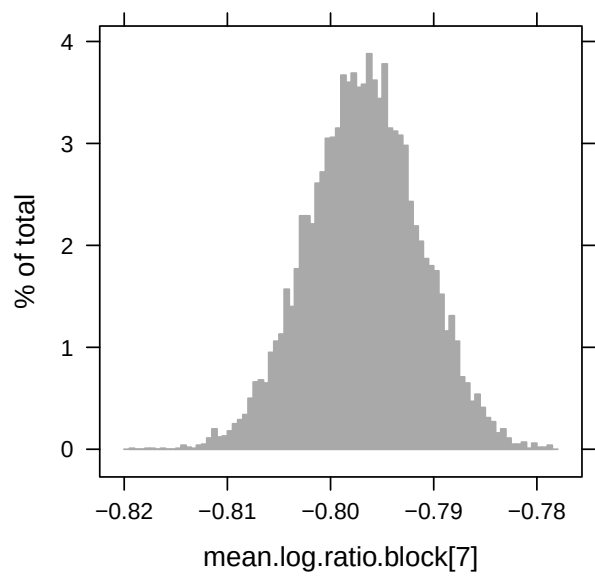
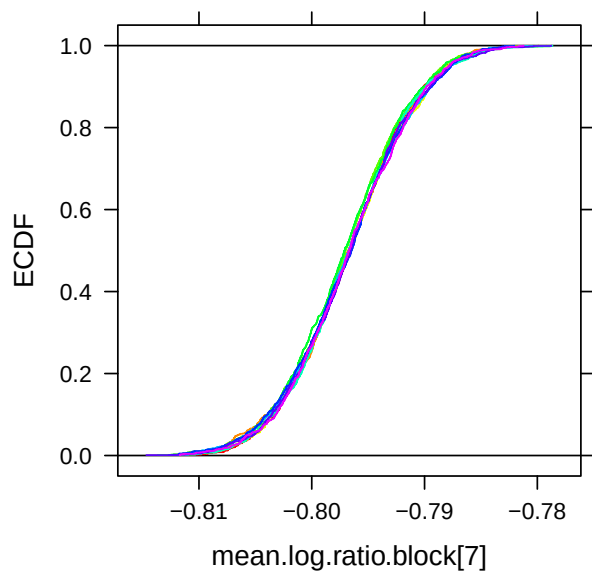
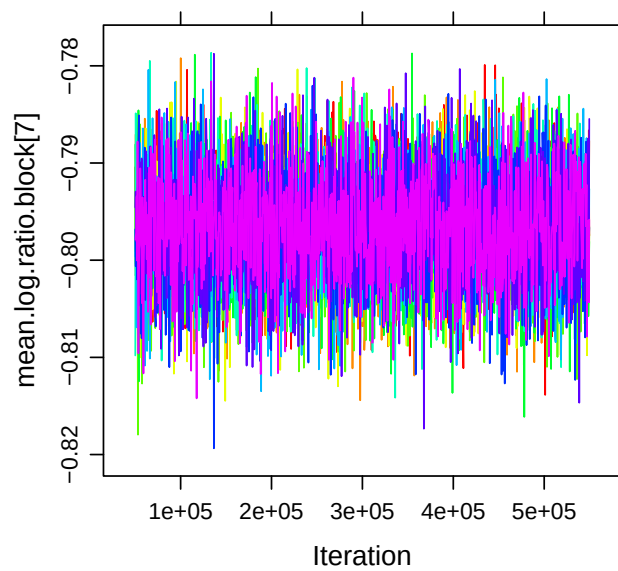


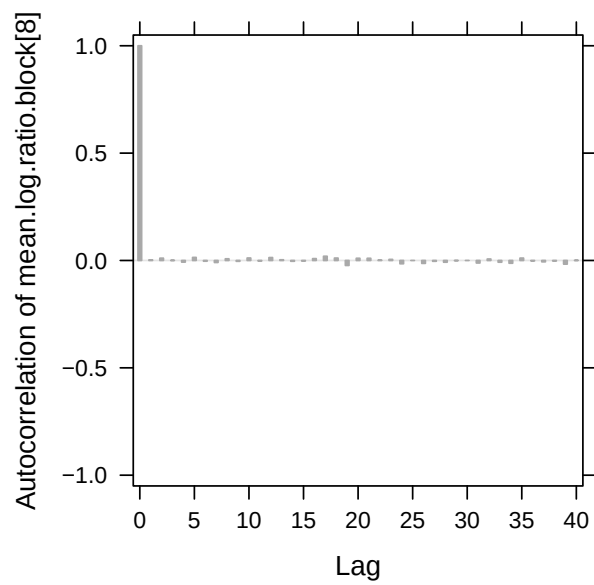
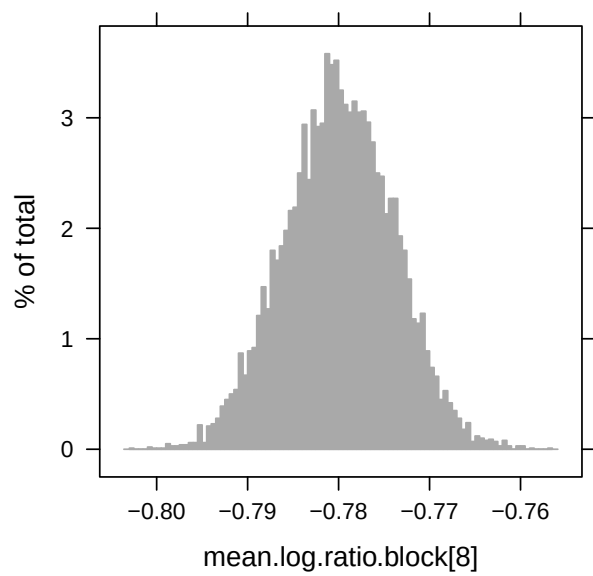
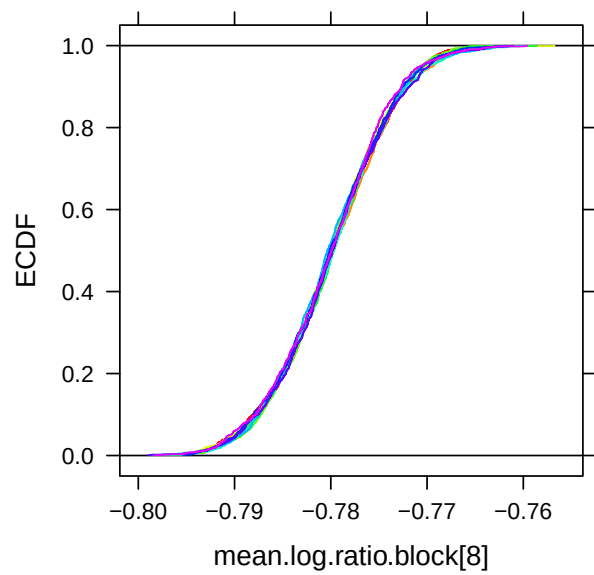
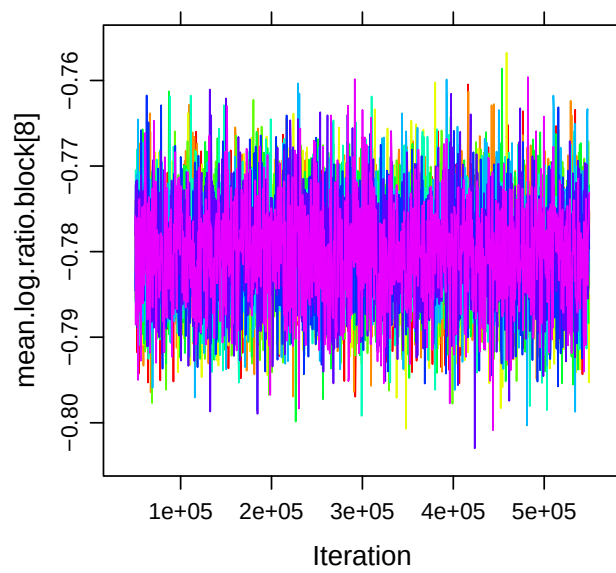


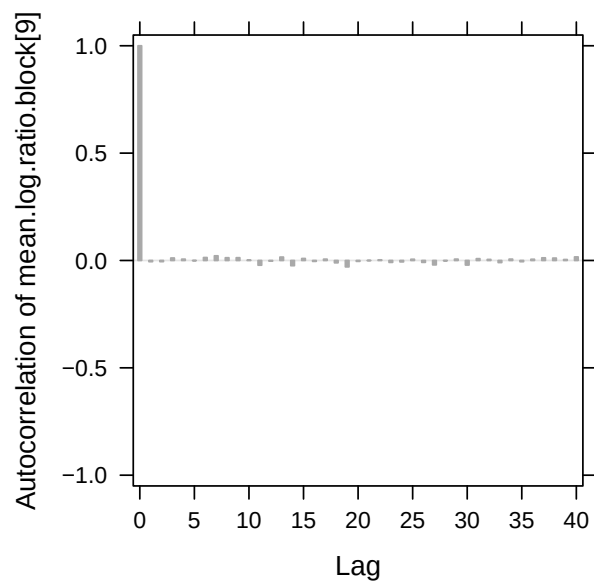
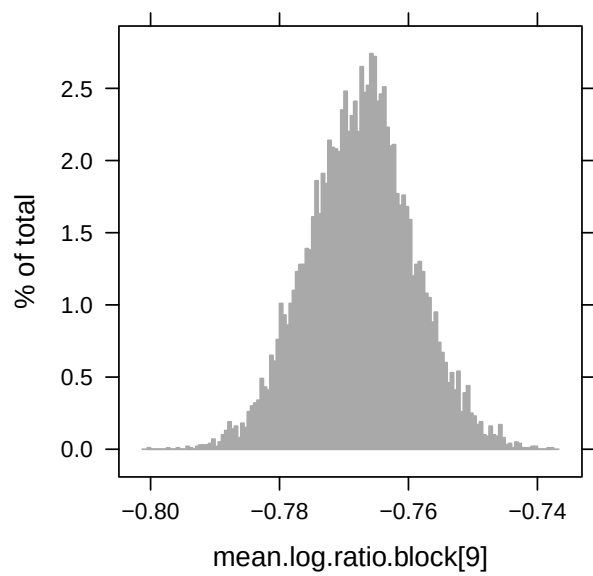
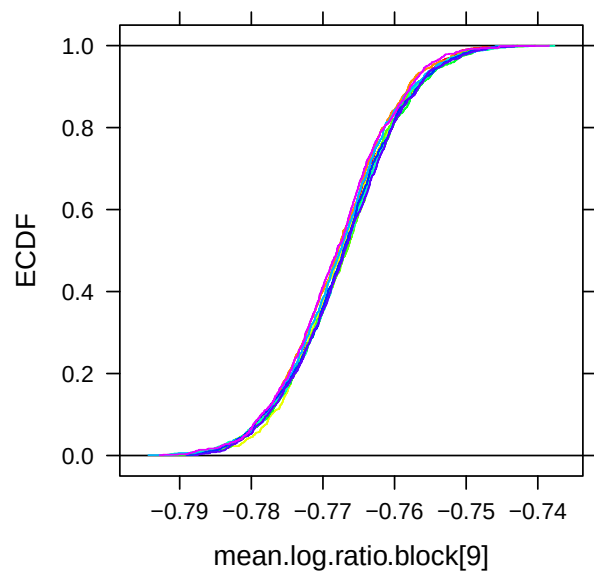
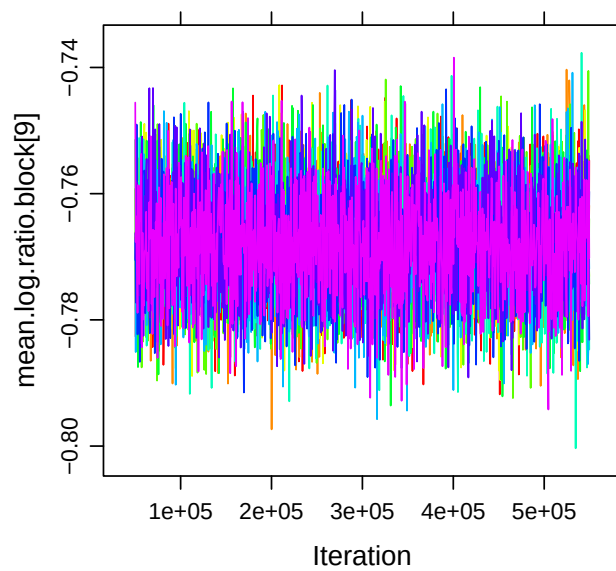


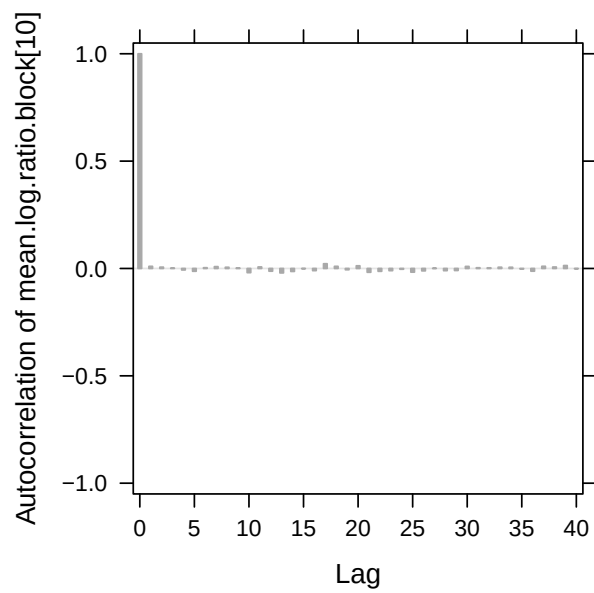
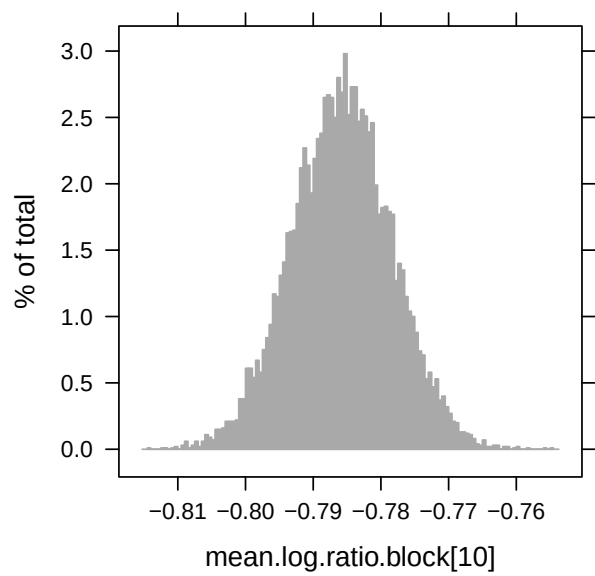
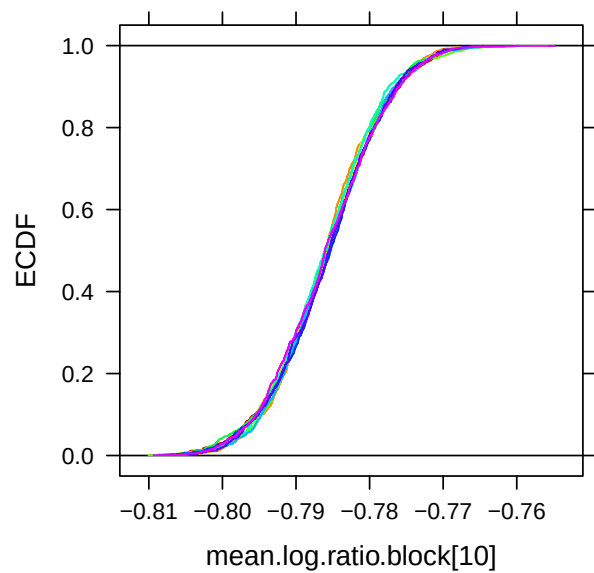
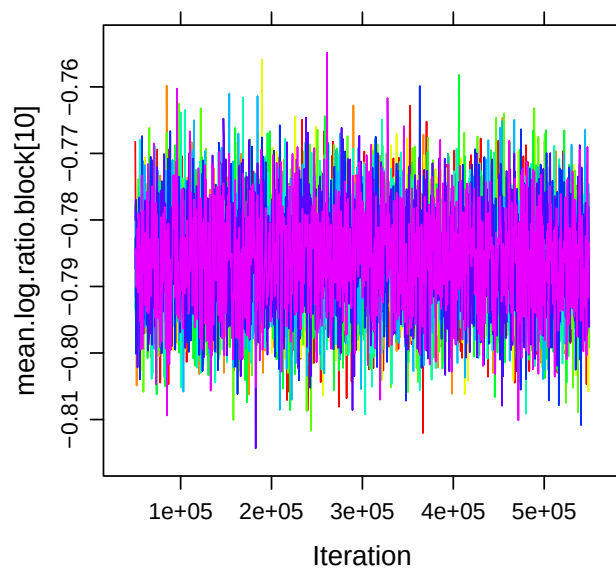


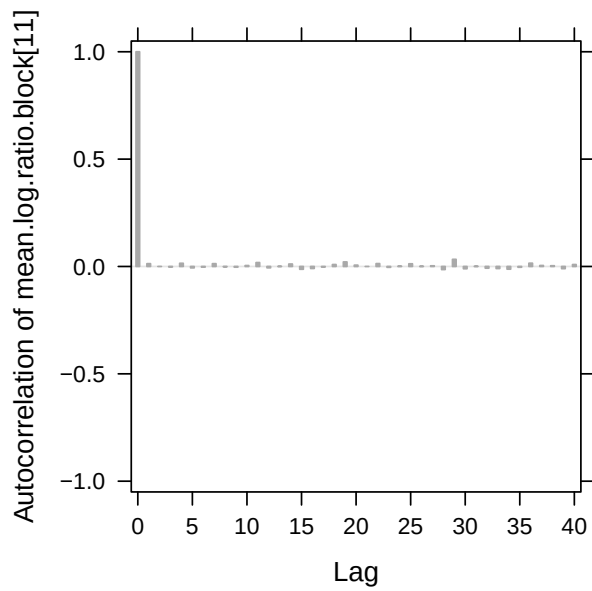
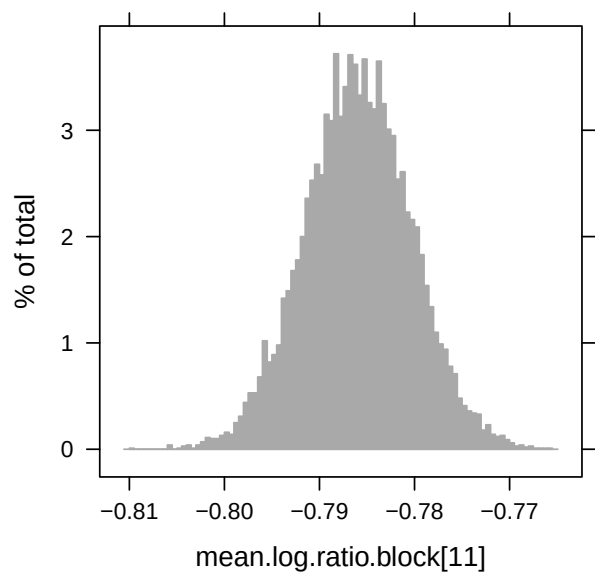
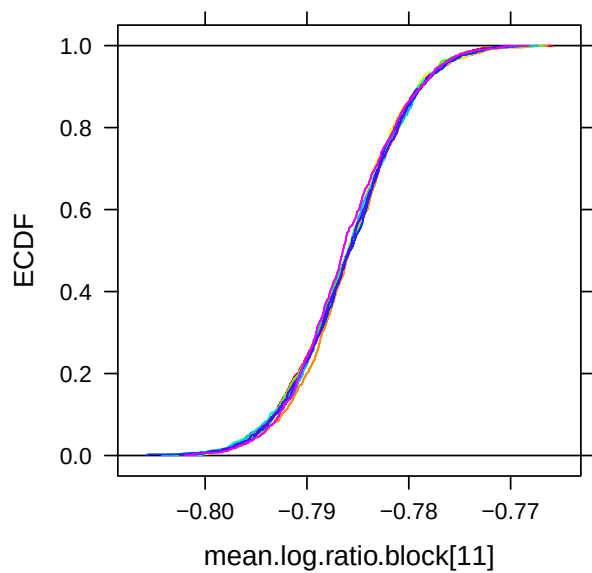
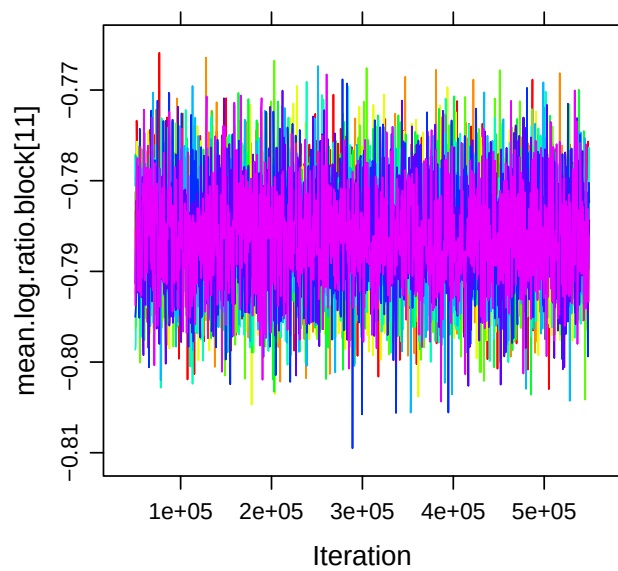


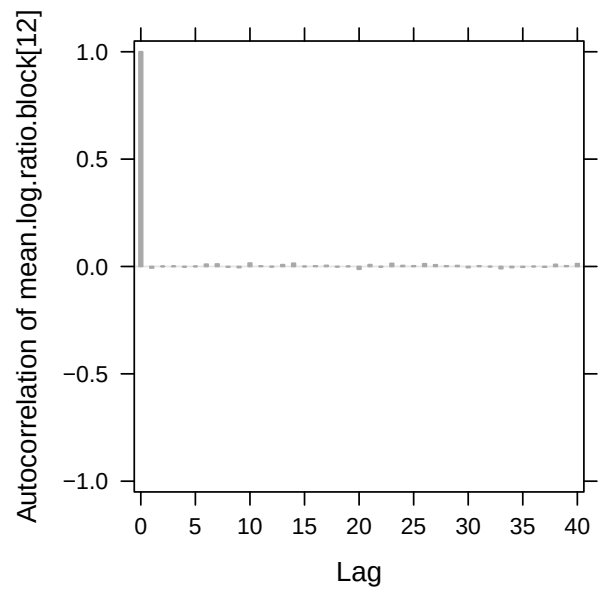
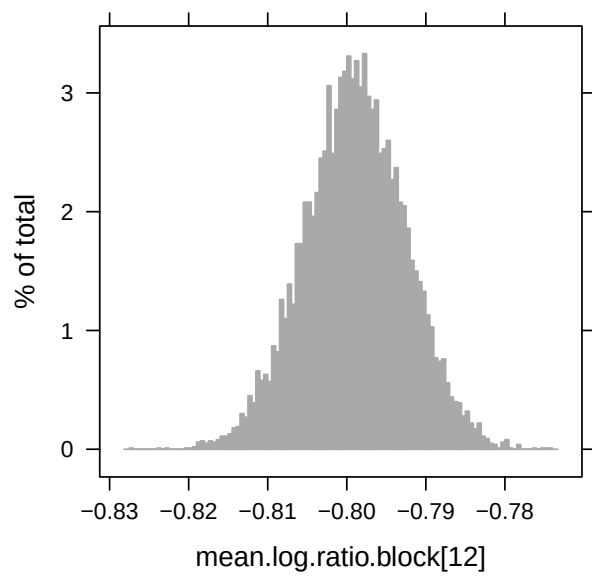
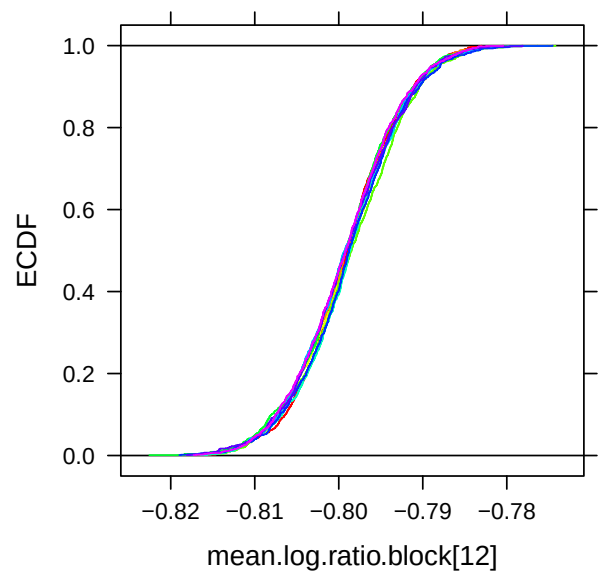
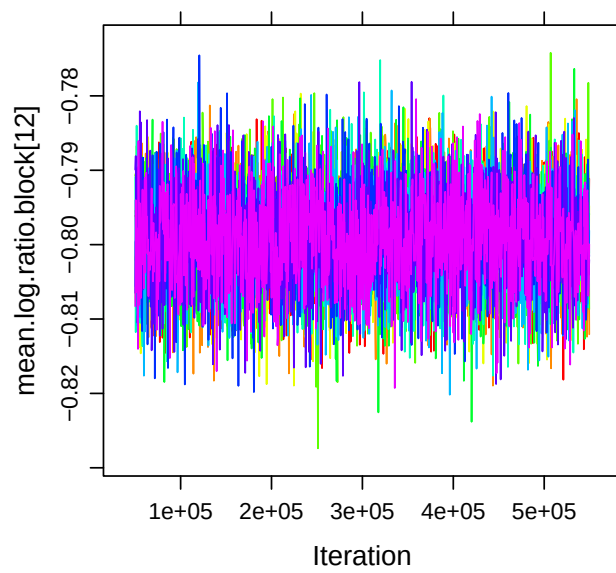


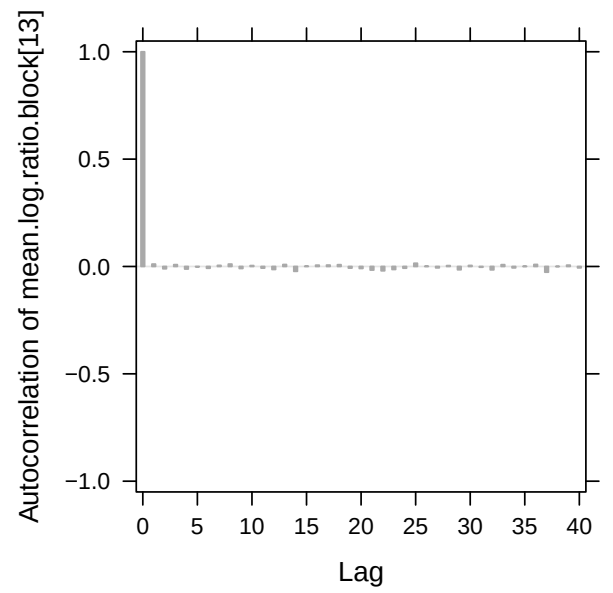
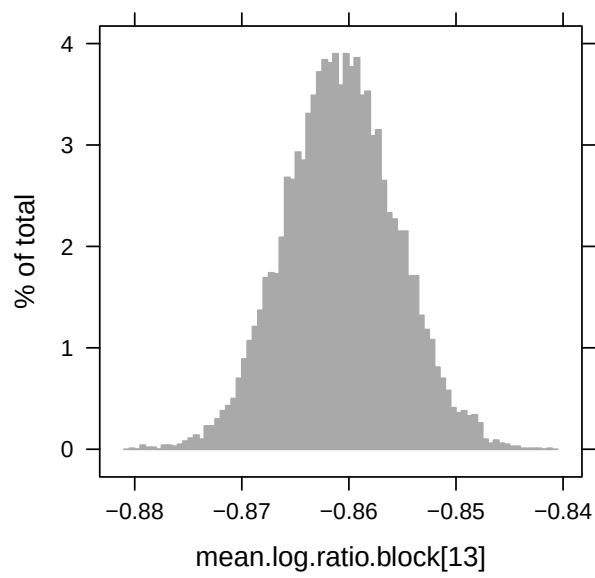
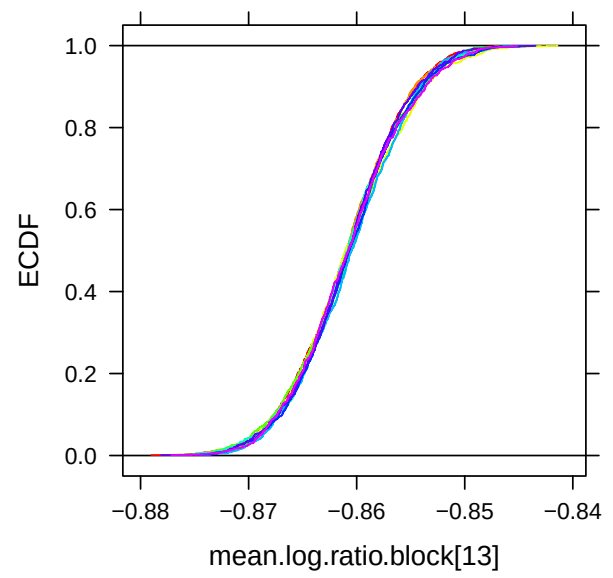
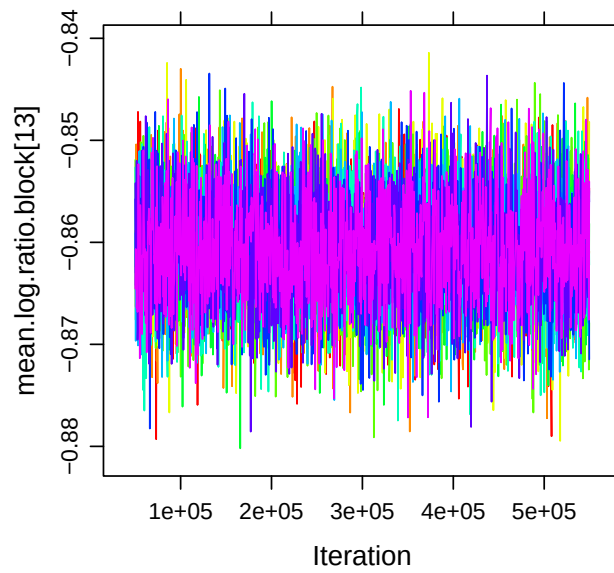


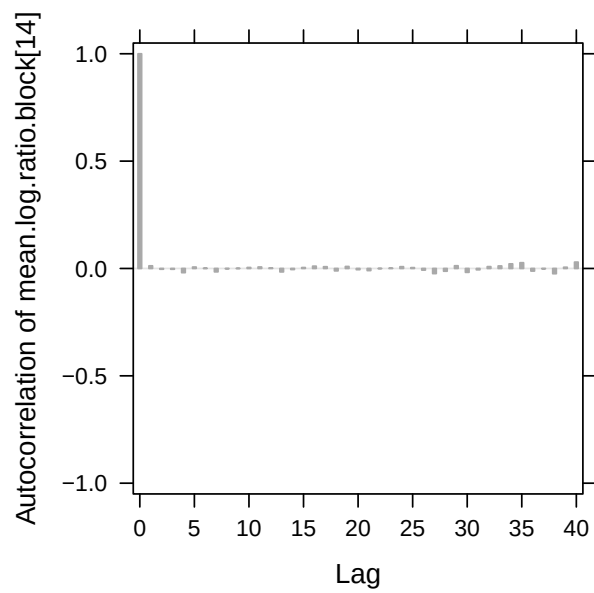
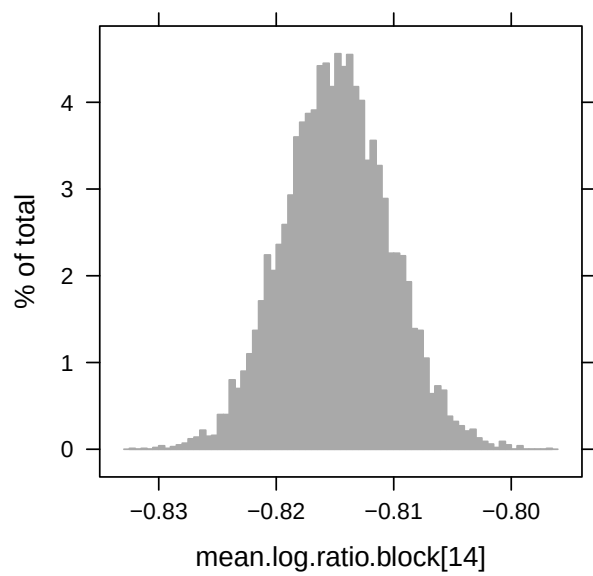
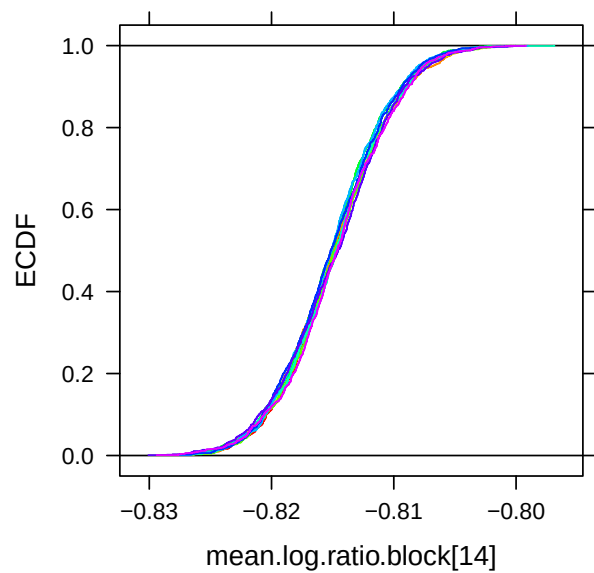
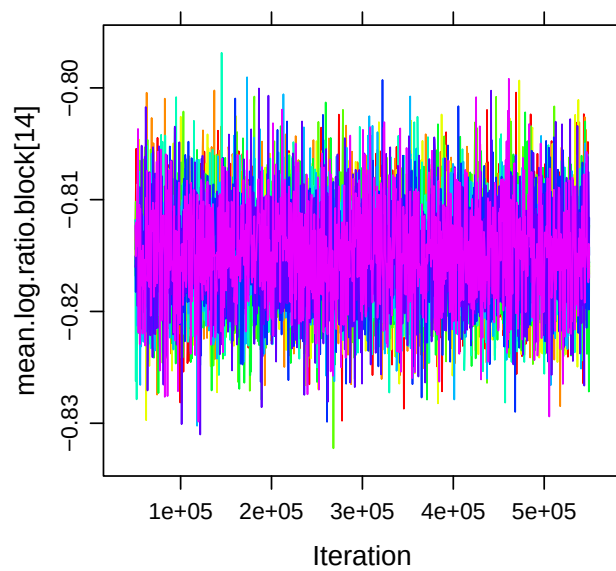


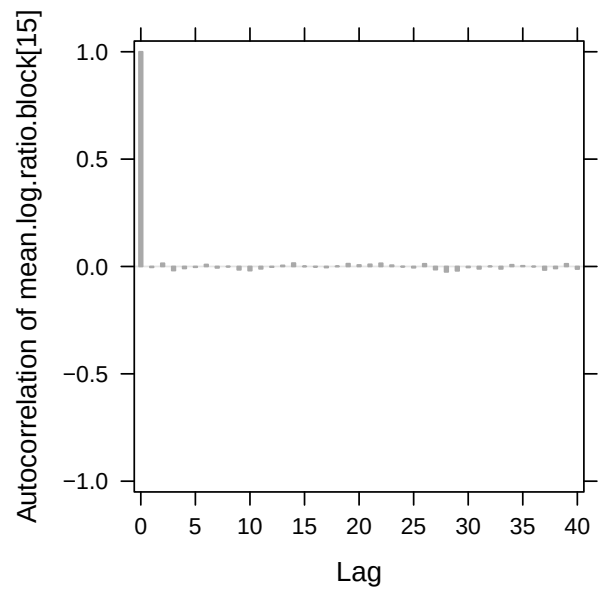
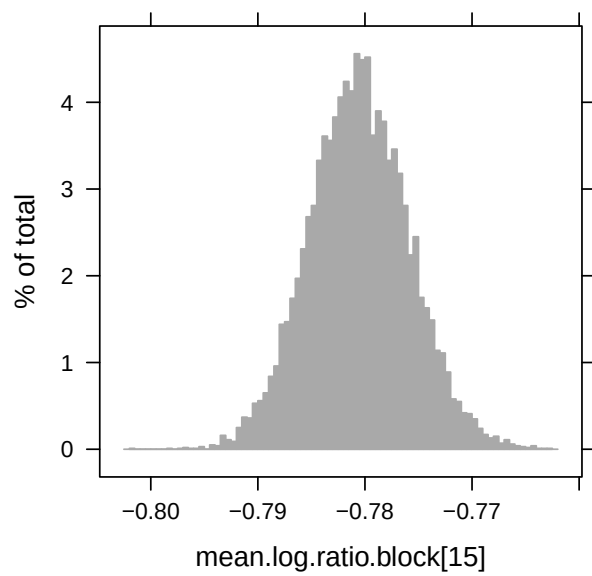
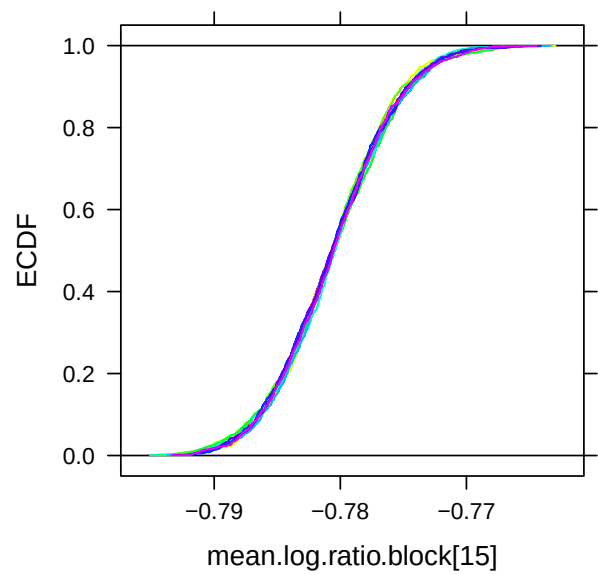
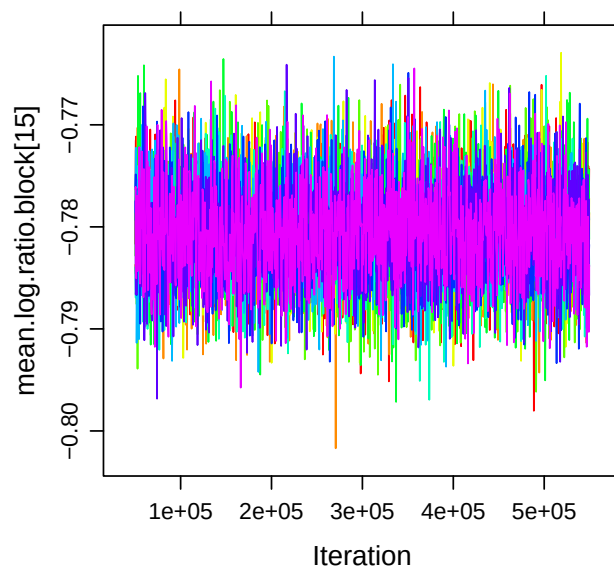


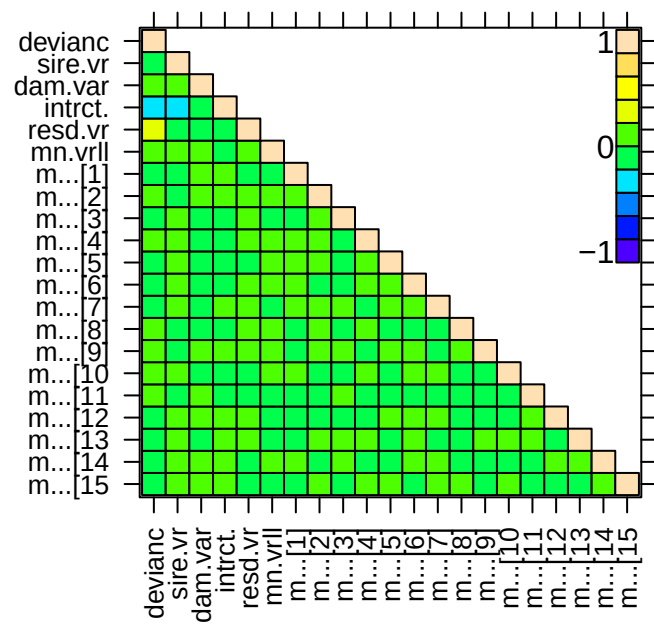












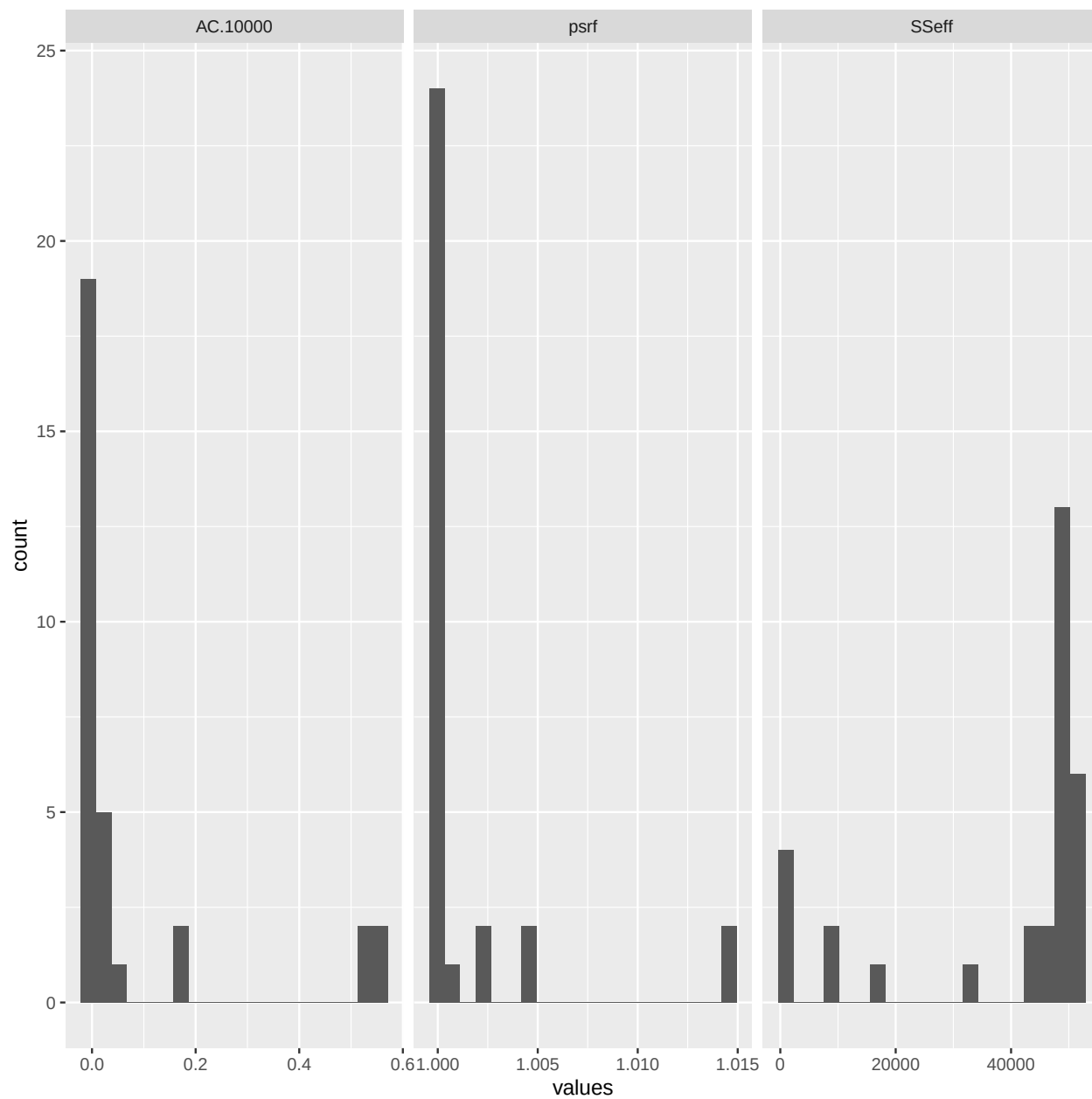
Model III

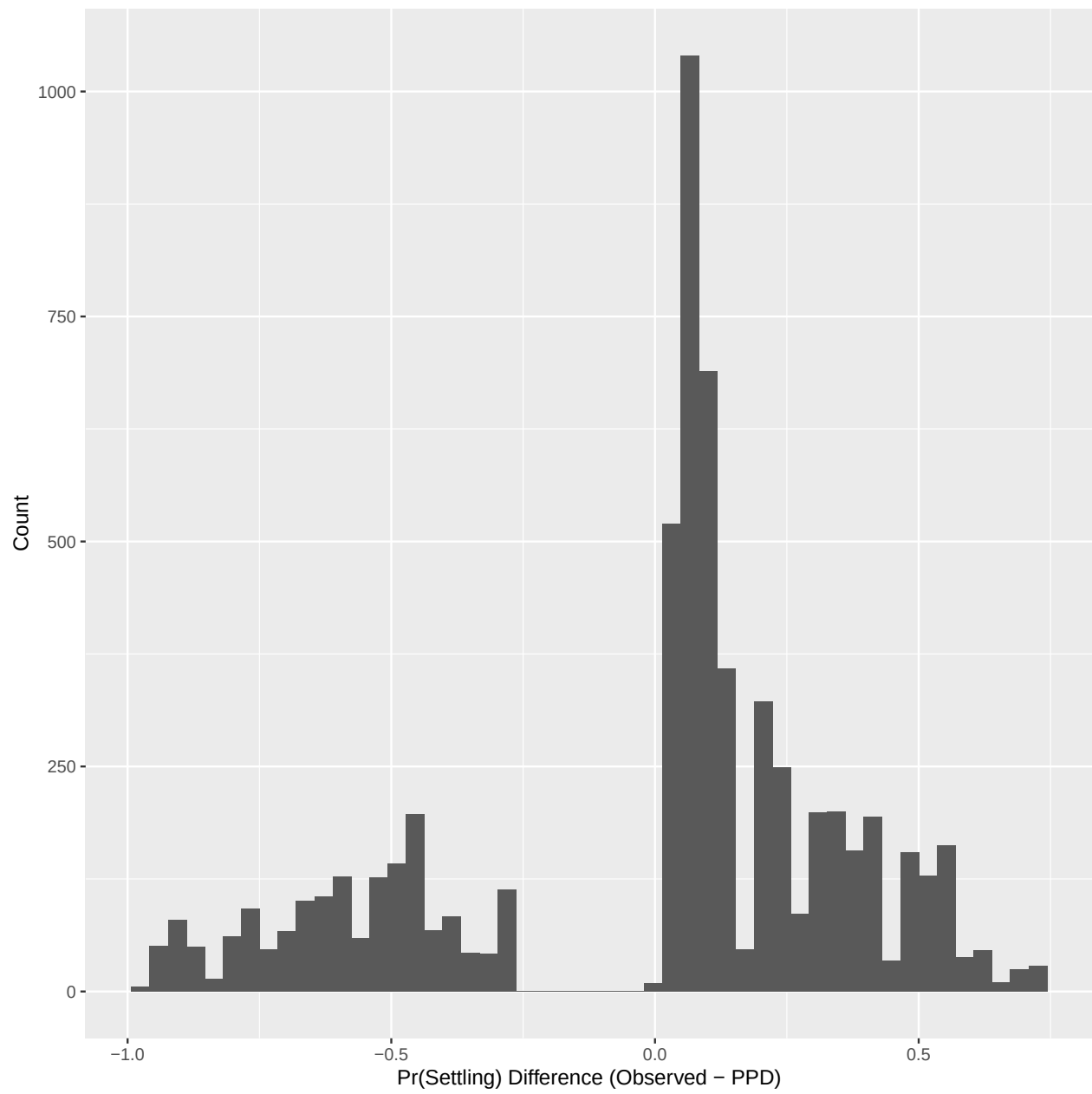
Number of chains: 10

Burnin: 1.001×10^5

Samples per chain: 5000

Total samples: 50000



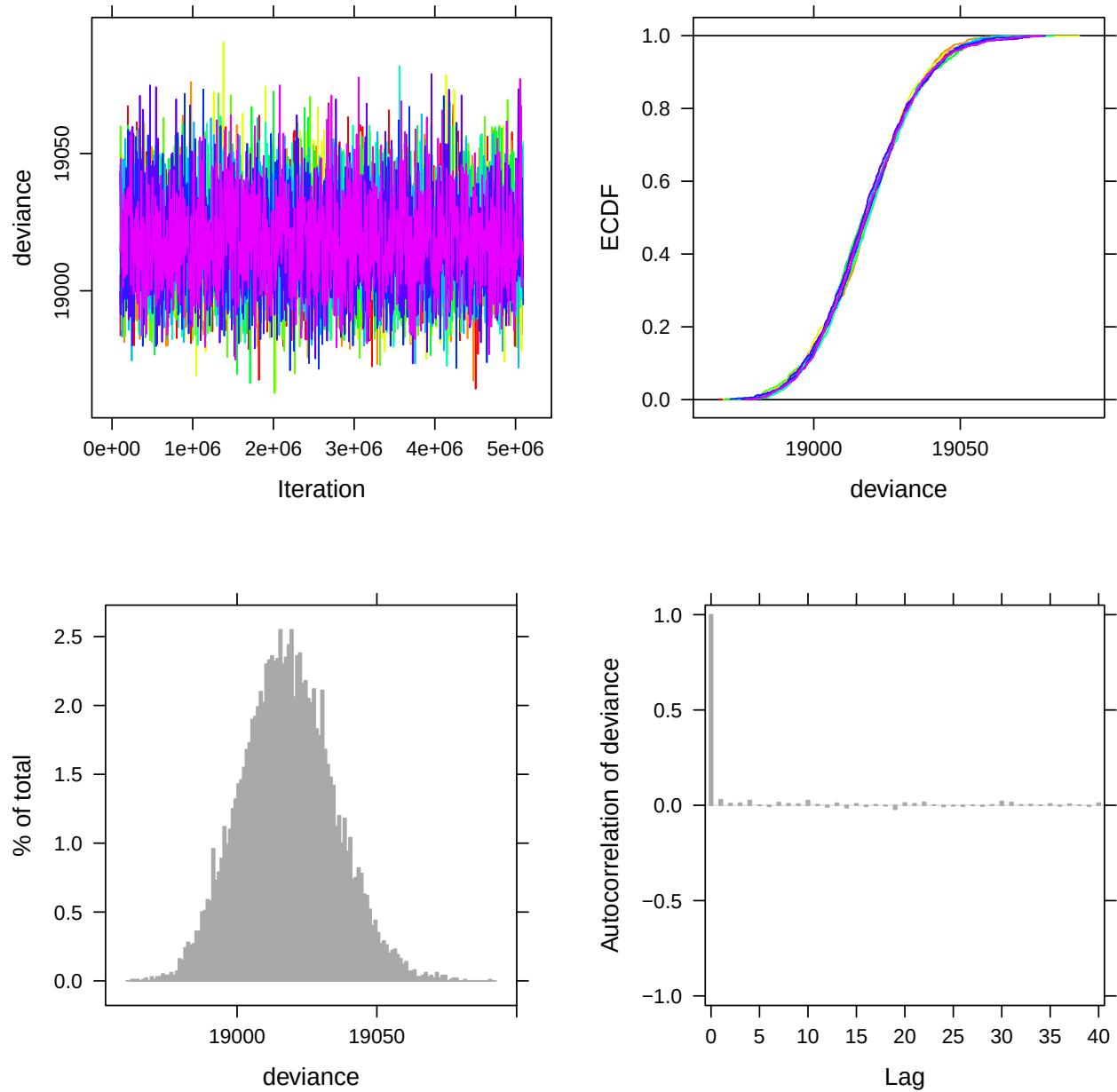


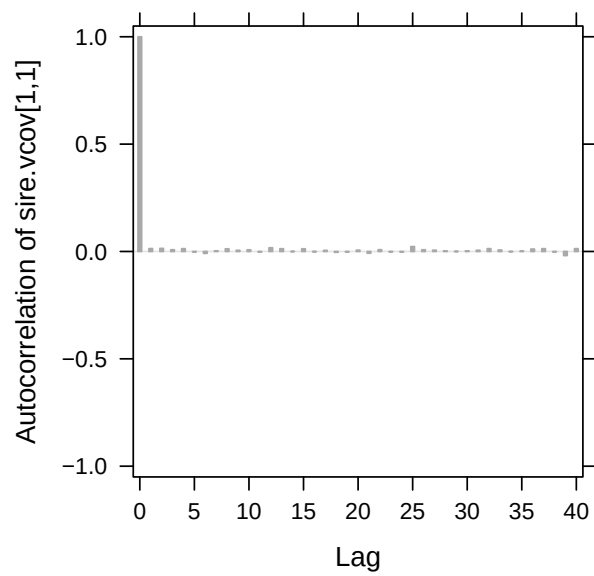
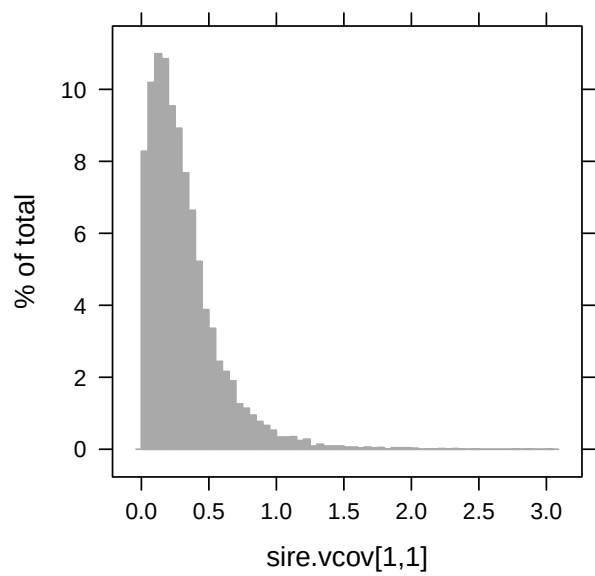
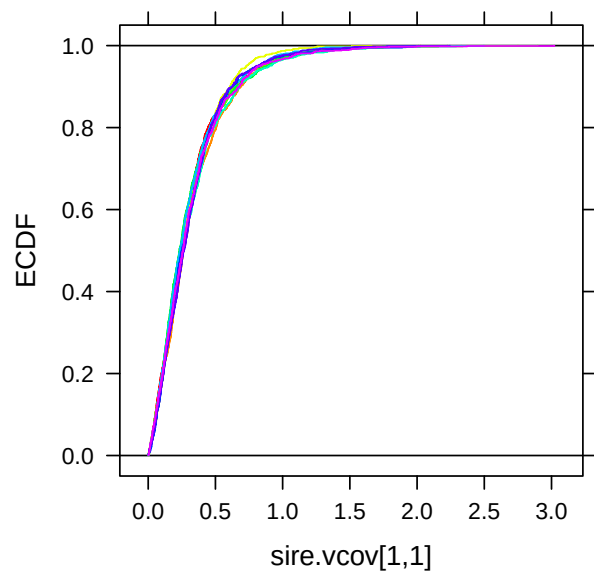
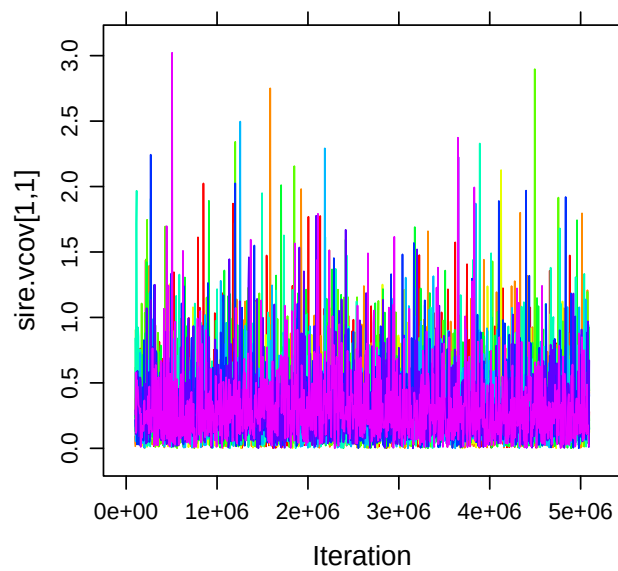
Generating summary statistics and plots (these will NOT be saved for reuse)...

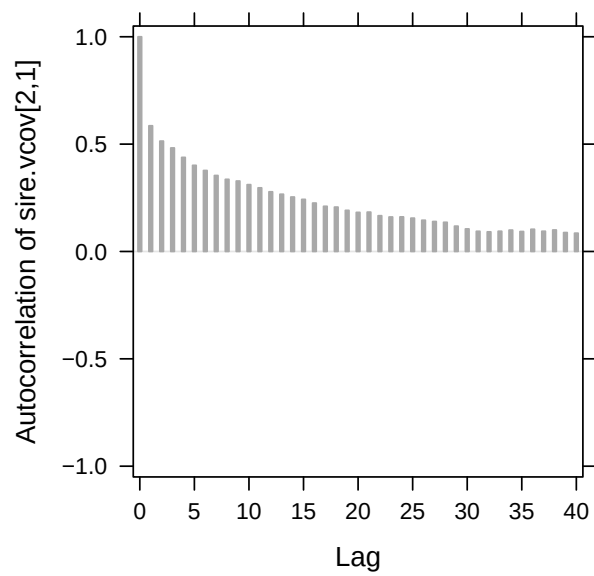
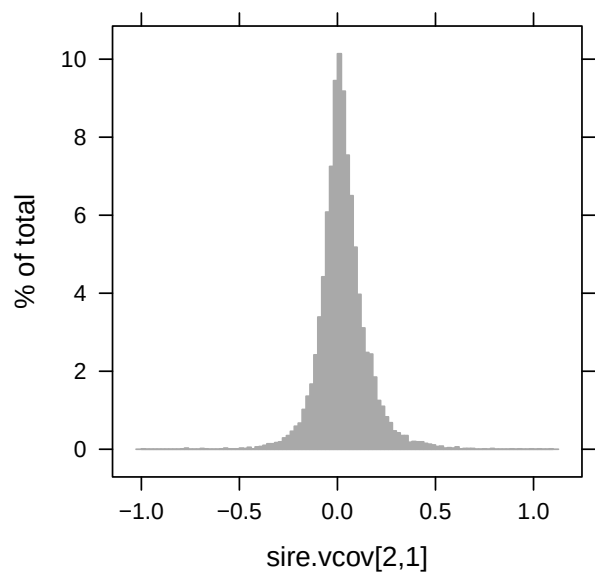
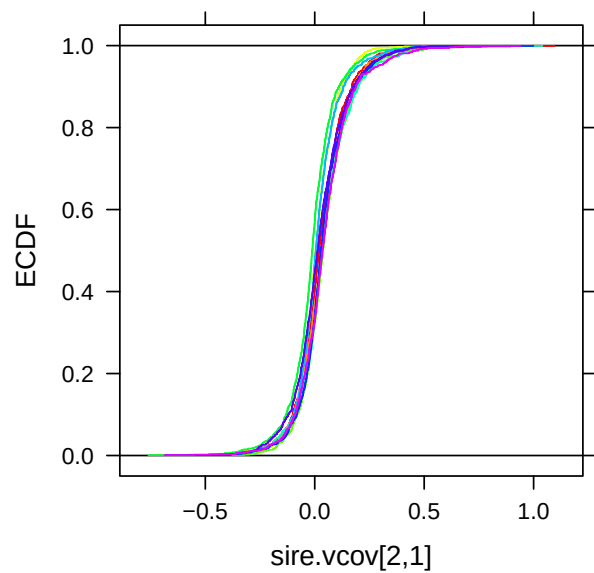
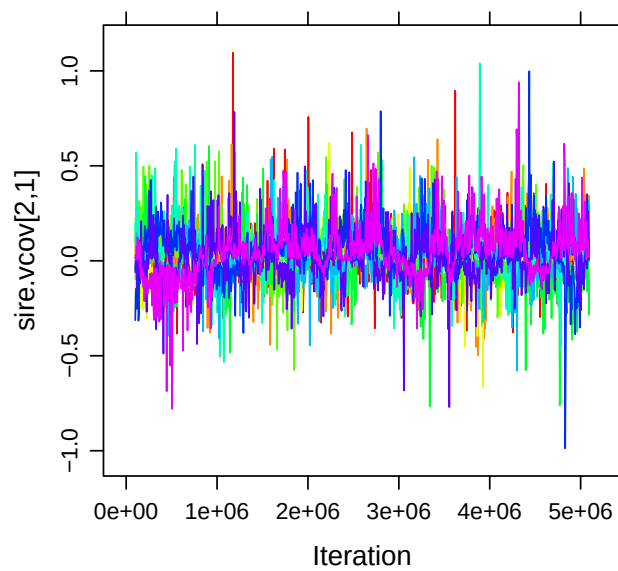
Calculating summary statistics...

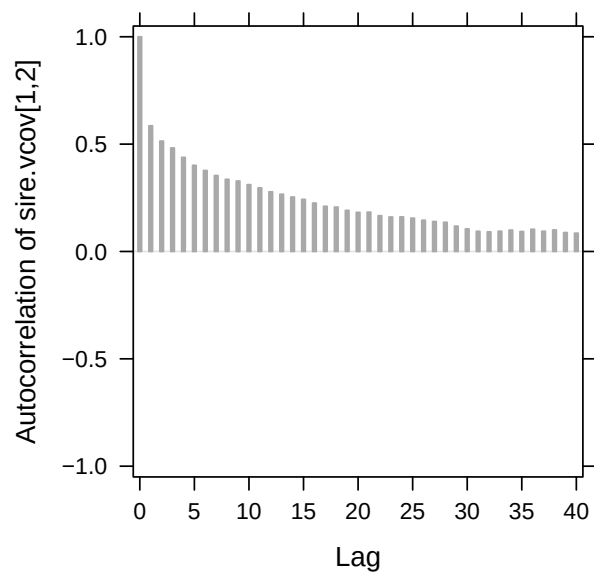
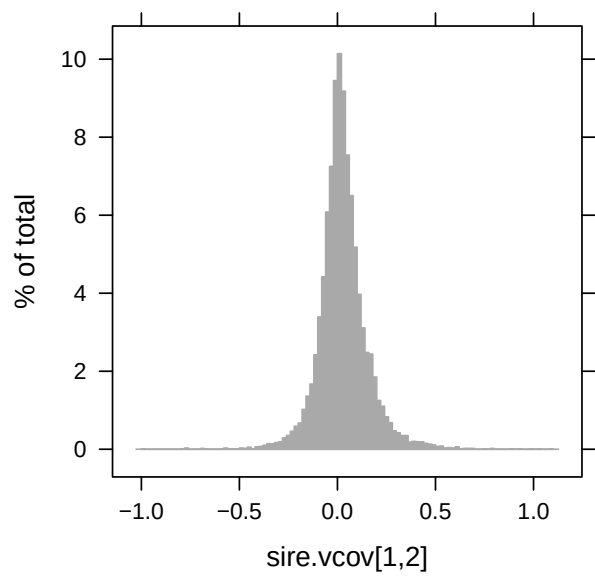
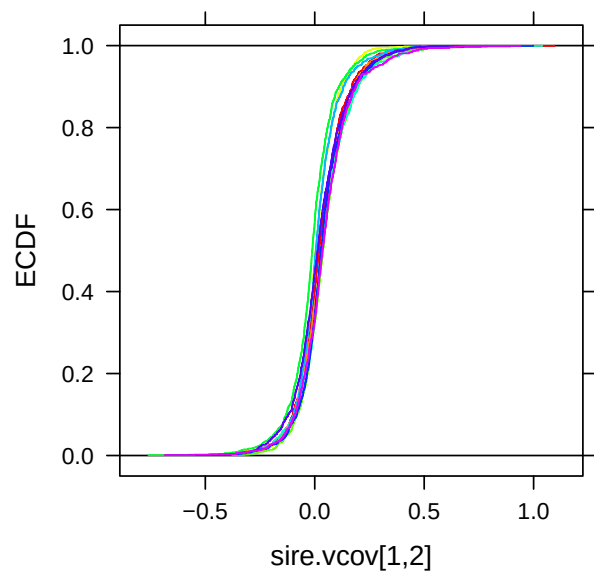
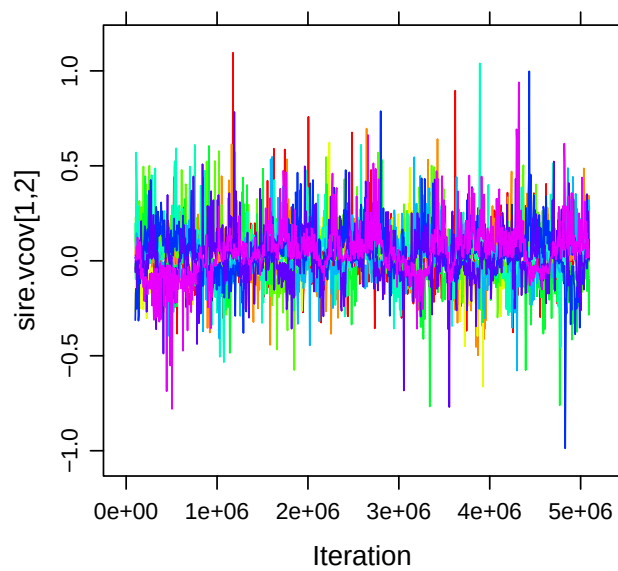
Calculating the Gelman-Rubin statistic for 31 variables....

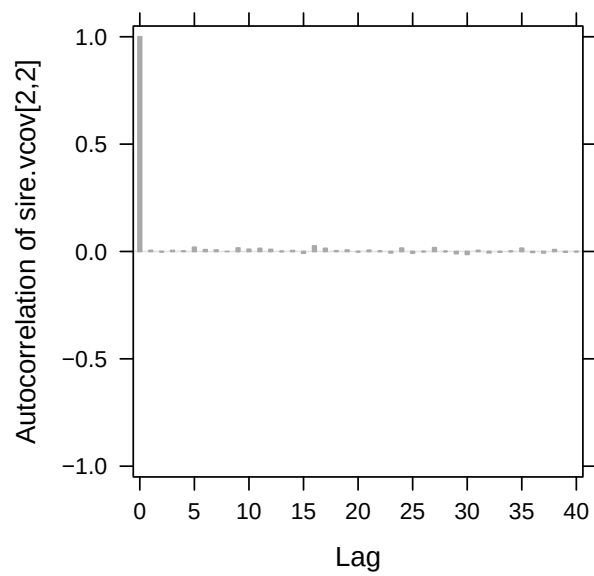
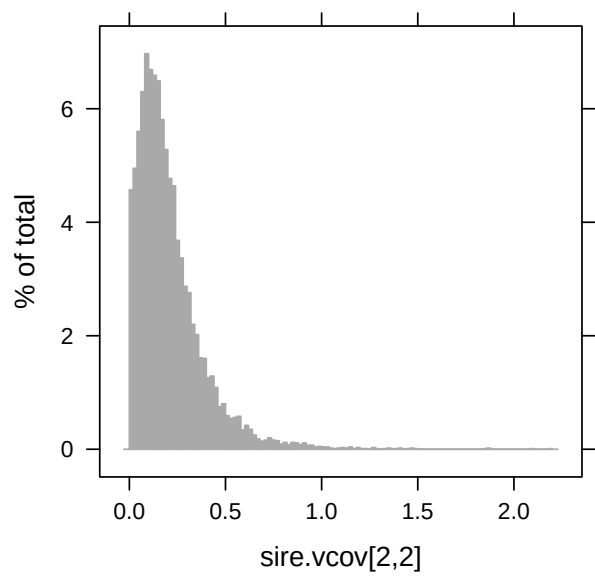
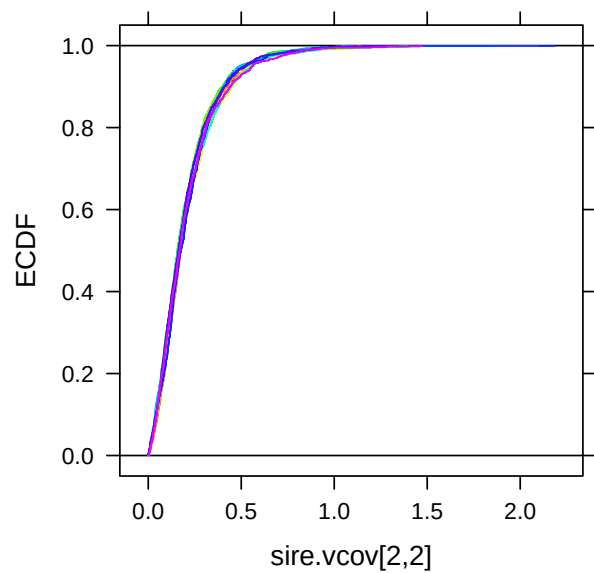
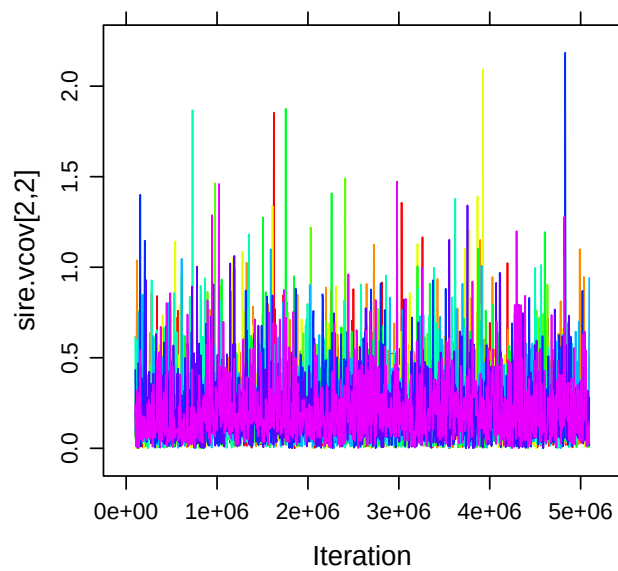
Note: Unable to calculate the multivariate psrf

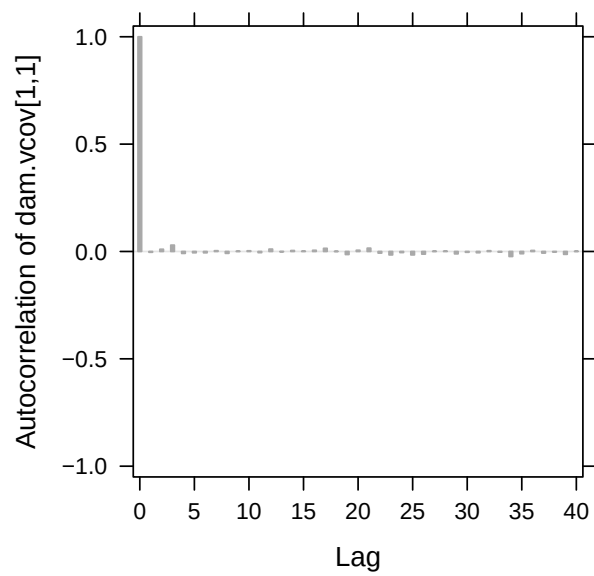
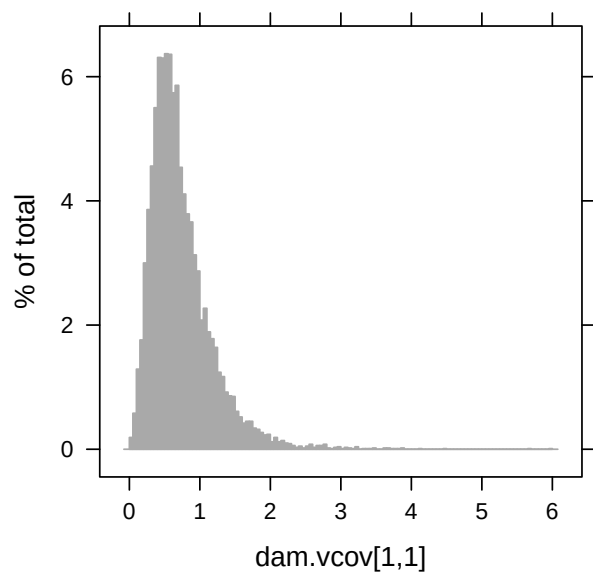
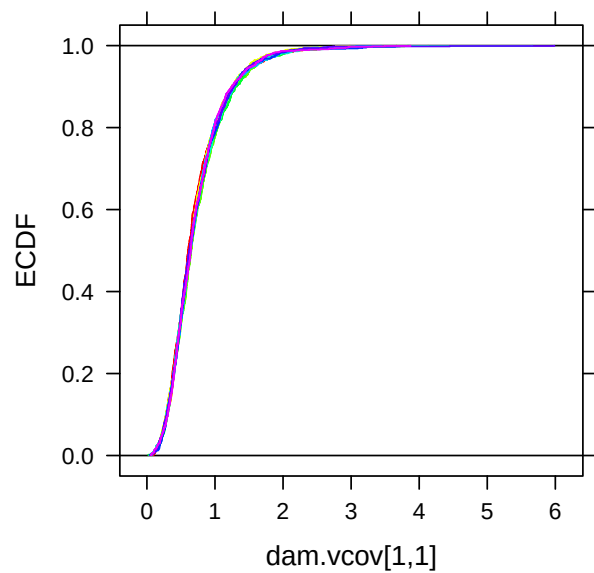
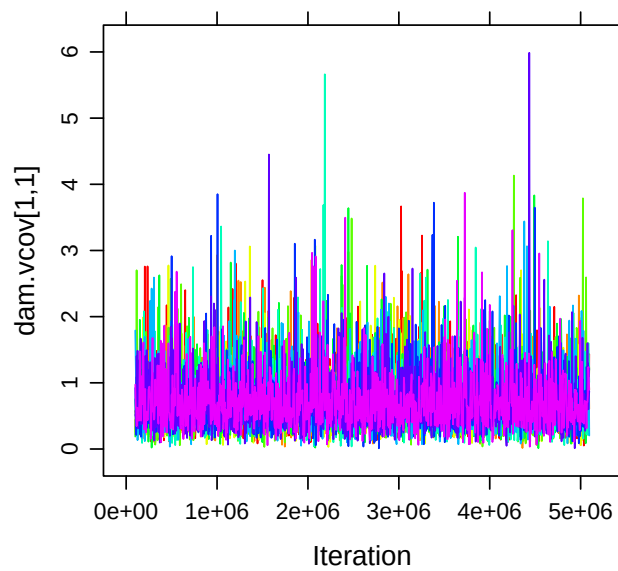


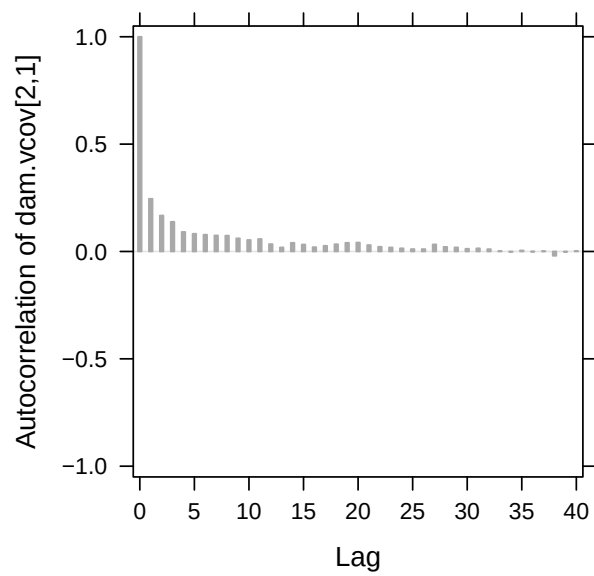
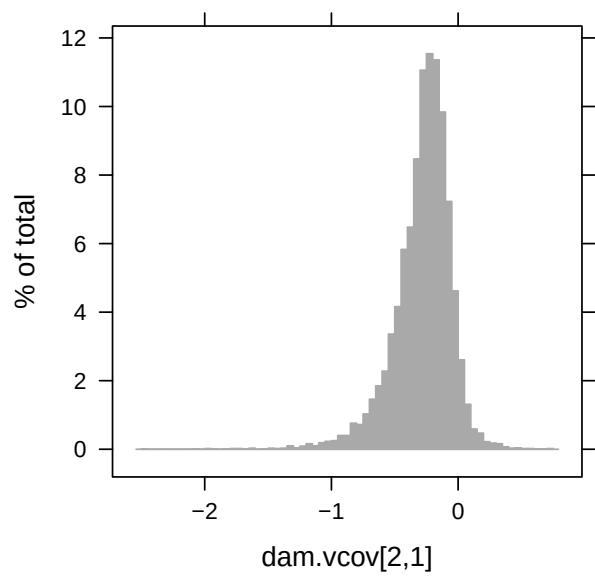
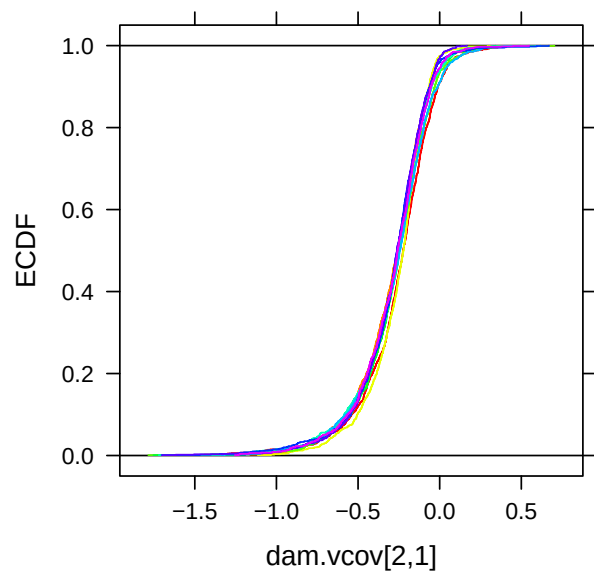
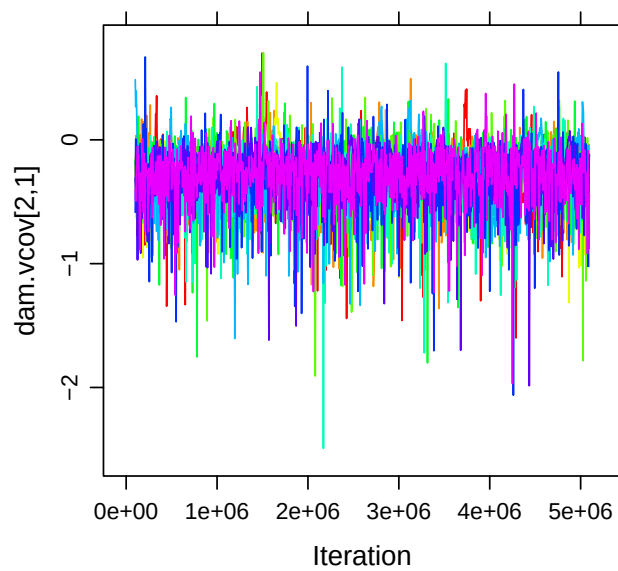


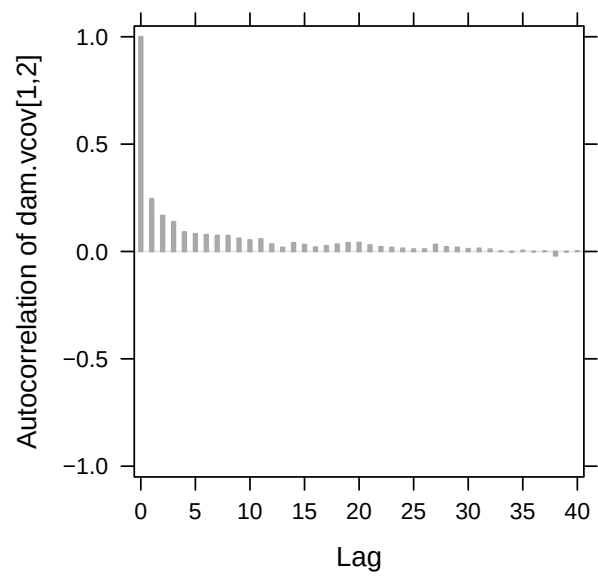
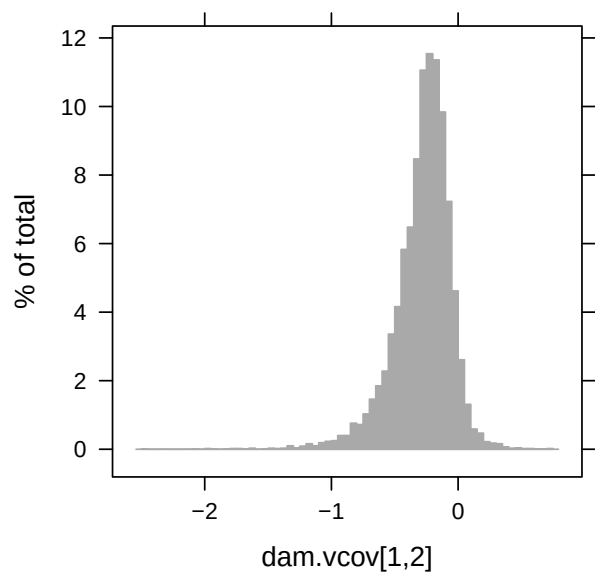
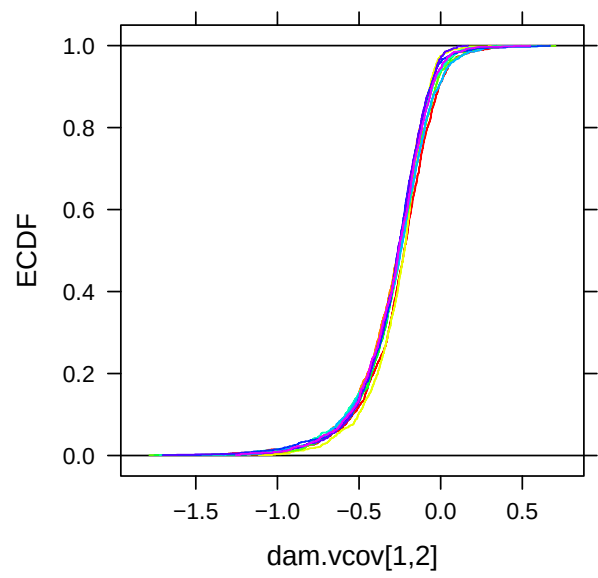
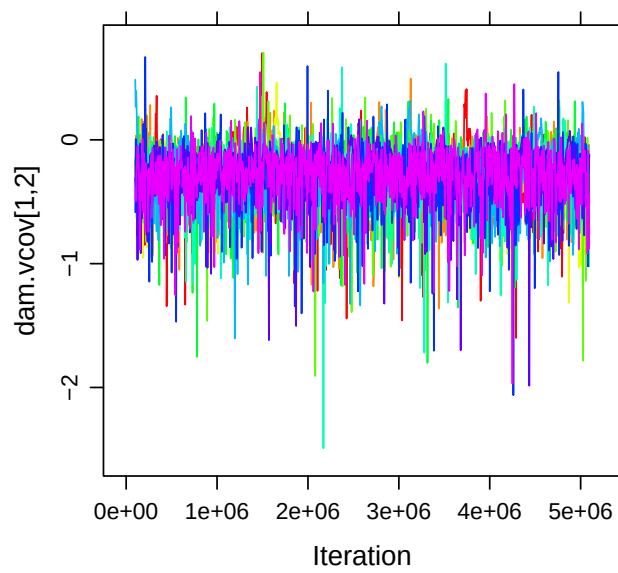


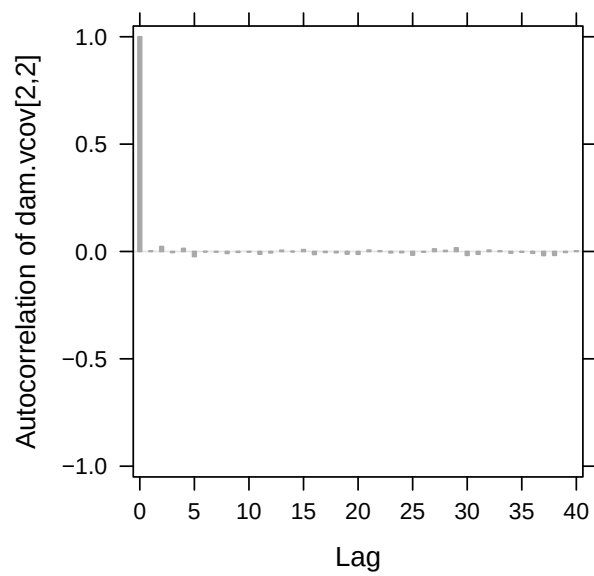
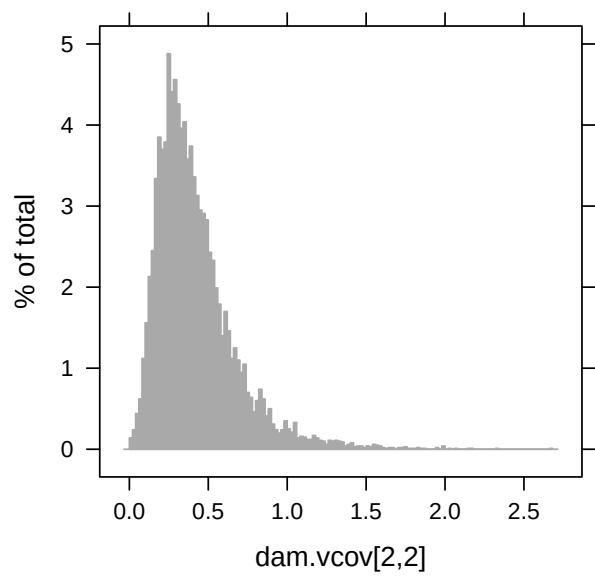
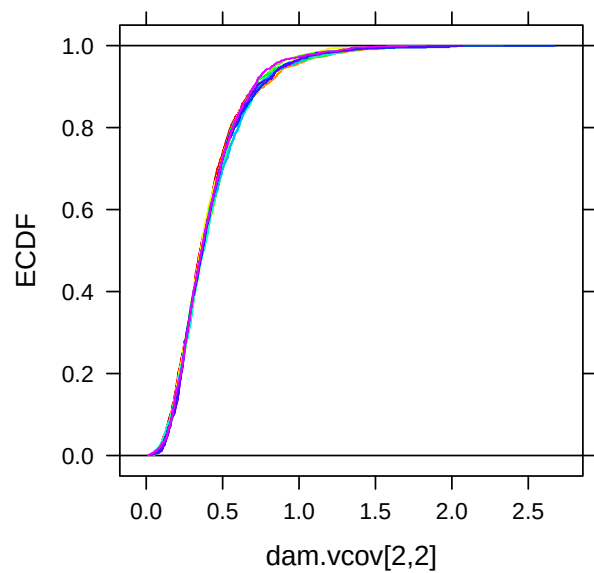
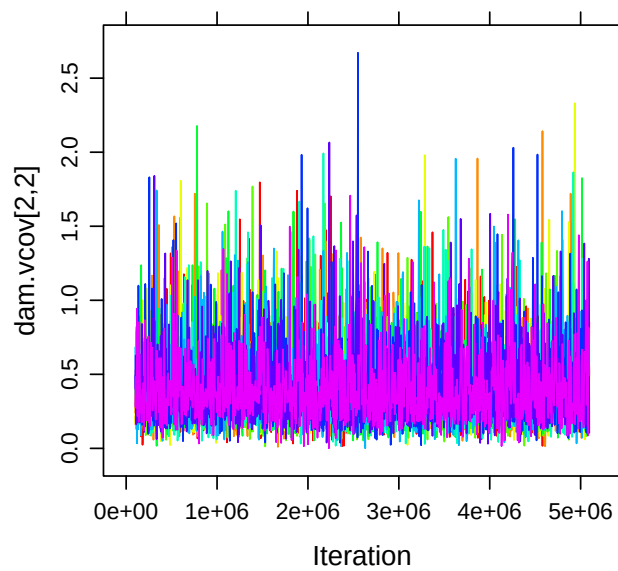


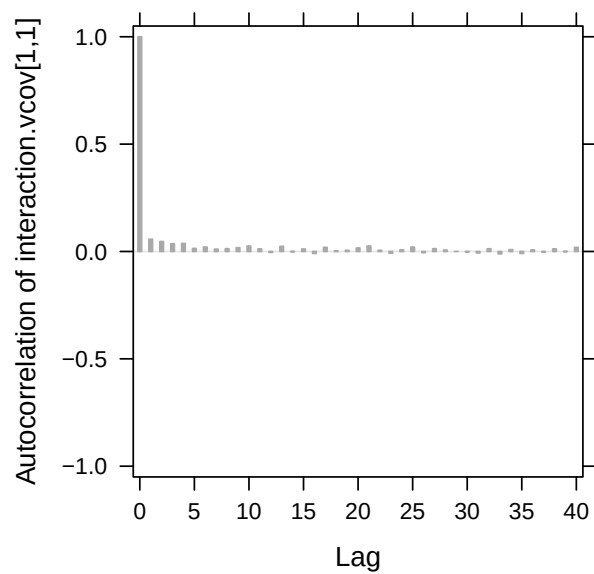
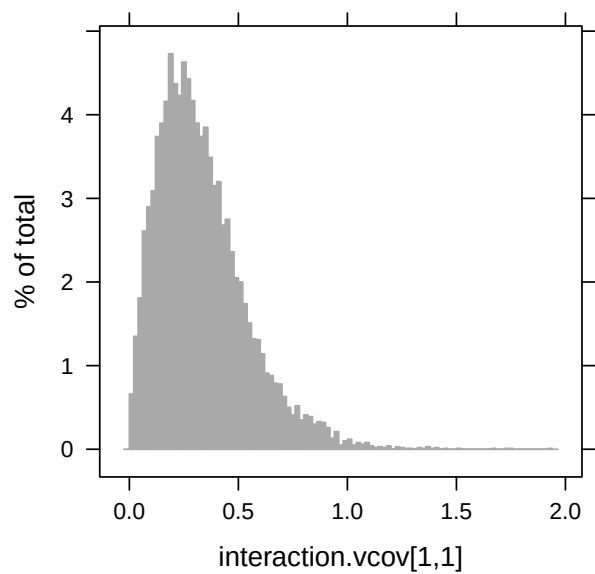
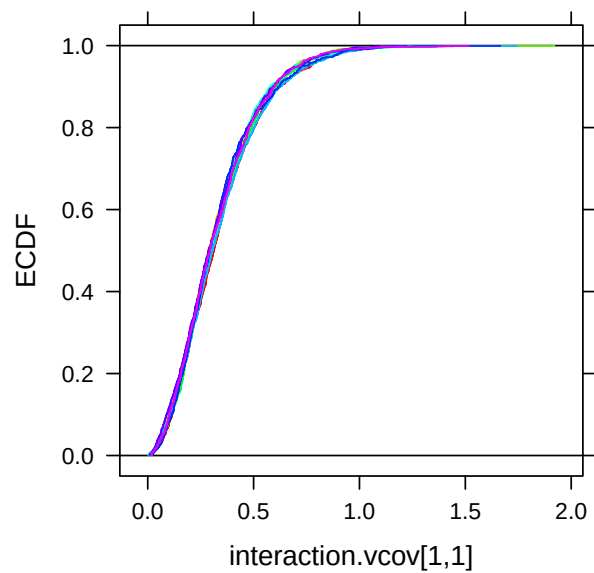
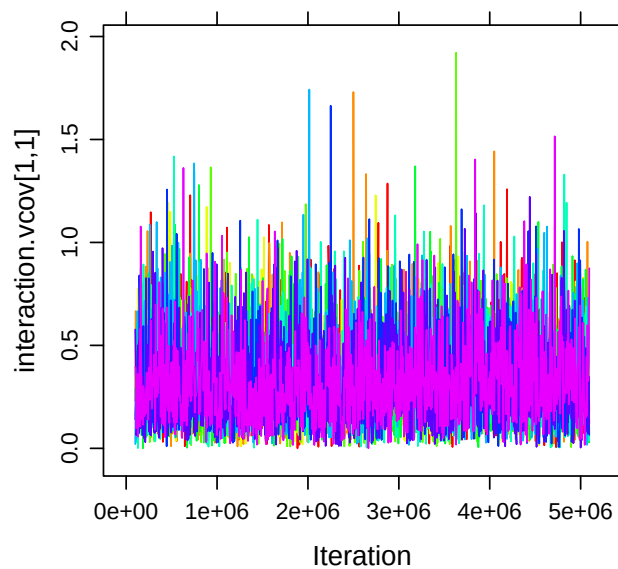


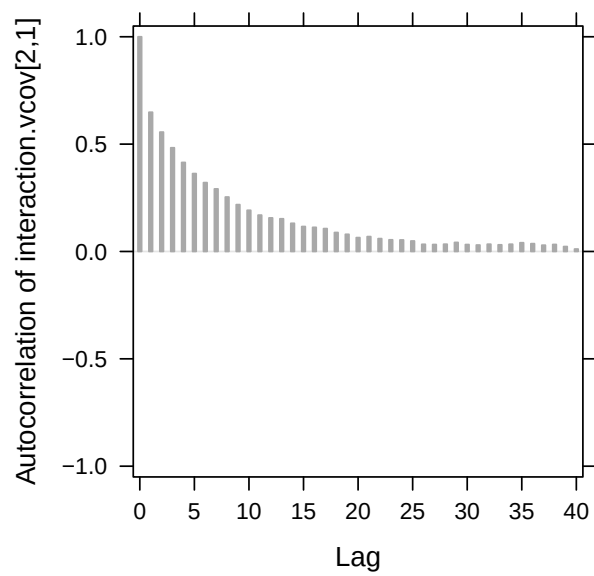
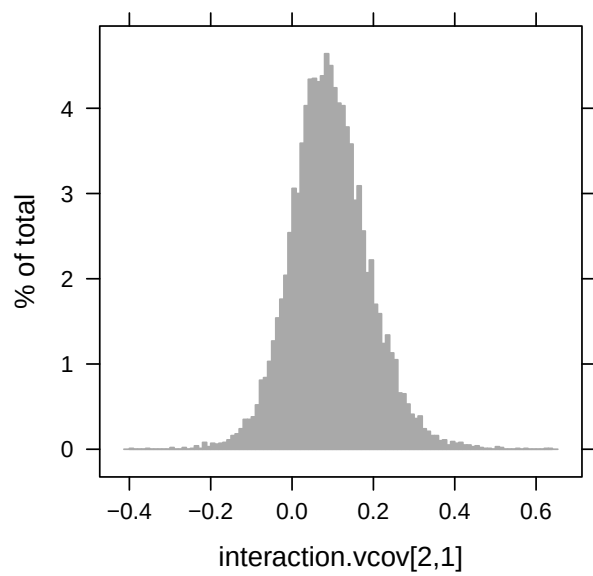
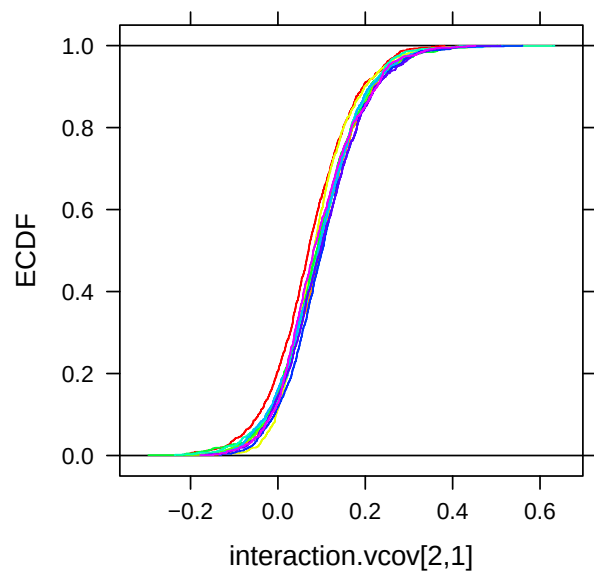
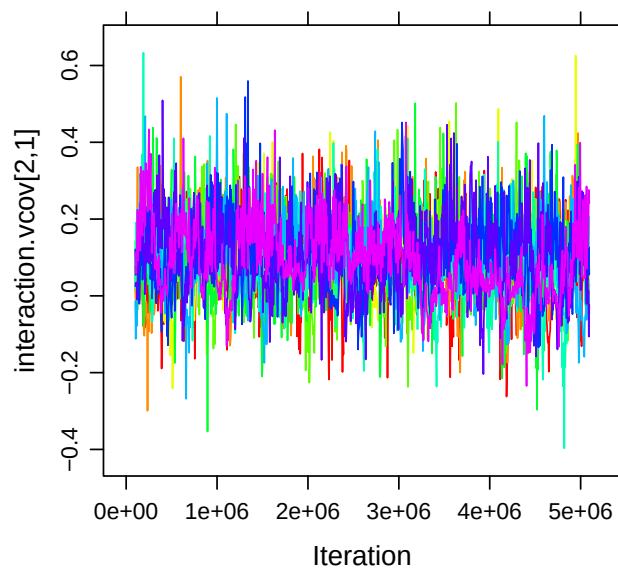


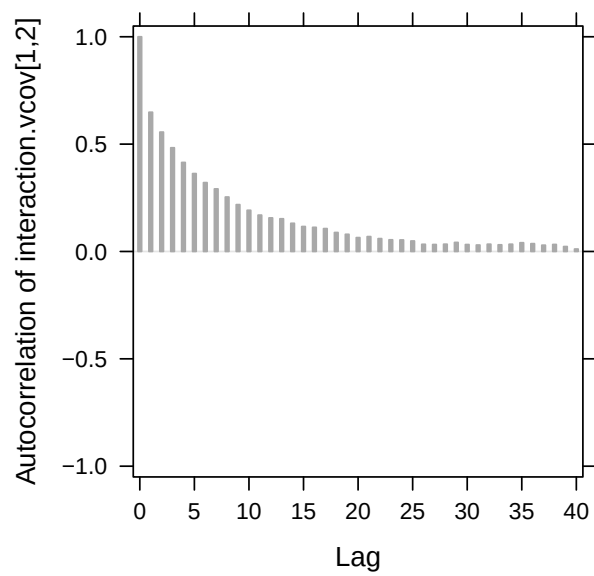
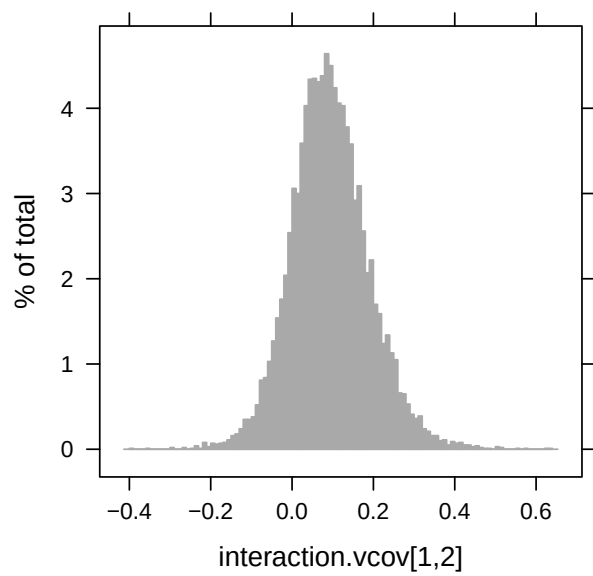
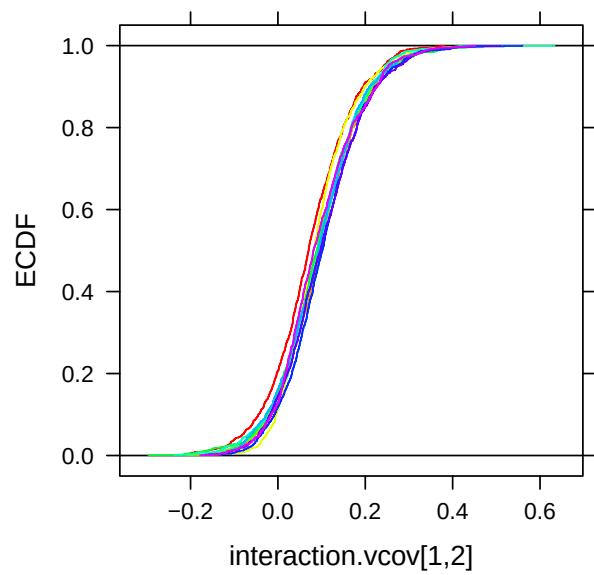
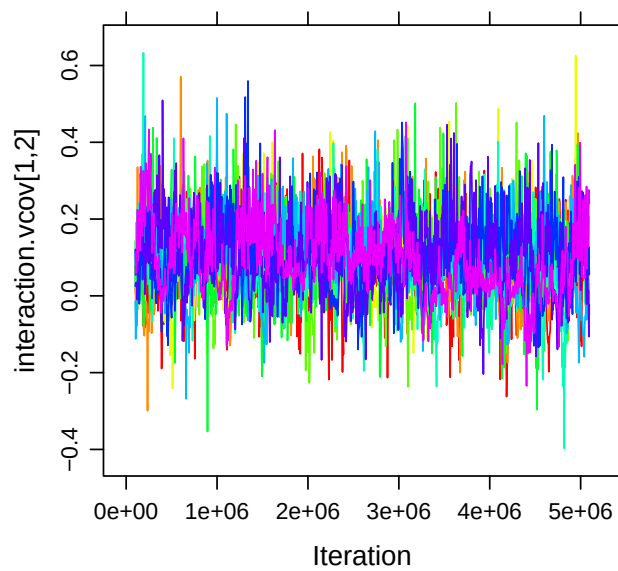


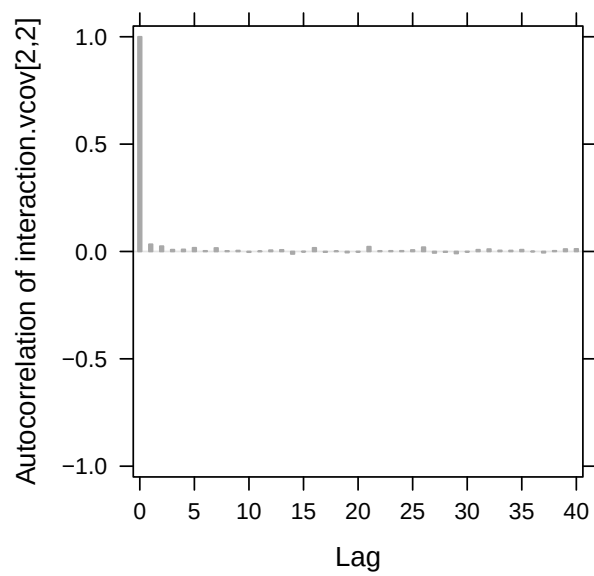
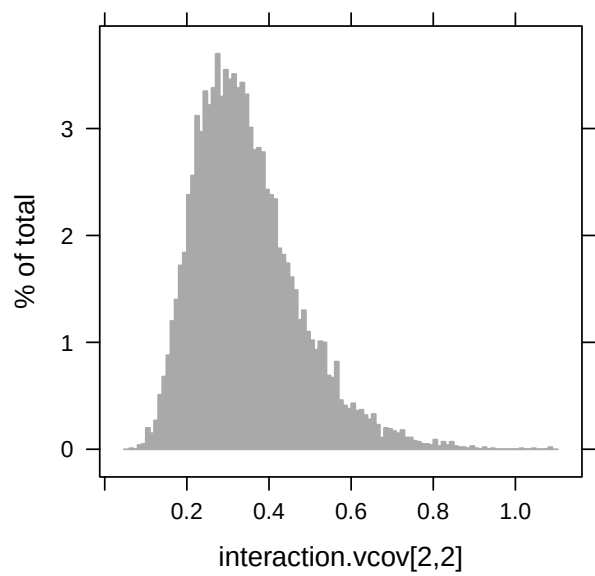
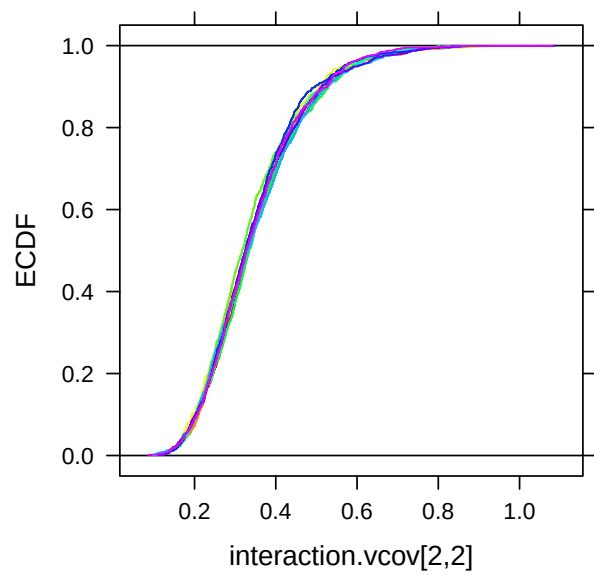
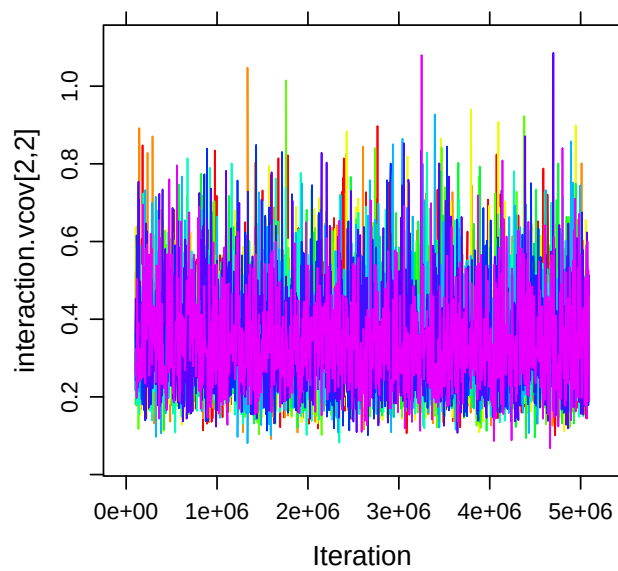


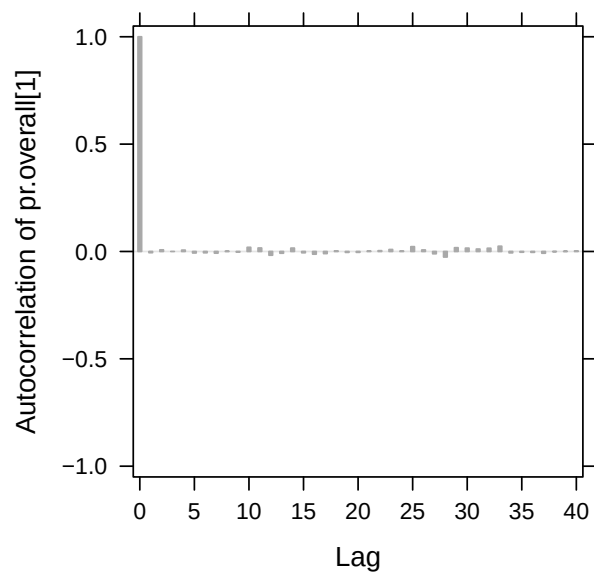
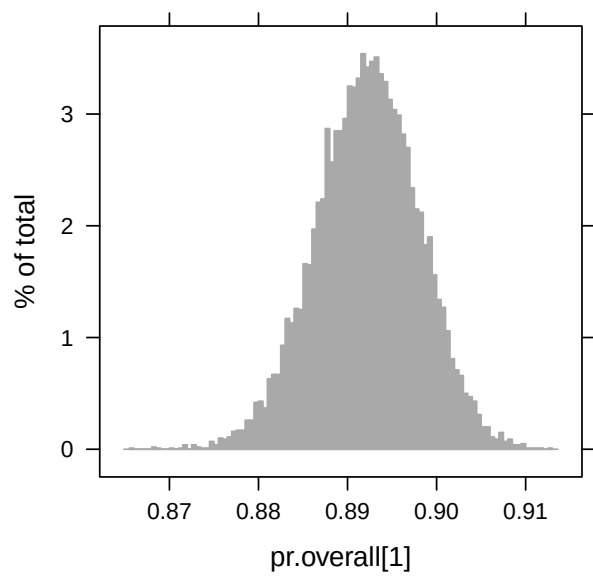
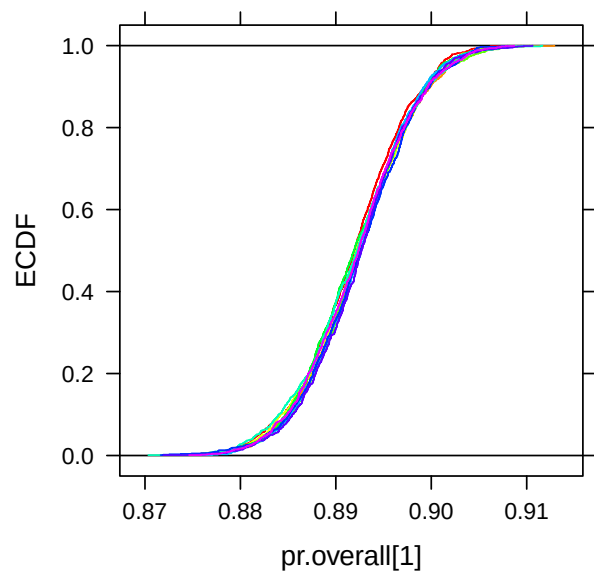
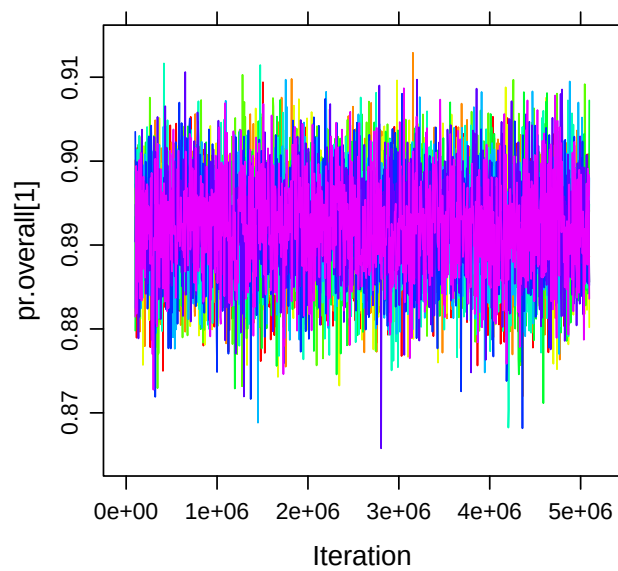


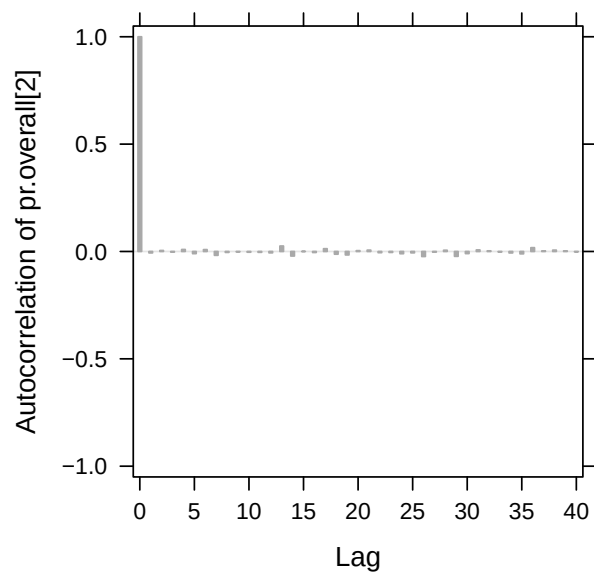
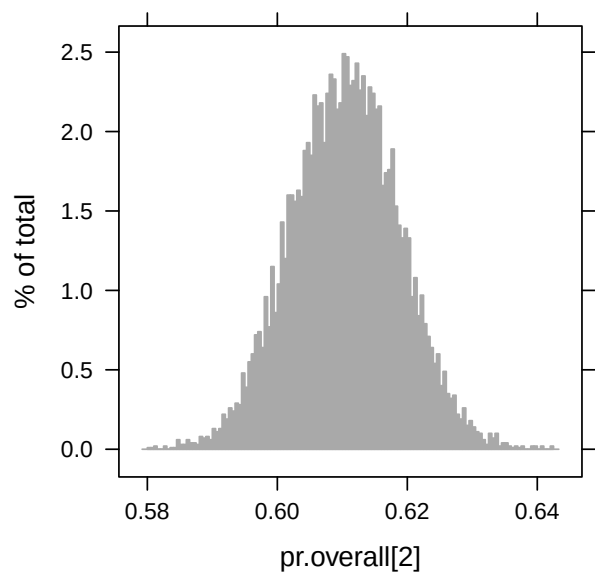
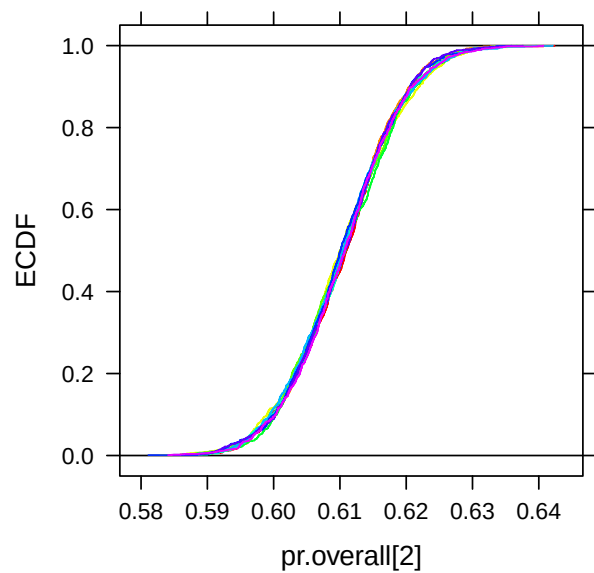
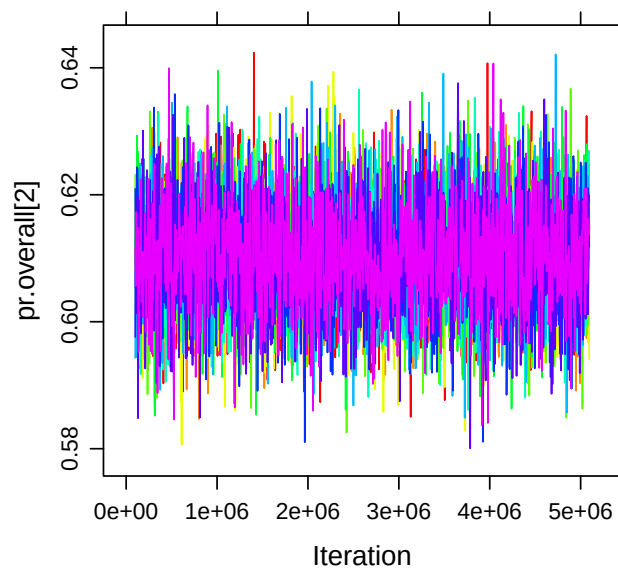


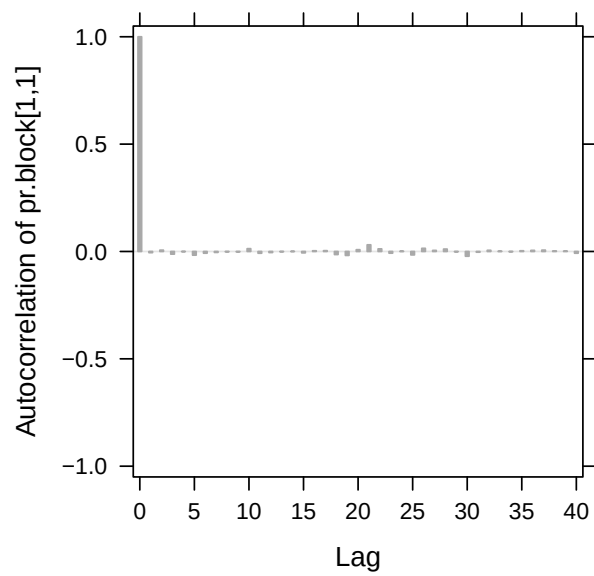
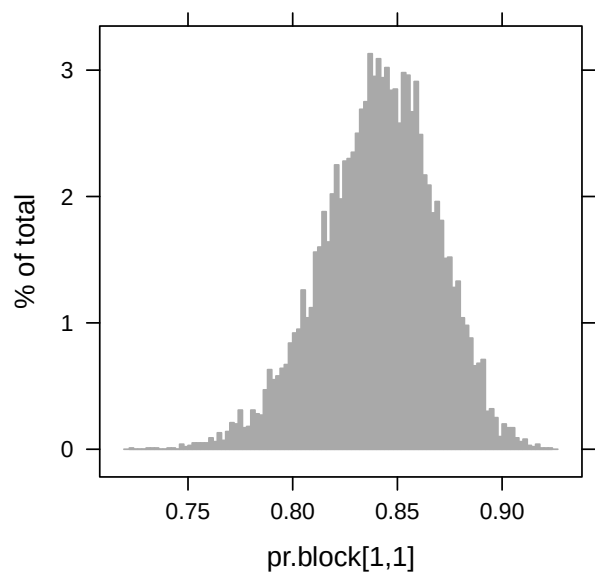
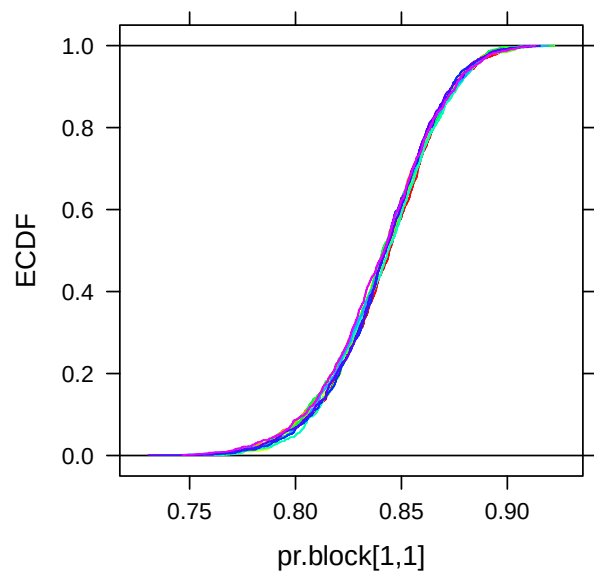
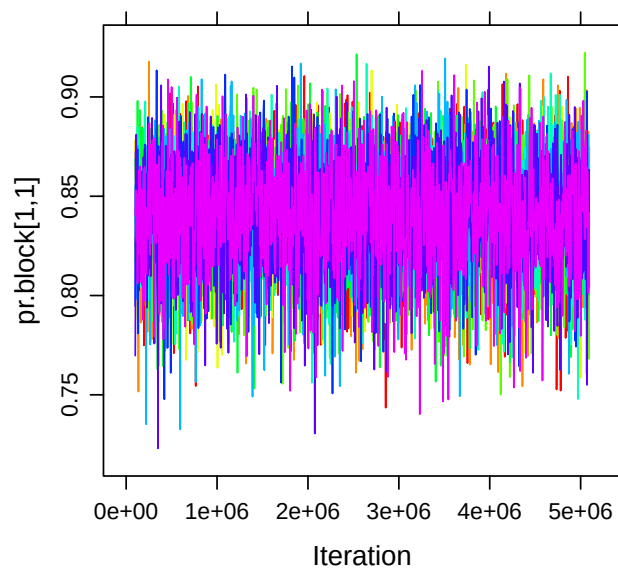


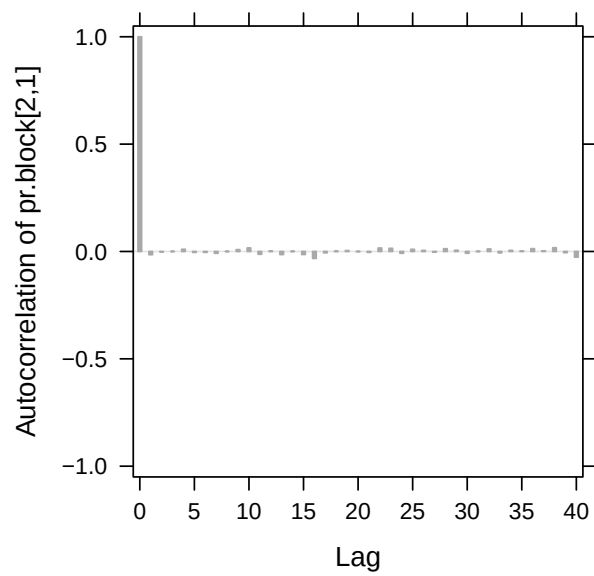
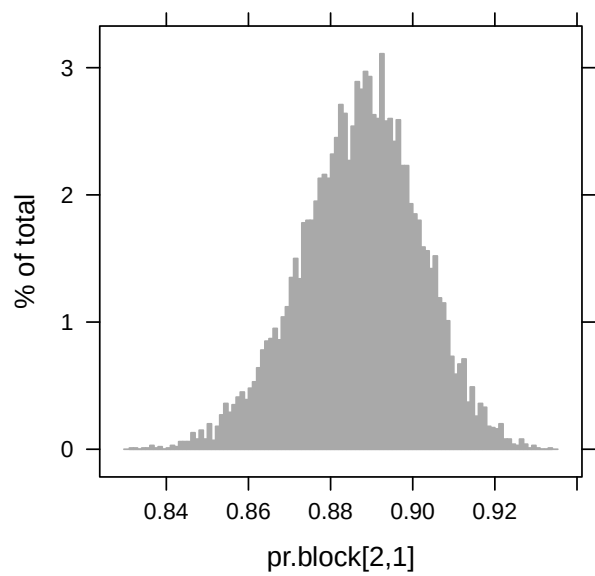
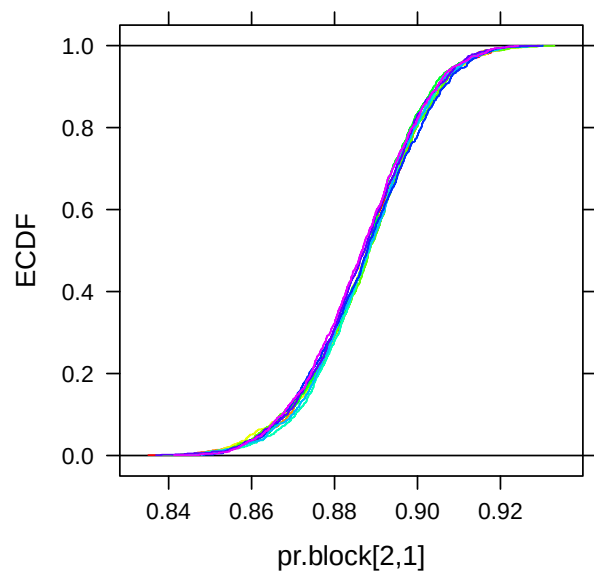
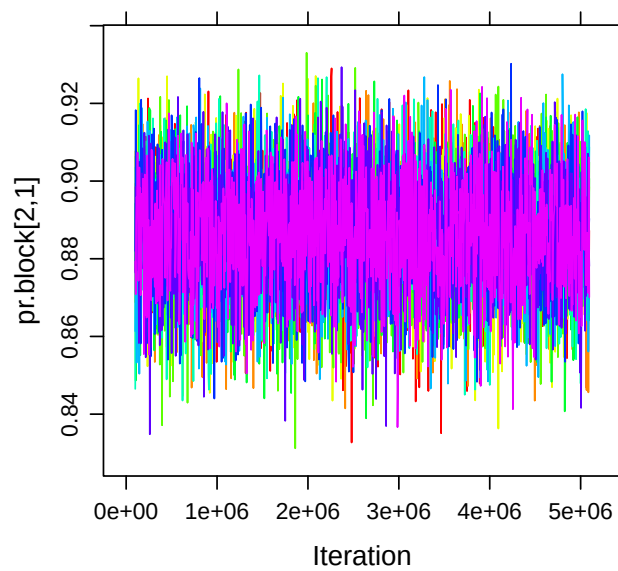


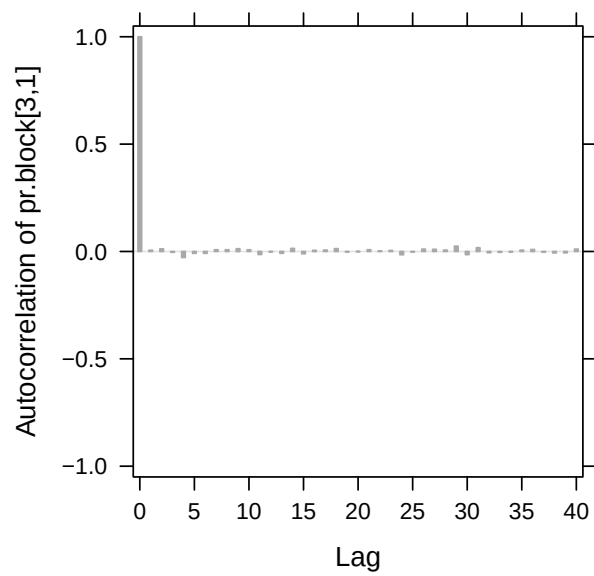
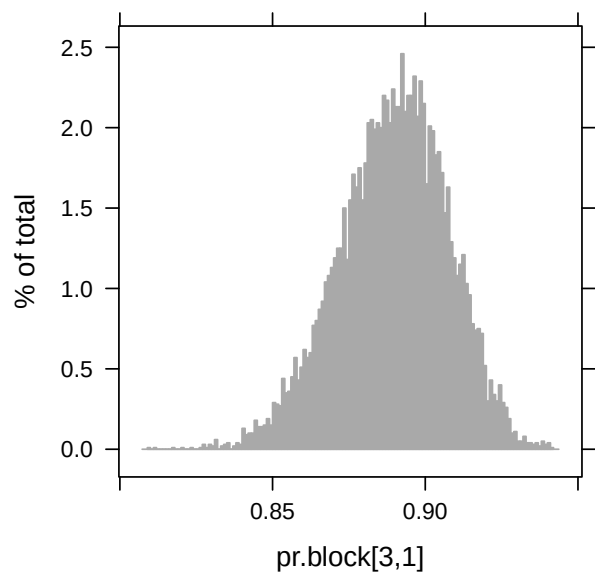
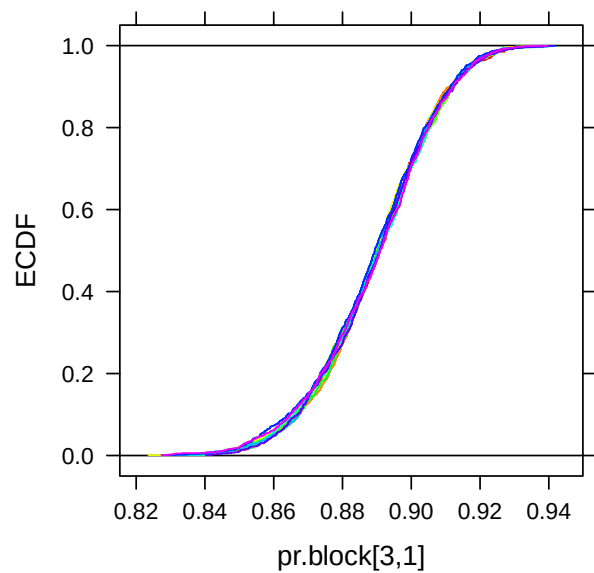
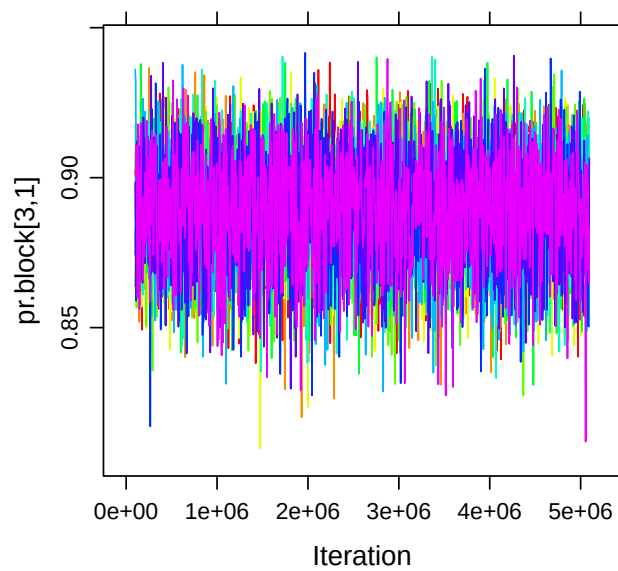


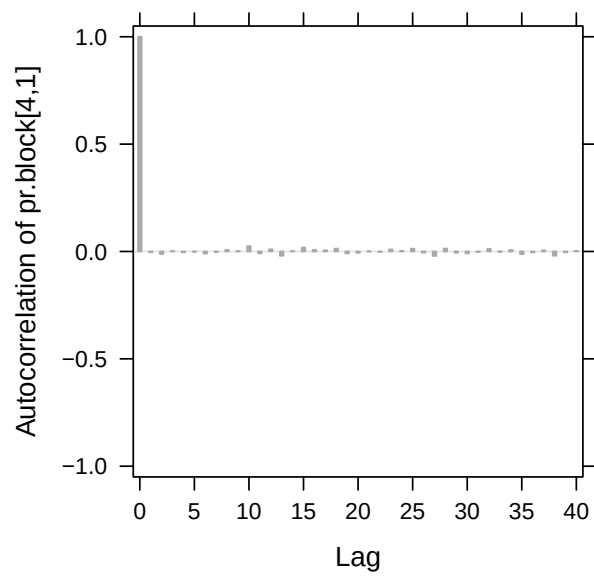
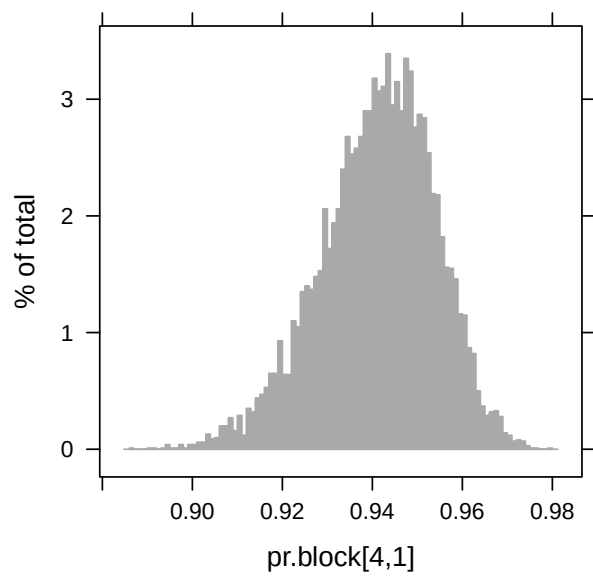
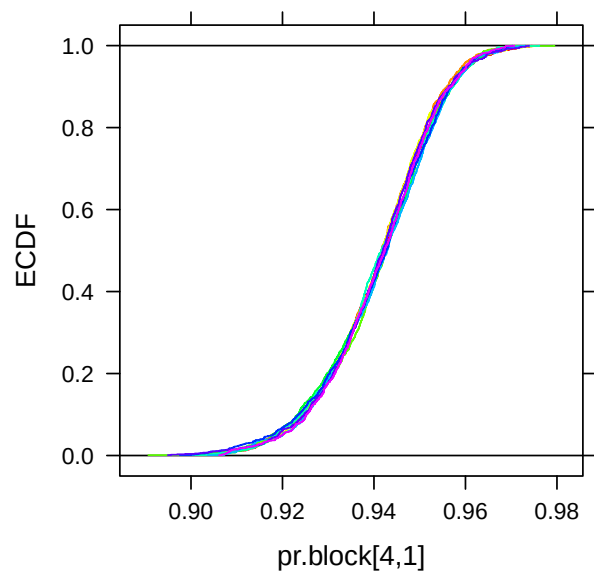
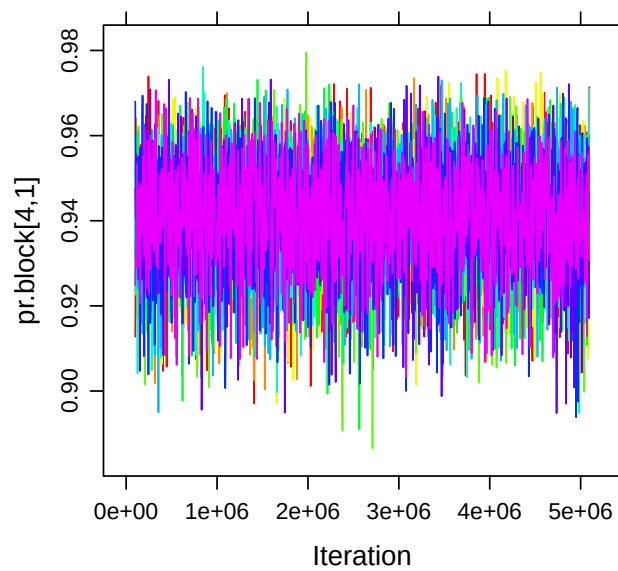


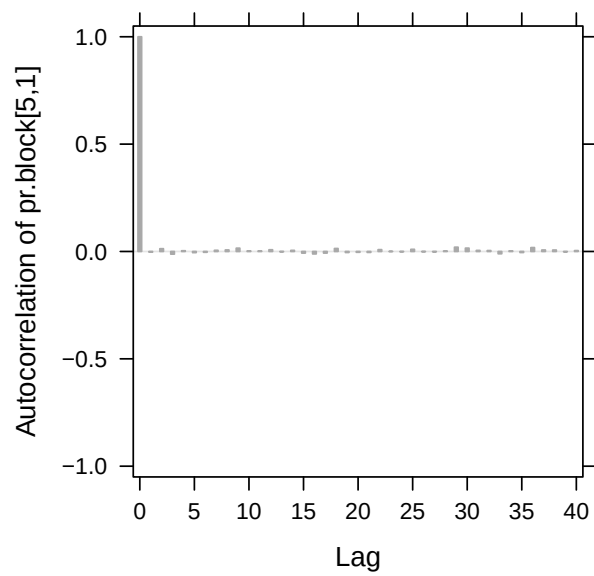
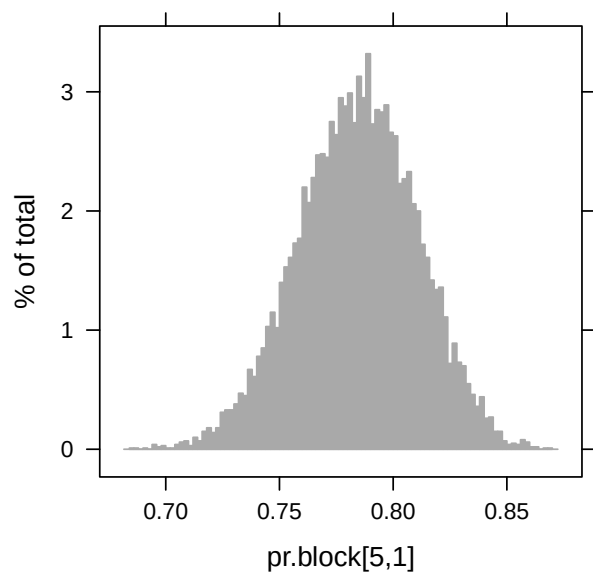
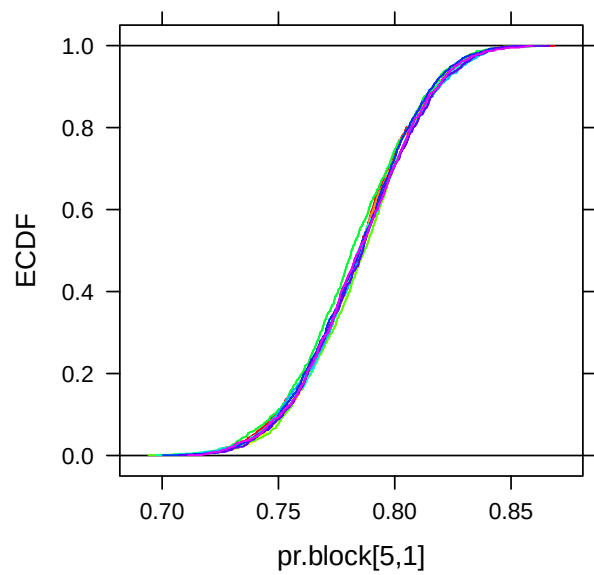
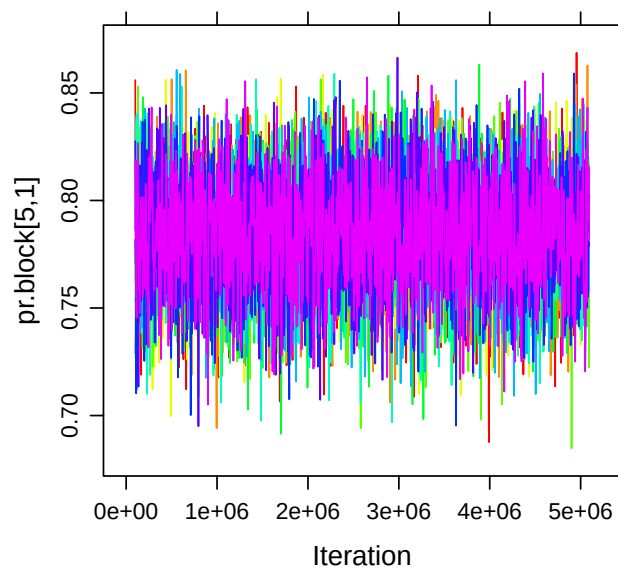


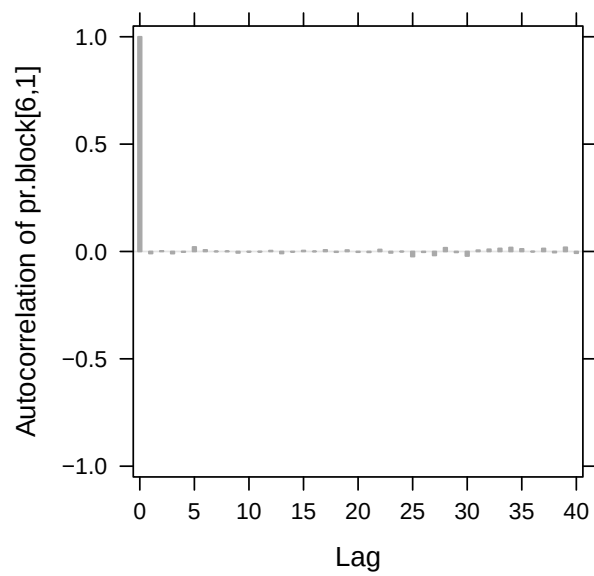
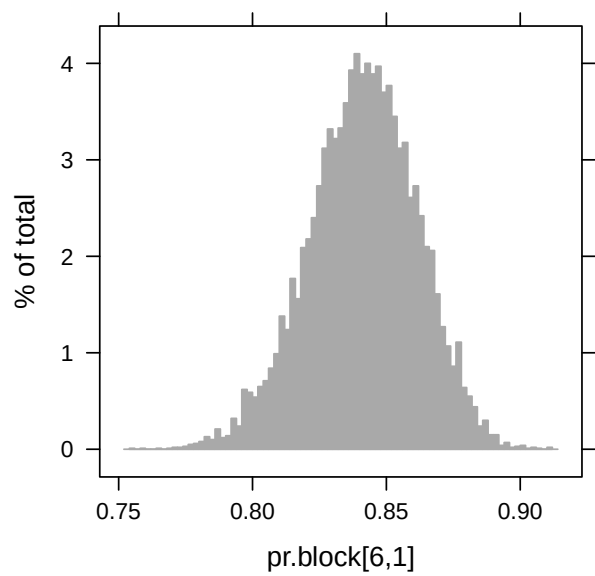
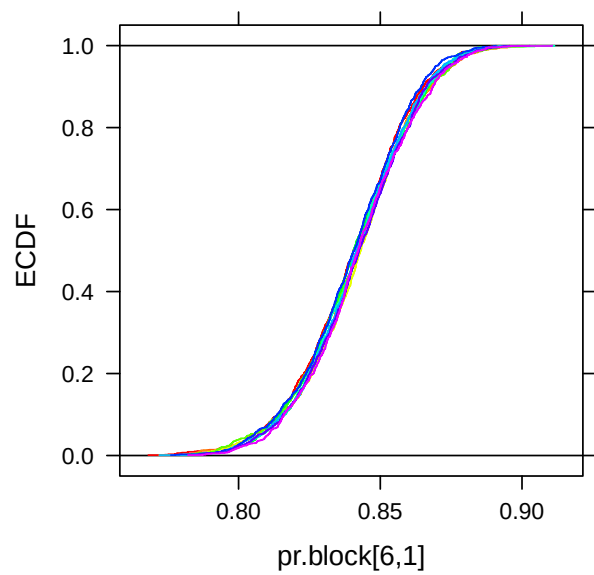
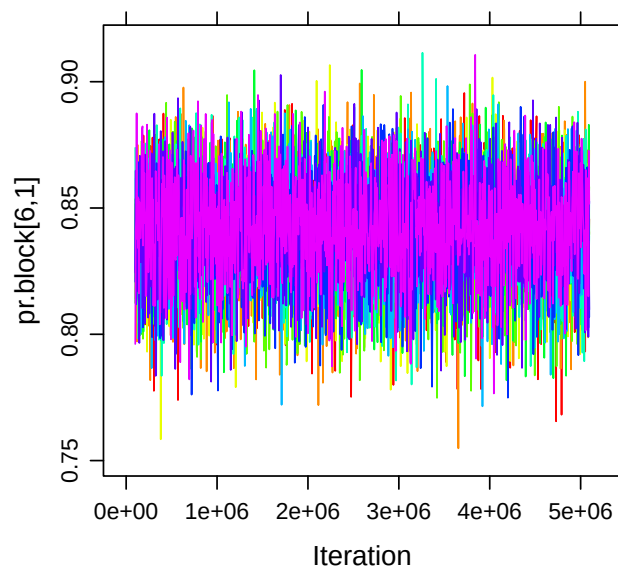


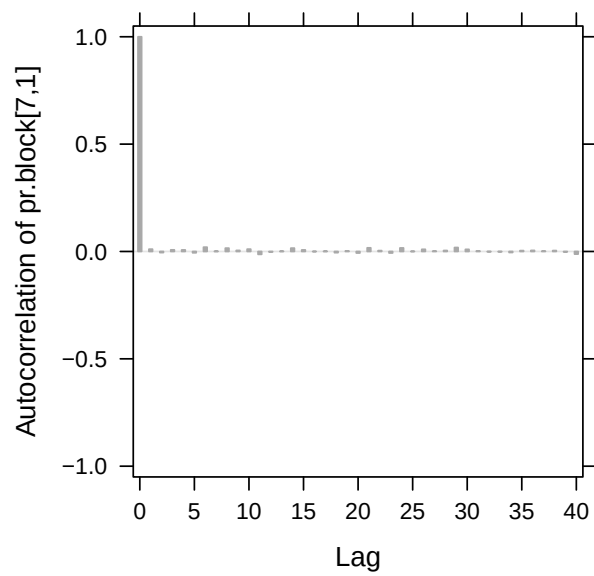
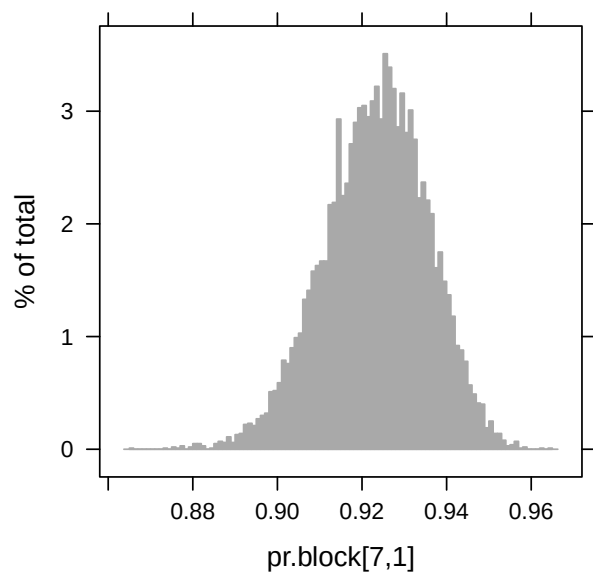
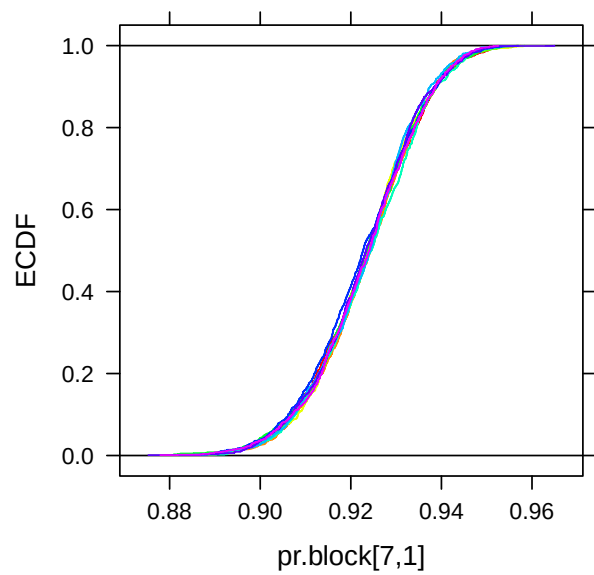
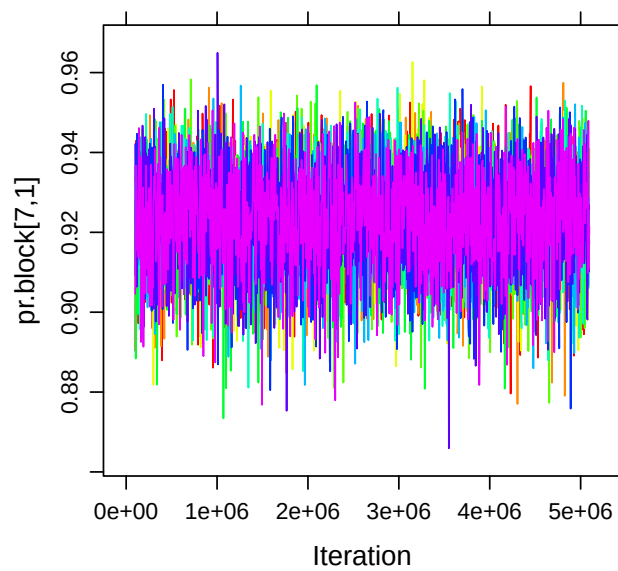


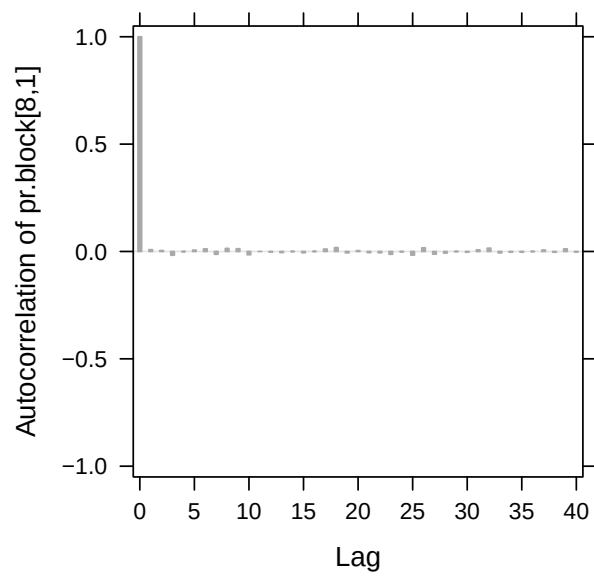
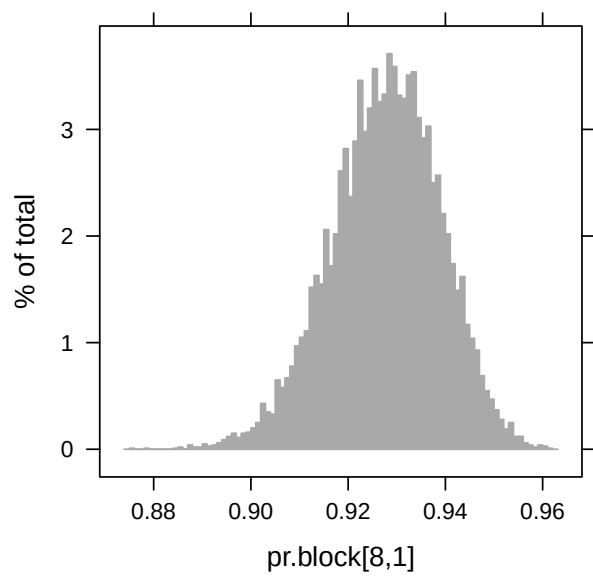
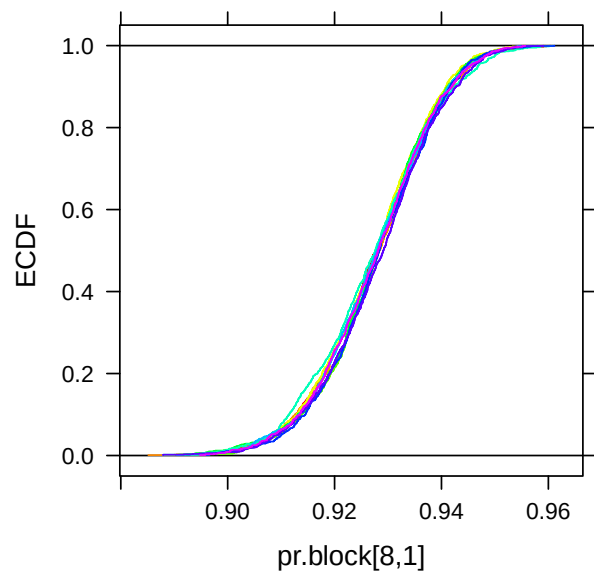
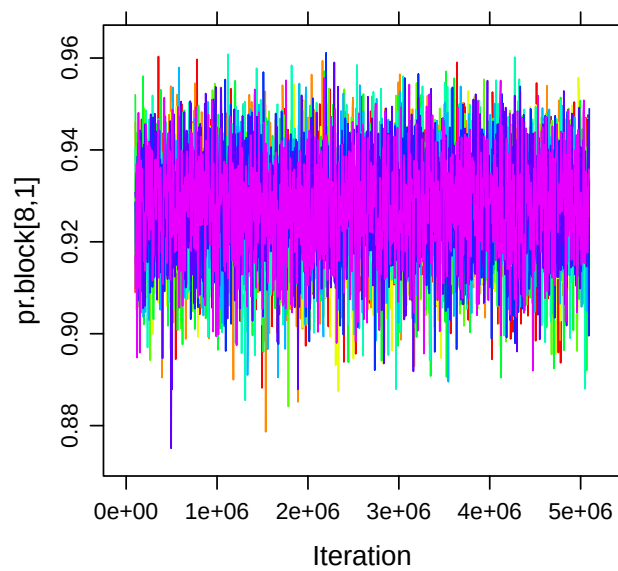


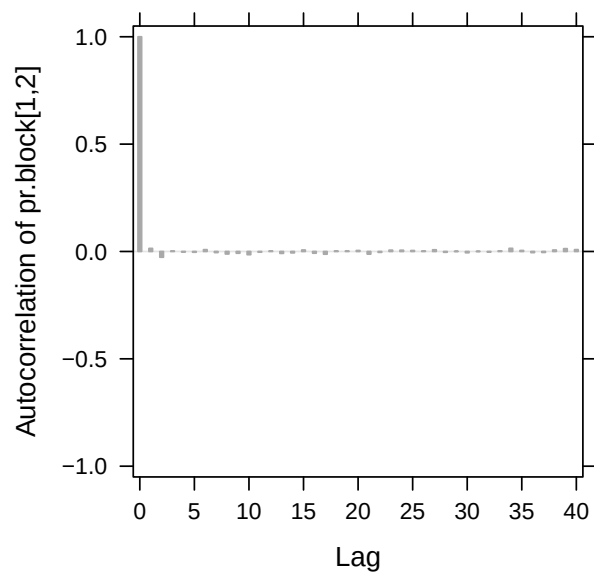
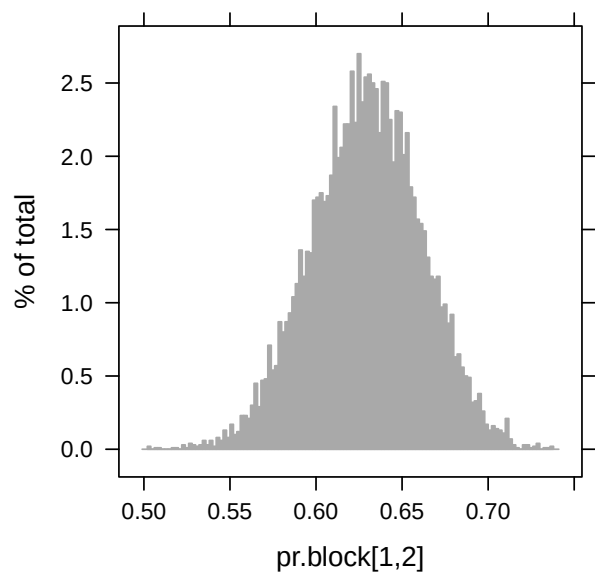
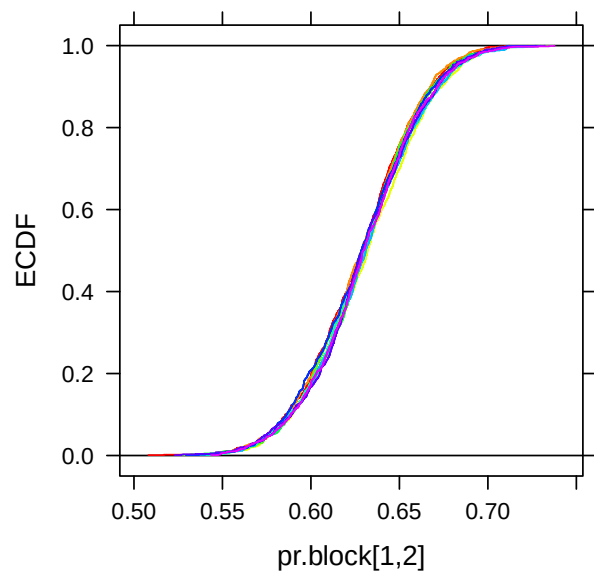
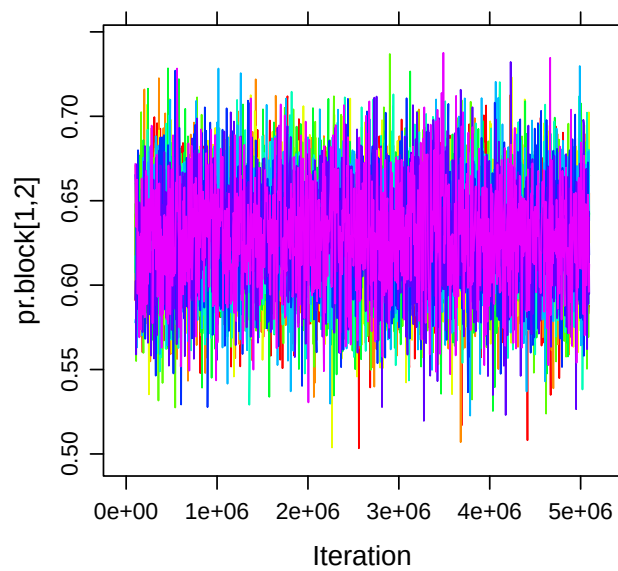


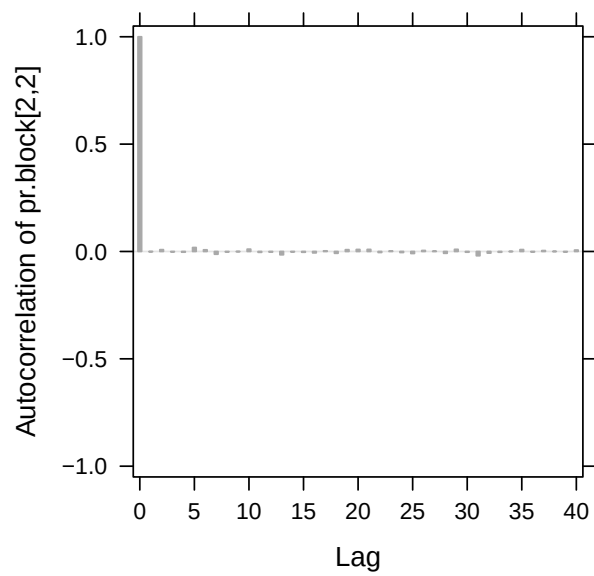
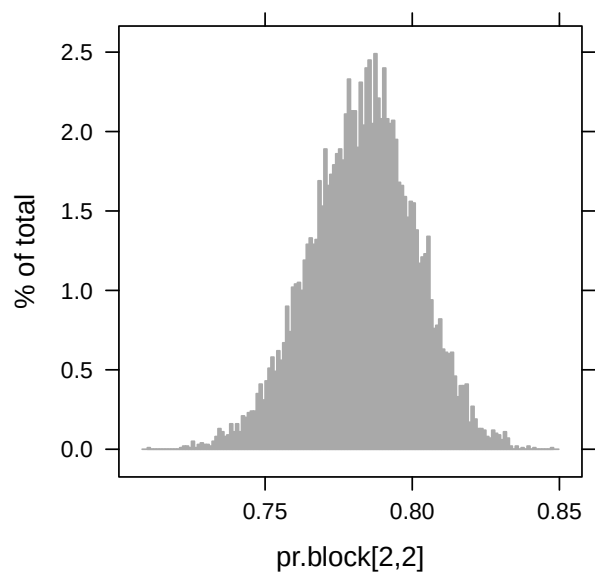
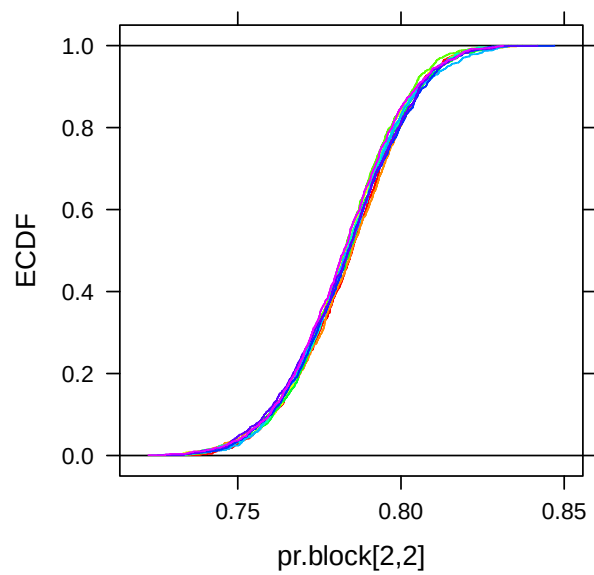
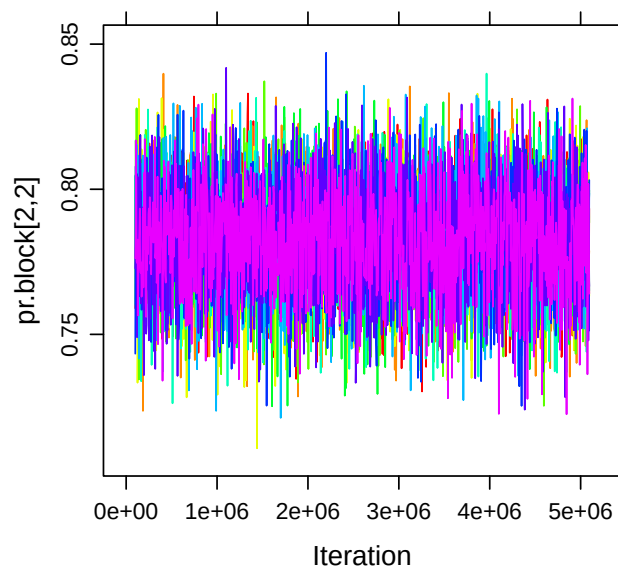


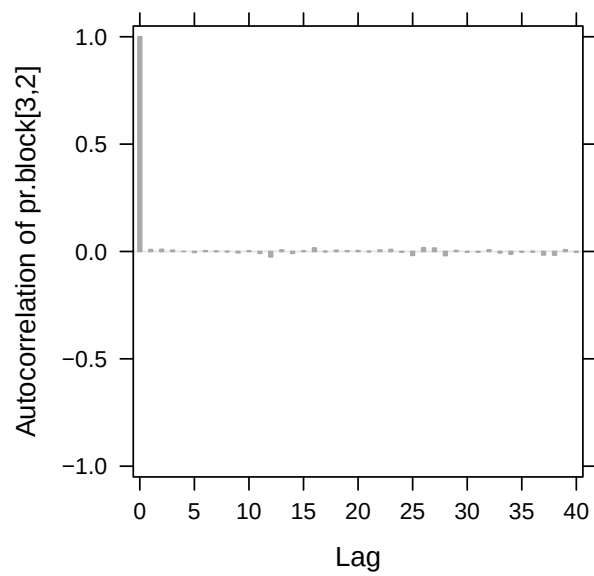
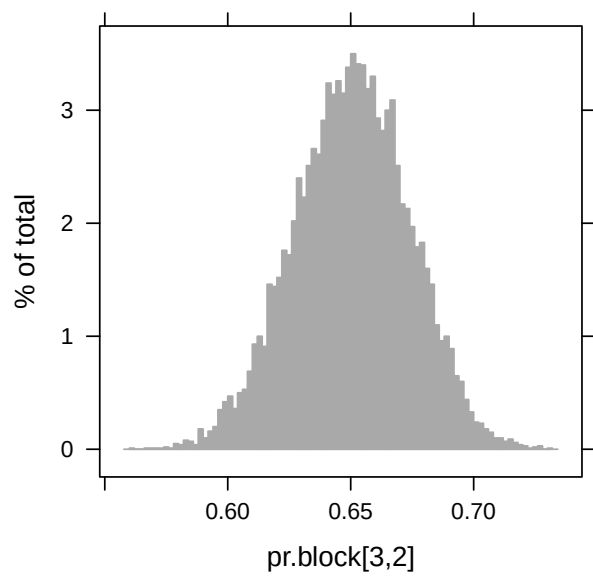
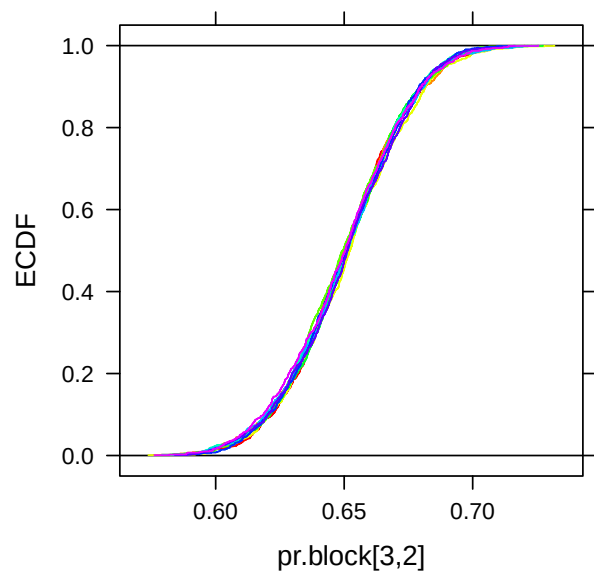
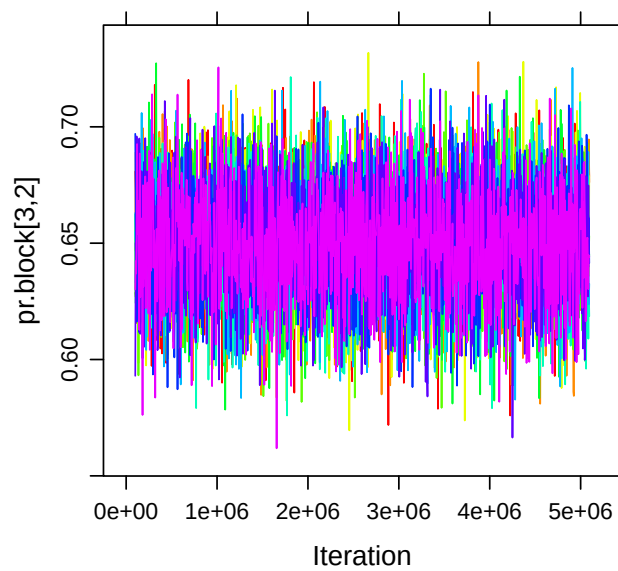


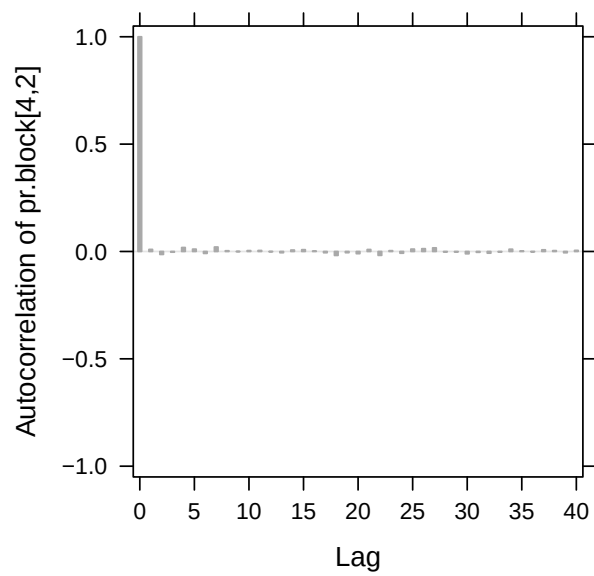
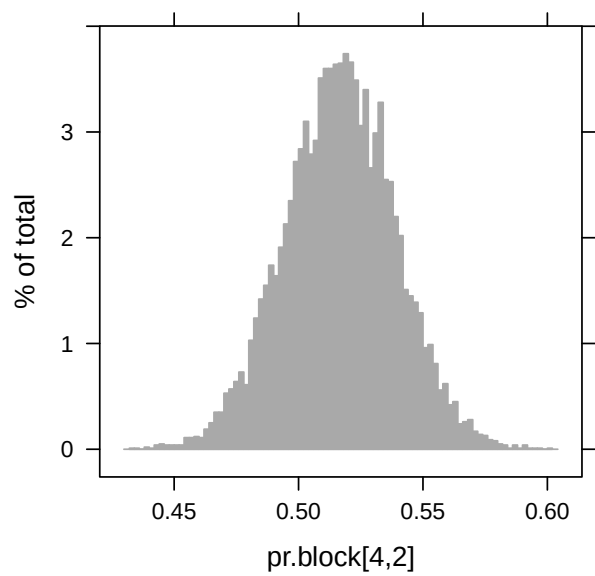
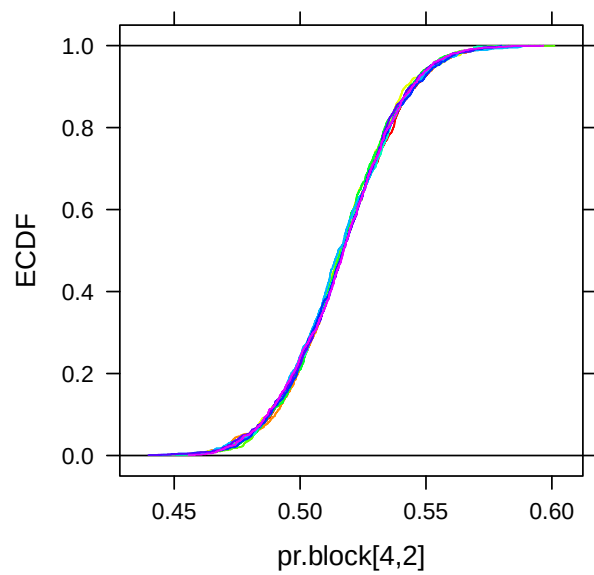
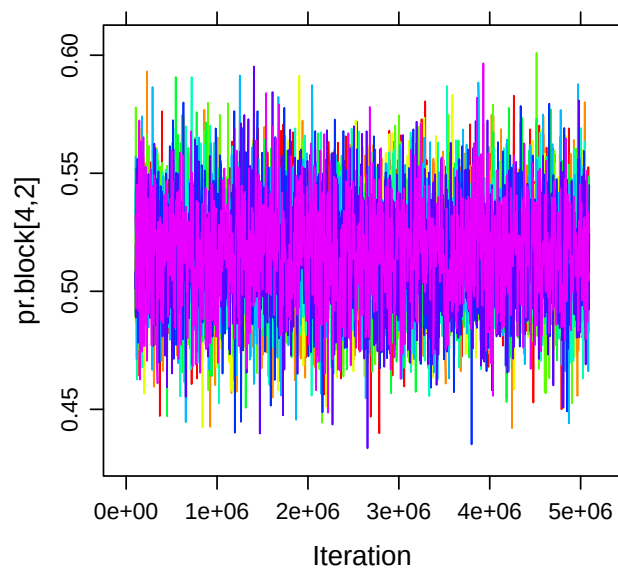


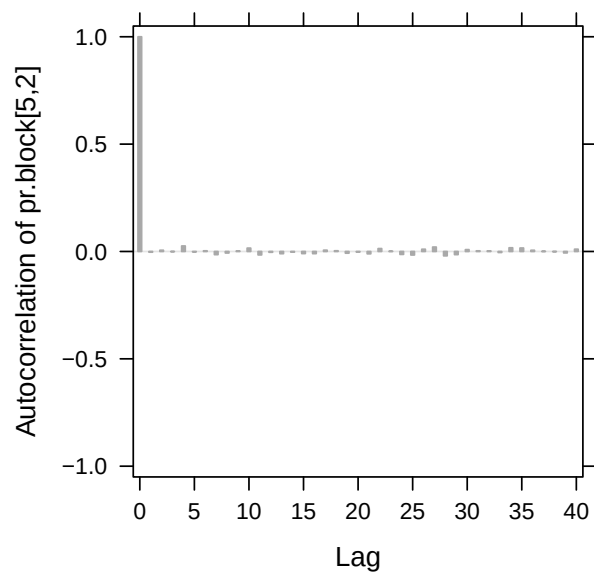
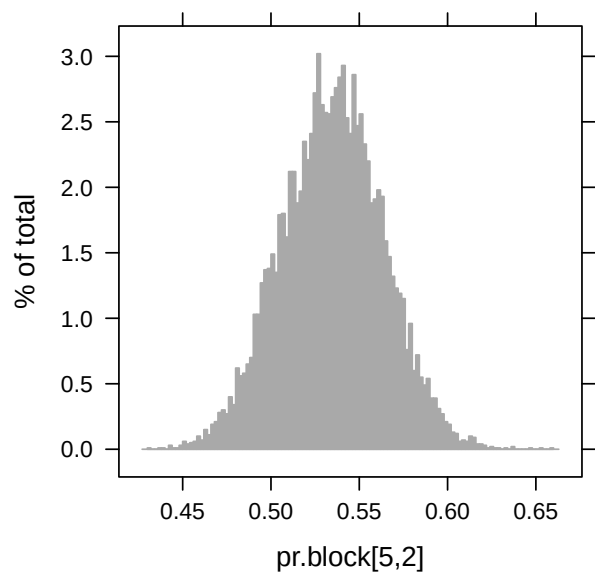
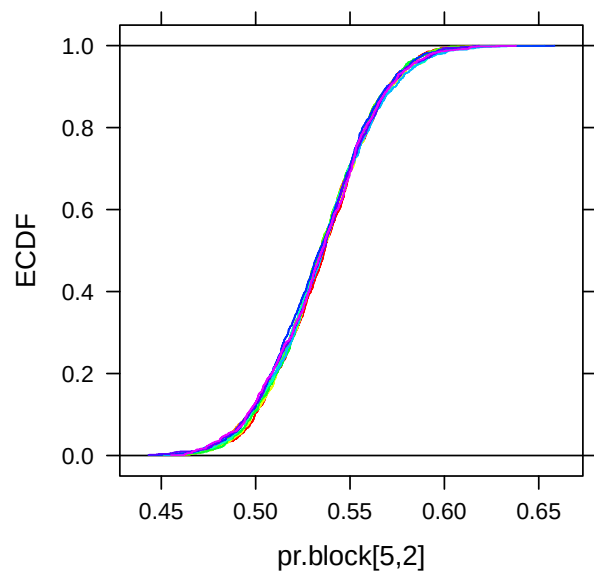
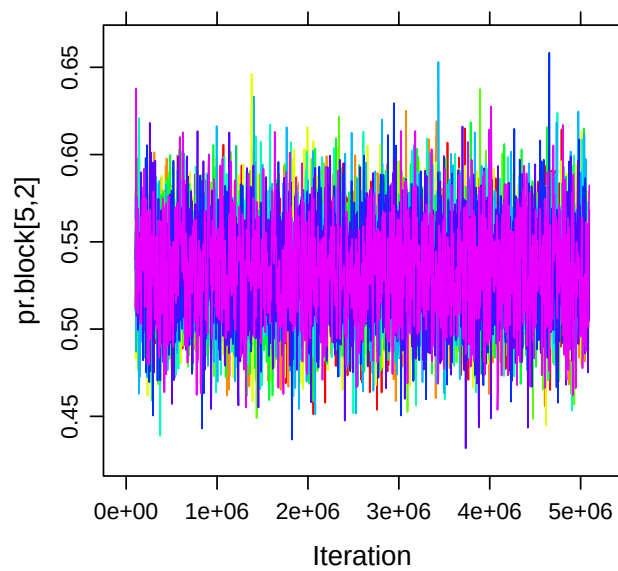


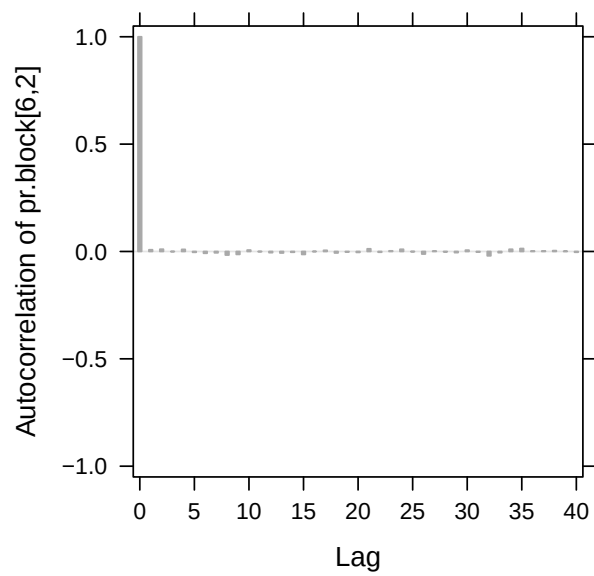
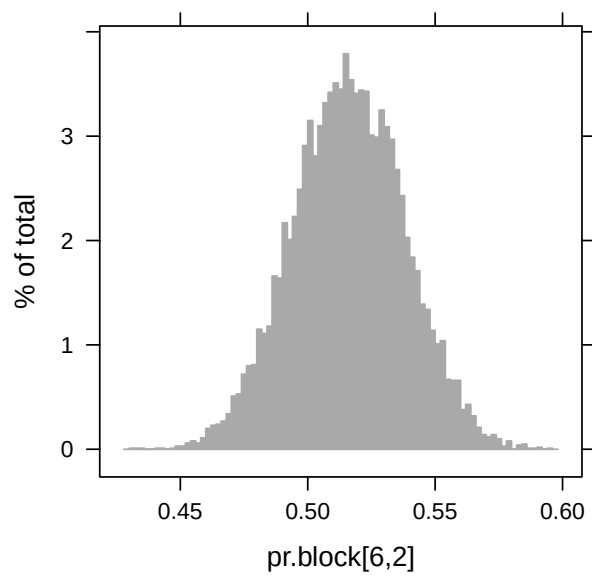
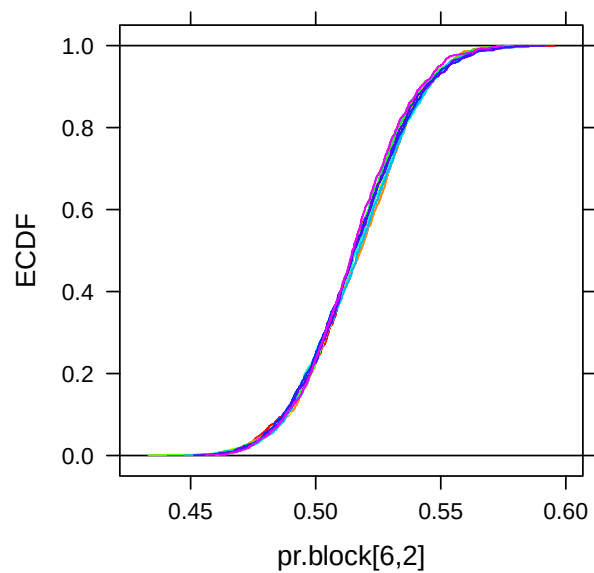
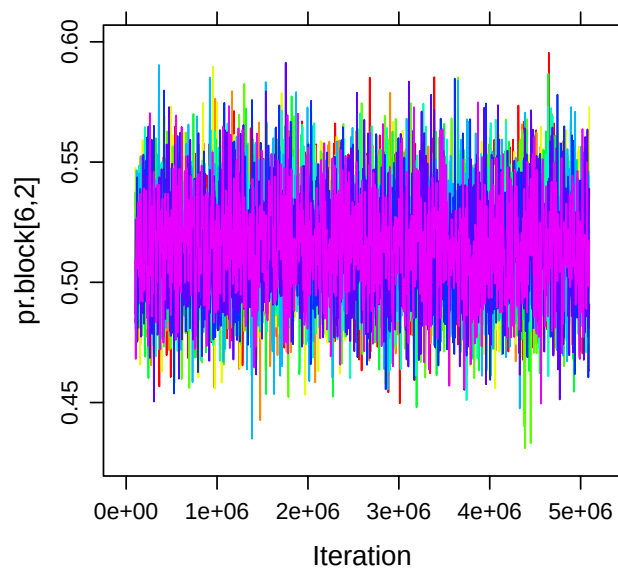


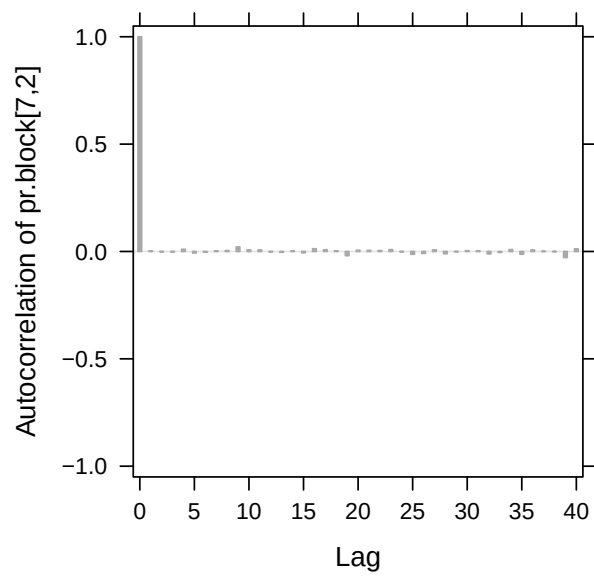
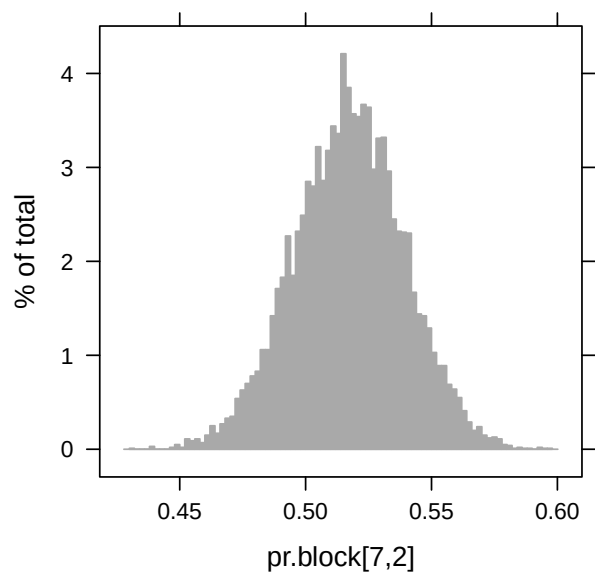
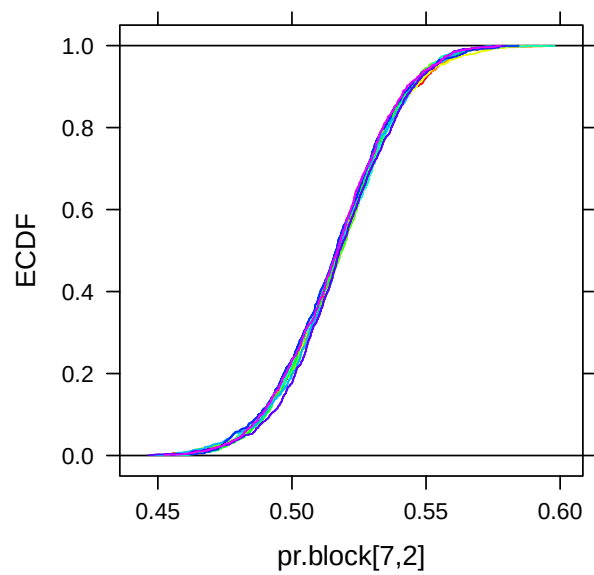
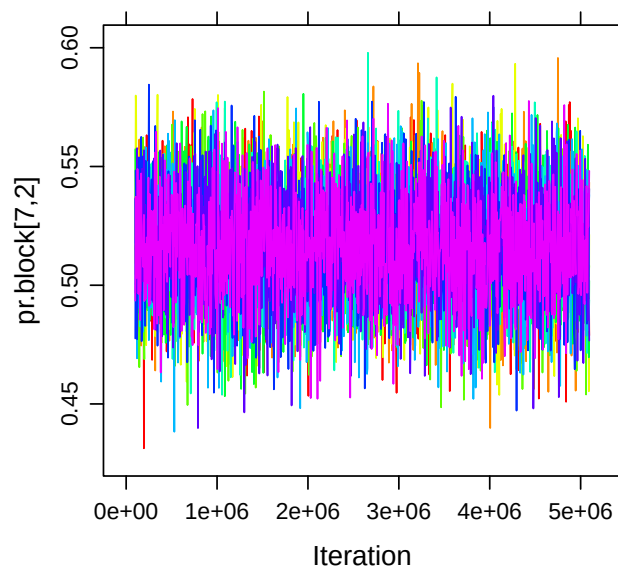


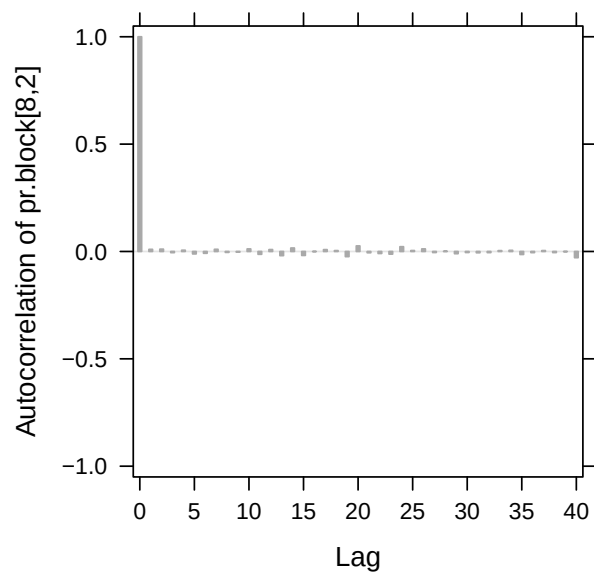
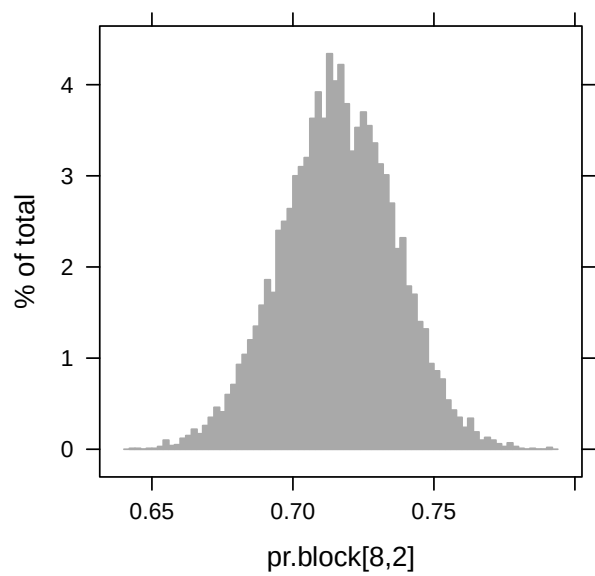
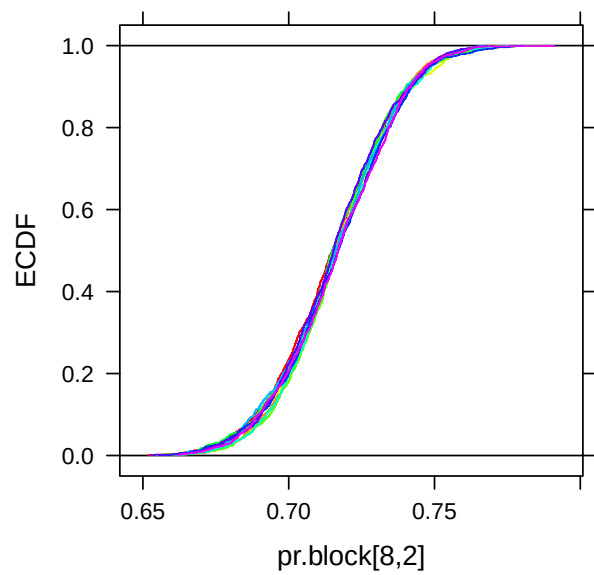
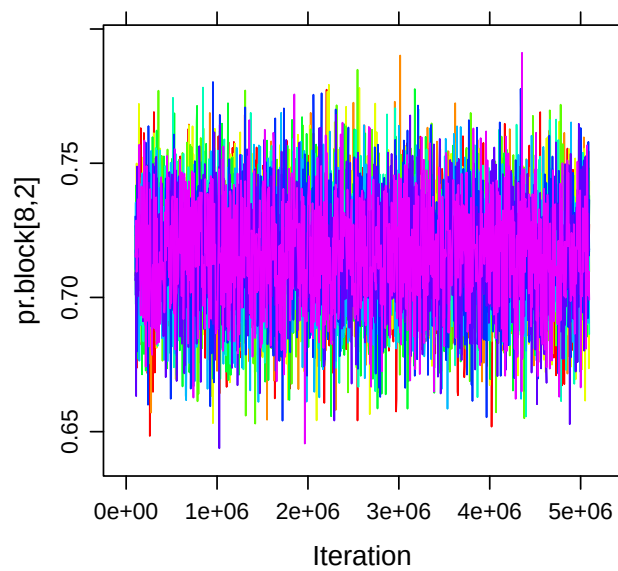


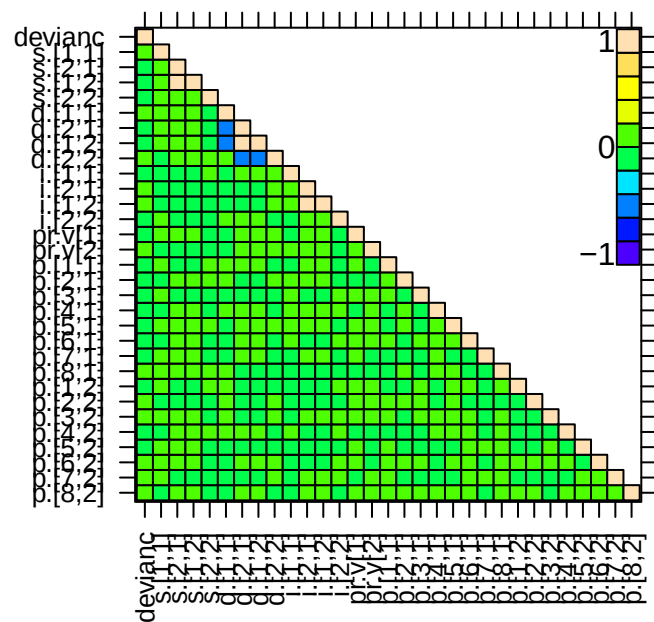












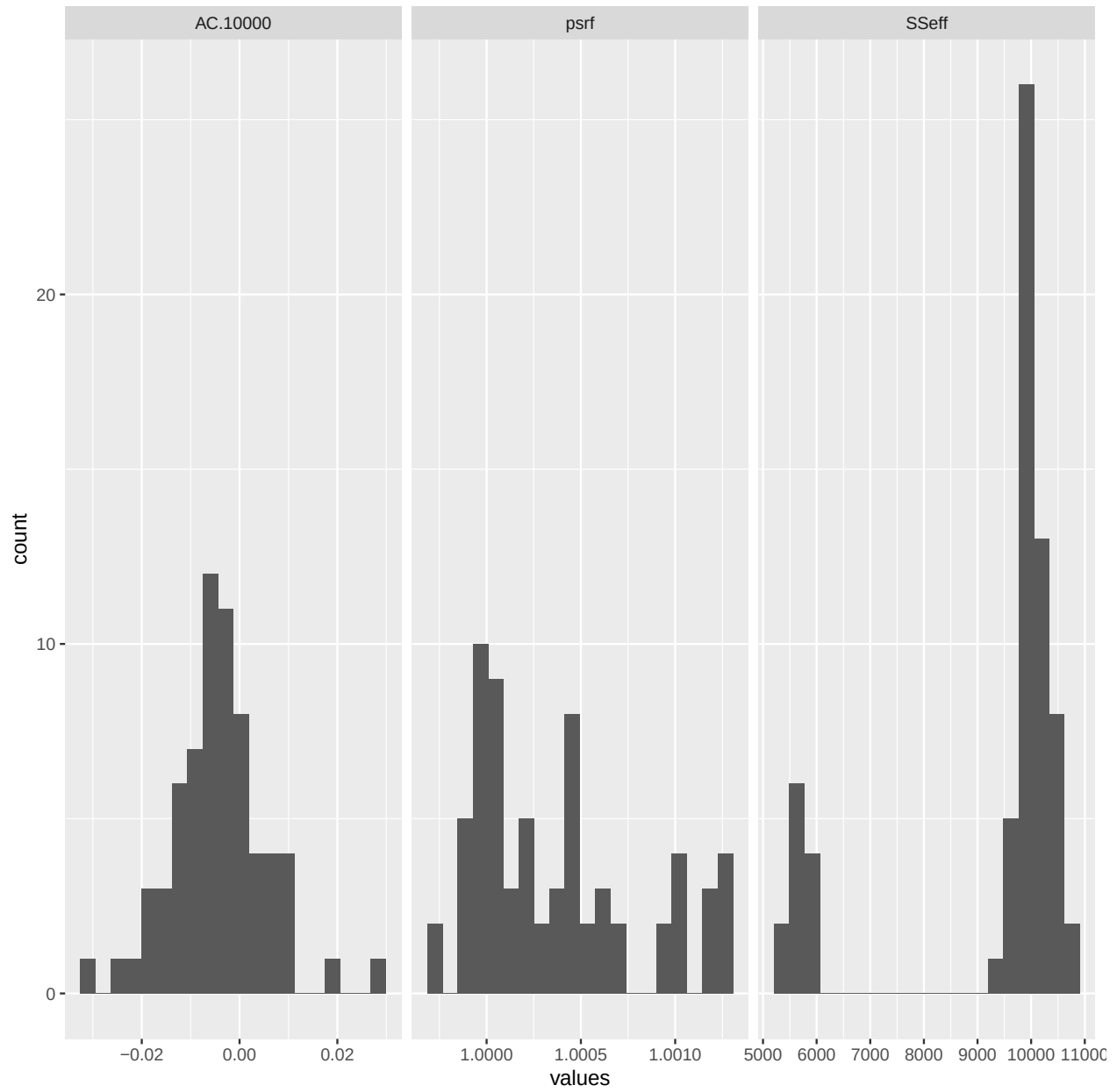
Model IV

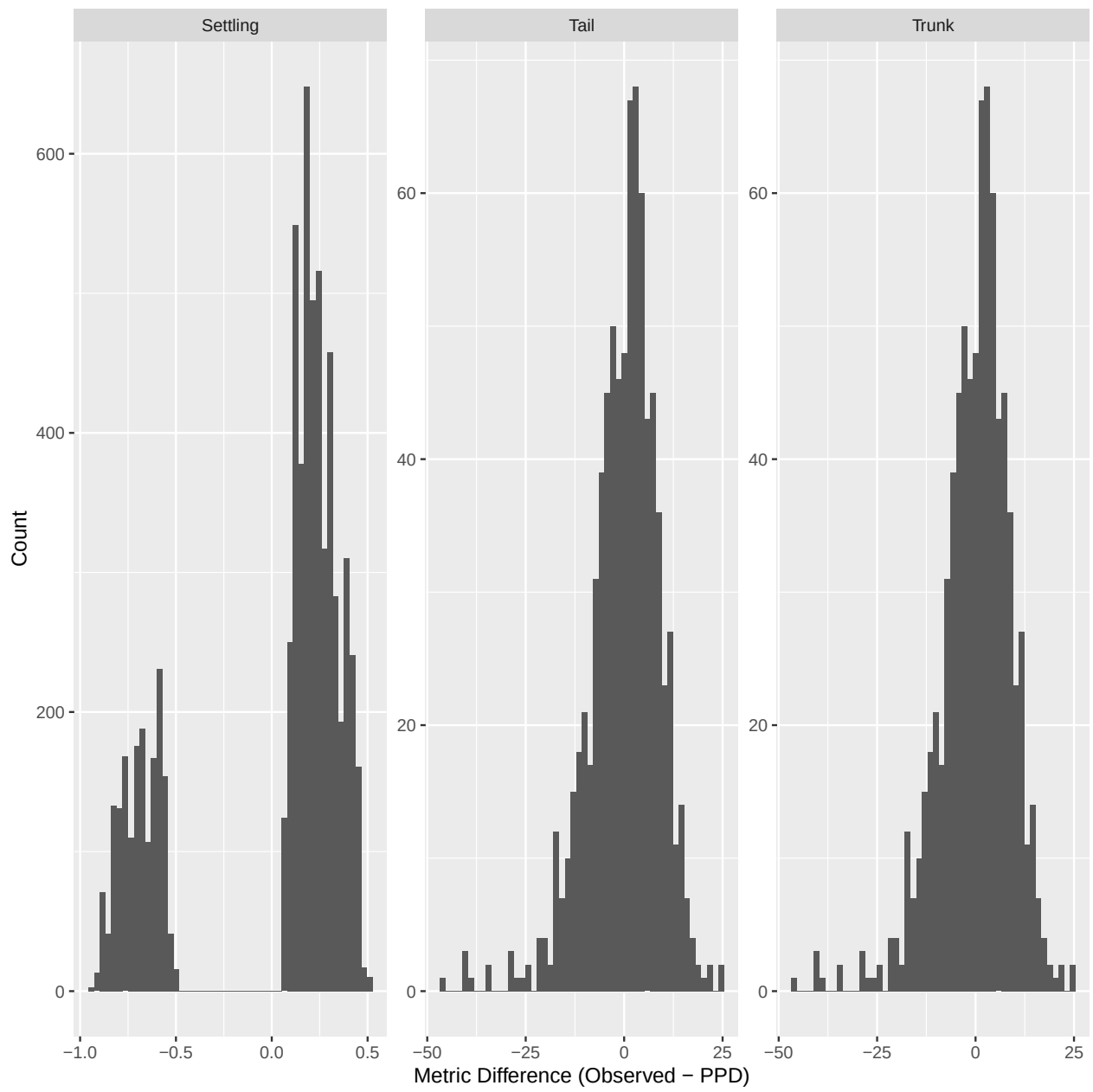
Number of chains: 10

Burnin: 1.001×10^5

Samples per chain: 1000

Total samples: 10000





Generating summary statistics and plots (these will NOT be saved for reuse)...

Calculating summary statistics...

Calculating the Gelman-Rubin statistic for 67 variables....

Note: Unable to calculate the multivariate psrf

